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Not all Emerging Markets are equal: Hormuz, triple deficits, and the new energy risk premium

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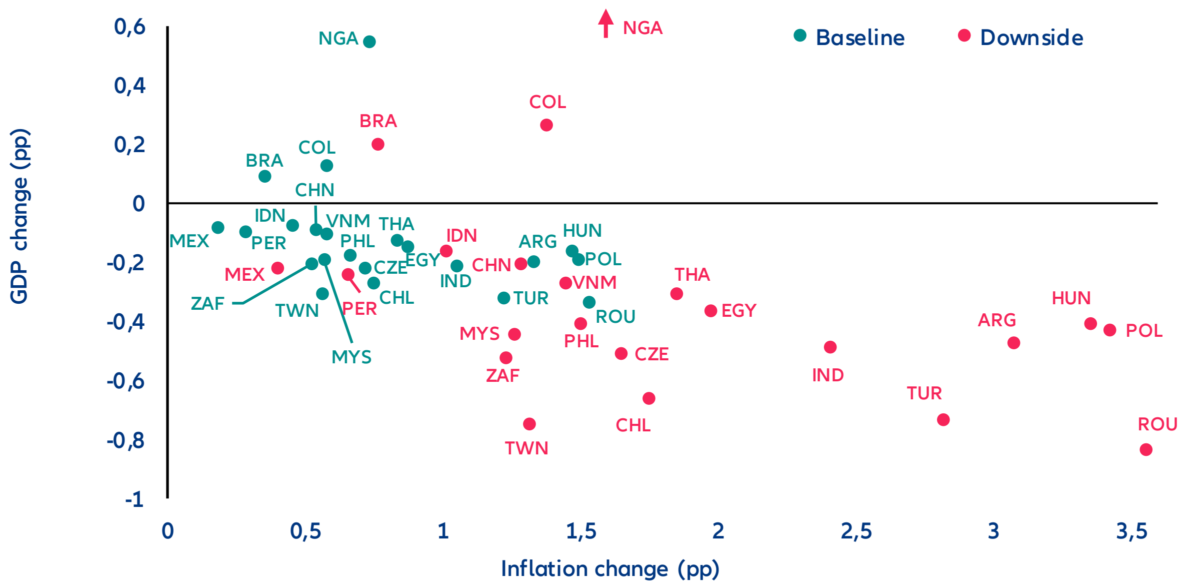
In Summary

- **Less than 2 months' disruption of the Strait of Hormuz could push average EM inflation higher by +0.8-1.0p with limited recessionary effects – apart from GCC countries.** We estimate that the closure of the Strait of Hormuz up to six weeks would result in a -1.6pps decrease in GDP for Saudi Arabia and -3.3pps for the UAE. Tourism – a key pillar of diversification across the Gulf – would also be hit in the short term, with knock-on effects on FDI and mega-project timelines including AI.
- **From price shock to supply shock? As the conflict drags on, Asian economies could face supply disruptions on top of stronger inflationary shocks, given that 56% of their oil imports and 30% of their total gas imports originate from the Middle East.** In Asia, Taiwan, Vietnam, Thailand, Pakistan, Bangladesh and Sri Lanka are more exposed to a supply shortage while in Africa, Egypt, Ethiopia, Kenya and Tunisia would suffer the most from a global oil supply shortage, given their exposure to Middle Eastern hydrocarbons. A temporary shortfall could be partially mitigated through adjustments in the energy mix (i.e. coal and renewables to a lesser extent), but energy supply disruptions for an extended period would require demand rationalization of -5 to -7% in the final energy consumption should prices double.
- **Beyond three months of closure for the Strait of Hormuz, many more EM countries are at high recession risk as they run triple deficits (fiscal, current account, energy).** GDP growth impact would average at least -0.5pp to 3.1% for emerging markets excluding China. Bangladesh, Egypt, Ethiopia, Jordan, Kenya, Morocco, Pakistan, Poland, Romania, Sri Lanka and Tunisia would be most at risk. Meanwhile, a second group of economies sits in moderately high risk as they have more room to maneuver to support their economies: Chile, China, Hungary, India, the Philippines, Taiwan, Thailand and Türkiye. By contrast, large commodity exporters such as Brazil and Mexico appear structurally resilient despite fiscal deficits as energy exports cushion the impact of higher prices.
- **The shock came at a supportive moment for EM carry. It calls for selectivity along a targeted energy risk premium.** Early market repricing has begun: FX markets reacted quickly, with the Egyptian pound experiencing the largest depreciation (-9.2%), followed by the Hungarian forint (-8%) and the Chilean peso (-4.9%). Repricing in local bond markets reveals a highly differentiated picture. In Mexico, the rise in nominal yields is largely explained by higher inflation expectations, while in Central and Eastern Europe – where the sell-off has been strongest – markets are pricing a larger share of risk and liquidity premia. The shock also complicates the policy outlook: With energy-driven inflation risks rising, many EM central banks are likely to remain on hold for longer despite slowing growth. A prolonged conflict could see inflation expectations repricing more forcefully across EM curves, steepening local yield curves and delaying monetary easing cycles. Overall, the adjustment is likely to remain selective rather than systemic. The most likely outcome is therefore the emergence of a targeted energy risk premium across vulnerable EM economies rather than a broad-based sell-off across the asset class.

From price shock to supply shock?

The 40% spike in energy prices in the first week of the conflict is the most rapid transmission channel to the global economy. On 9 March, the oil benchmark briefly reached USD120/bbl, later dropping to USD80/bbl. At the time of publication, oil remains around 15% above its pre-conflict level. Even if energy transportation is disrupted for only a short period, the conflict will still have a lasting impact on energy prices as it would take several weeks for production and supply to go back to pre-conflict levels. We expect prices to return closer to USD70/bbl, which would still be 16% above the pre-conflict baseline. This shift could push average EM inflation higher by 0.8-1.0pp, but the magnitude will depend on countries' energy exposure (particularly across Asia and energy-importing EMs). CEE economies – particularly Hungary and Romania – face the largest GDP hit given acute energy import dependency, followed by Thailand and Chile. Oil exporters such as Nigeria, Colombia, and Brazil would be relatively shielded, while Indonesia stands out among importers as more resilient given significant domestic energy production capacity.

Figure 1: Change to inflation and GDP growth rates in 2026 in selected EMs, baseline vs downside scenario



Sources: Oxford Economics, Allianz Research

The most immediate economic consequences are concentrated in the countries closest to the conflict — across the Gulf and the wider Middle East — where normal economic activity has been directly disrupted by the conflict. Early estimates of a short-term conflict show a GDP drop for Saudi Arabia and the UAE of -3pps and -4.3pps, respectively, considering major economic disruptions, the fall in tourism and a temporary drop of energy exports due to the effective closure of the Hormuz Strait. However, a short-lived conflict would also bring a quick rebound, with rates of +6.5% in Saudi Arabia and +7.6% in the UAE. The GDP loss in the entire Gulf Cooperation Council would be -3.3pps, with a considerable rebound of +6.4% of GDP growth in 2027. Dubai's real estate sector — one of the world's most dynamic property markets in recent years — took a significant hit on the Dubai financial market in the week since the conflict began, with main listed developers repricing sharply between -13% and -17%: The rest of the region, specially Kuwait and Bahrain, will be the most impacted given its dependence on the Hormuz Strait for both exports and imports. Bahrain's sovereign CDS spreads have widened by nearly 40% since the outbreak of hostilities — the sharpest move within the GCC complex and a clear market signal that investors are differentiating sharply between Gulf credits on the basis of fiscal resilience and hydrocarbon self-sufficiency.

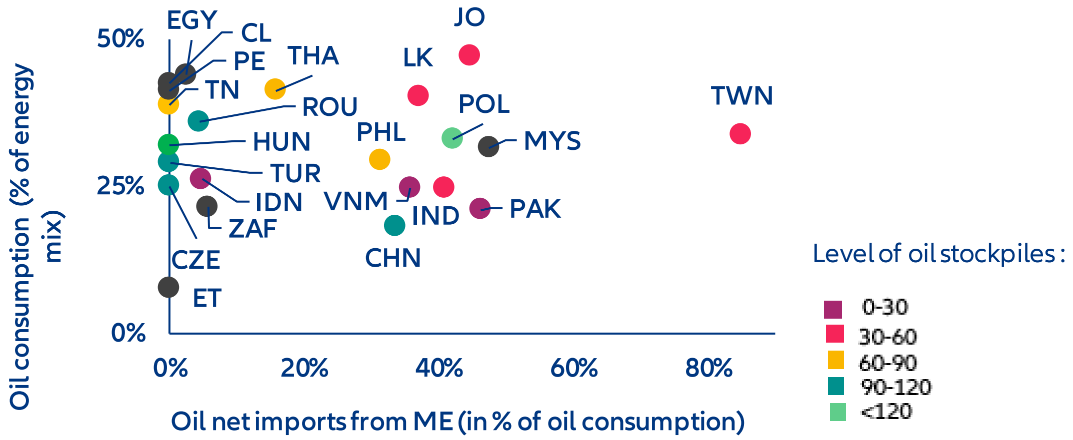
The long-term war scenario would result in harsher consequences via a large drop in hydrocarbon exports and reduced confidence in the region. History indicates several scenarios for how the conflict could impact the region. The Iraq-Iran tanker war in the 1980s, which threatened the closure of the Strait of Hormuz, had large long-term implications for Saudi Arabia, with the Kingdom growing at an average of +0.5% between 1981-1988 due to lower exports of oil. While both Saudi Arabia and the UAE both have some capacity to continue oil exports through different pipelines that connect their oil fields to ports outside of the Gulf, a sustained closure of the Hormuz Strait would greatly reduce both countries' hydrocarbon exports. Kuwait, Qatar, Bahrain and Iraq, which do not have much capacity to export without circulating the Strait of Hormuz, would suffer the most. Saudi Arabia and the UAE have both also enjoyed large inflows of investment and people, especially since the pandemic due to their open borders, tax exemptions for residents and increasing leisure and tourist attractions. Historical precedent suggests Gulf tourism could recover within one to two years. But if the conflict drags on and reshapes the region's security perception durably – as happened with Egypt between 2013 and 2017 – the recovery timeline extends to five to seven years, with compounding effects on FDI, mega-project timelines, the broader Vision 2030 ambitions and the region's ambition to become a global AI hub. The physical risk of data centers (Iran has targeted at least three data centers operated by Amazon in the UAE and Bahrain) and the consequences that this brings to economic activity in the country as it could paralyze banks, government offices, etc. Beyond the GCC, the rest of the region could be further impacted, with countries as far as Egypt or Türkiye recording drops in tourism.

A second wave of impact would come from hydrocarbon shortages from a longer closure of the Strait of Hormuz. This would turn into a significant supply shock – a qualitatively different and more damaging scenario that in most cases would lead to a recession in countries dependent on energy from the Gulf. Asia would be the most impacted region, given the high dependence on Middle Eastern hydrocarbons, with 56% of the region's oil imports coming from the Gulf. Gas prices could potentially double, which could trigger a fall in gas consumption of 5-7% in Asia. Building from the European precedent, we estimate that every -1% drop in gas consumption would equal 0.08–0.10pp GDP growth loss. Hence, for Asia, this could mean an impact above -0.5pp of GDP for the most impacted countries such as South Korea (-0.7pp) or a drop below 0.2pp as it is the case for India (-0.1pp).

The impact across Asian economies would be asymmetric, with Taiwan, Pakistan and Vietnam among the most impacted while Malaysia, Indonesia and China would remain shielded. Taiwan emerges as the most exposed country through its high dependence on Middle Eastern hydrocarbons, high dependence on oil in the energy mix and very low levels of oil stockpiles domestically. Pakistan and Vietnam are also in a delicate situation, with around 40% of oil consumption dependent on Middle East sources and very low reserves. However, their energy mix is slightly less dependent on oil than other EMs with a similar dependence on the Middle East but higher reserves. Despite reserves of around a month, India, and more particularly Sri Lanka, present similar characteristics and are also highly exposed. Thailand is another economy at risk: Despite relatively well-filled reserves and a low dependence on the Middle East, the country's high reliance on oil in its energy mix could result in a higher inflationary impact as it will need to buy more oil compared to peers. Similarly, if the conflict and closure of the Hormuz strait were to last, the Philippines and Indonesia could also start seeing some inflationary impact as more and more countries start competing for new partners. Despite a relative dependence on imports from the Gulf, China benefits from one of the highest levels of reserves in the region combined with a relatively low dependence on oil in its energy mix. And Indonesia should be shielded from an inflationary shock despite low reserves, given its low dependence on the Middle East, but also on the rest of the world.

Non-Asian emerging markets are comparatively better insulated from the Strait of Hormuz disruption. European emerging markets are relatively shielded by their low exposure to Middle Eastern crude – with the exception of Poland – relatively low oil dependency in the broader energy mix and solid strategic reserves. Despite higher reliance on oil, Latin America countries are amongst the most isolated from the shock due to a very low reliance on Middle East imports.

Figure 2: Oil dependency on the Middle East in selected EMs



Note: Oil dependency is computed as net imports of crude oil from Middle East divided by total oil consumption.

Sources: IEA, UN Comtrade, Allianz Research

The final macroeconomic impact would also depend on the scale of government support measures implemented to cushion households and firms. Policy responses could significantly mitigate the inflation shock, as seen during the 2022 energy crisis in Europe. At that time, European governments deployed energy-support measures equivalent to roughly 4% of GDP over a year, including price caps, tax cuts and transfers. Assuming the shock remains roughly half the size of the 2022 European energy shock, Asian economies, given their higher dependence on imported energy, would likely require subsidies of around 0.25% of GDP (or about 2% annualized) to smooth the impact on households and firms. In the CEE region, Hungary has already introduced fuel price caps, increasing pressure on the fiscal deficit running already at -5.1%. In Asia, Indonesia, South Korea, and Japan are providing continued support to their populations through energy subsidies and oil price caps. In the Middle East, Egypt sits in a precarious position, subsidies remain substantial, with the 2025/26 budget already allocation 0.6% of GDP, with a fiscal deficit already at -10% and a matched external deficit, Cairo could fall into fiscal slippages. During the last shock, UAE and Qatar supported Egypt with large transfers of capital, but with new priorities in the Gulf capital's due to the war, it remains a question how Egypt could sustained a long-period of high energy prices.

Despite potential mitigations channels, energy supply disruptions would require demand rationalization. While part of the shortfall could be mitigated through adjustments in the energy mix – for instance via a temporary ramp-up in coal generation – and roughly half of the disrupted volumes could potentially be rerouted or offset by higher oil production elsewhere (notably in the US or Russia), supply substitution would only partially compensate for the disruption. In that scenario, energy-demand rationalization would become unavoidable, affecting both companies and households. Several Asian economies — from Japan to Thailand — are feeling the strain of the Hormuz closure and have moved quickly to manage supply, implementing measures ranging from fuel export bans and energy-saving policies such as four-day workweeks and remote working mandates, to ramping up oil purchases from Russia. The European experience in 2022 illustrates the scale of adjustment that may be required. At the peak of the crisis when gas prices tripled to 330 EUR/MWh, and for a dependency to Russia of more than 40% of total EU gas imports, Europe managed to reduce gas consumption by almost -20% driven by high prices, supply shortages and emergency policy measures.

The Middle East represents 30% of total natural gas imports to Asia, again with some disparities across individual countries. With 80% of its imports coming from the Gulf, India is the most exposed to the repercussions of the conflict on gas prices, followed by Vietnam and Indonesia. However, as natural gas does not represent a high share of these countries' energy supply, the impact should remain relatively muted. Though the shock will be much less than that of oil, Thailand, Taiwan and Pakistan should be carefully monitored as they present a larger share of natural gas in their consumption mix compared to the region combined with a relative dependence on imports form the Middle East, especially for Taiwan and Pakistan where the Gulf represents more than 25% of their consumption..

More severe and systemic spillover effects would materialize if the Strait of Hormuz remains closed beyond the short-term threshold, transforming what began as a price shock into a structural disruption. Each additional week of Hormuz closure compounds recessionary pressure as reserves deplete and supply shortages broaden. In a scenario extending beyond three months, EM GDP takes an average hit of -0.5pp or more, with the most energy-dependent economies bearing a disproportionate share. In that scenario, emerging markets running twin deficits – fiscal and current account – would face acute pressure as deteriorating energy import bills compound already-stretched external balances and financing needs. Those carrying a triple deficit – where an energy balance deficit layers on top – would be most severely impacted, facing simultaneous pressure on their currencies, sovereign spreads and growth trajectories, with limited policy space to respond.

Countries combining fiscal deficits, external imbalances and energy import dependence emerge as the most exposed to a prolonged disruption in Middle Eastern energy supply. Economies such as Romania, Poland and Tunisia display the clearest triple-deficit configuration, where sizeable fiscal deficits coincide with current account gaps and structurally negative energy balances. In these cases, higher oil prices would simultaneously widen external deficits, worsen fiscal dynamics through energy subsidies or weaker growth and weaken currencies through deteriorating terms of trade. Several frontier and lower-income importers – notably Egypt, Sri Lanka, Kenya, Ethiopia, Jordan, Morocco and Bangladesh – also appear particularly vulnerable given already elevated financing needs and limited macroeconomic buffers. In contrast, commodity exporters such as Nigeria, Colombia and Indonesia remain comparatively insulated as positive energy balances provide a natural hedge against higher oil prices, supporting both fiscal revenues and external accounts.

A second group of economies sits in an intermediate zone where vulnerabilities stem primarily from energy dependence rather than broad macroeconomic imbalances. Chile, China, India, the Philippines, Hungary, Taiwan, Thailand and Türkiye fall into this category. These economies run structurally negative energy balances and would therefore face pressure on external accounts if prices remain elevated, but the overall macro impact depends on policy space and external buffers. In several cases, relatively manageable current account positions, diversified economies or stronger reserve coverage could mitigate the shock. By contrast, large commodity exporters such as Brazil and Mexico appear structurally resilient despite fiscal deficits, as energy exports cushion the impact of higher prices. Overall, the table highlights that the transmission of a prolonged oil shock across emerging markets would remain highly uneven, with risks concentrated in economies where external imbalances, fiscal constraints and energy dependence overlap.

Table 1. Emerging markets exposure, market reactions and coping mechanisms

	Exposure to the Middle East crisis				Market reaction since Feb 27		Coping mechanisms					
Country	Energy balance (% of GDP)	Strategic oil reserves (days of consumption)	Vulnerability to energy price shock	Vulnerability to ME supply shock	FX (% change vs USD)	Bond yields (bps)	Fiscal balance (% of GDP)	Current Account (% of GDP)	FX reserves (months of imports)	FX outlook	Central Bank stance (by mid-	Risk of fiscal slippage throughout
Argentina	0.6	NA	Medium	Low	-0.04	11.0	-1.5	-0.4	3.8	Neutral	On hold	Low
Bangladesh	-3.5	49	High	High	-0.41	32.1	-3.9	-0.9	4.9	Negative	On hold	High
Brazil	0.9	NA	Low	Low	-3.77	88.3	-7.5	-2.3	12.1	Positive	On hold	Medium
Chile	-4.4	25	High	Medium	-4.94	18.5	-0.6	-2.2	6.7	Negative	On hold	Low
China	-2.4	100	Medium	Medium	-0.56	38.0	-5.0	2.8	12.3	Neutral	On hold	Medium
Colombia	3.9	NA	Low	Low	1.31	N/A	-5.7	-2.6	9.6	Negative	Hike	Medium
Czechia	-2.4	117	High	Medium	-4.47	58.8	-2.1	0.4	9.3	Negative	On hold	Medium
Egypt	0.0	NA	Medium	High	-9.16	40.0	-10.7	-4.3	4.3	Negative	On hold	High
Ethiopia	-4.0	NA	High	High	-0.47	N/A	-1.7	-2.6	3.7	Negative	On hold	Medium
Hungary	-3.6	215	High	Medium	-8.02	94.0	-5.1	0.9	3.7	Negative	On hold	High
India	-3.7	30	High	Medium	-1.61	2.0	-1.8	-1.4	9.1	Neutral	On hold	Medium
Indonesia	1.1	20	Low	Medium	-1.04	40.7	-2.7	-1.2	7.1	Negative	On hold	High
Jordan	-8.0	45	High	High	0	-50.7	-3.5	-5.5	7.7	Negative	On hold	Medium
Kenya	-4.5	NA	High	High	-0.27	10.0	-5.6	-3.4	5.7	Negative	On hold	High
Malaysia	-0.4	NA	Medium	Medium	-1.23	7.6	-3.6	1.8	5.3	Negative	On hold	High
Mexico	0.7	NA	Low	Low	-4.25	48.5	-4.1	-0.3	4.3	Neutral	On hold	Medium
Morocco	-5.0	30	High	Medium	-3.28	21.5	-4.5	-1.5	7.4	Neutral	On hold	Medium
Nigeria	12.2	NA	Low	Low	-1.91	61.3	-3.7	3.6	12.0	Neutral	On hold	Medium
Pakistan	0.0	28	High	High	0.02	66.0	-4.1	-0.4	11.6	Negative	On hold	High
Peru	-0.5	NA	High	Low	-3.1	30.7	-2.2	1.2	14.8	Negative	On hold	Low
Philippines	-4.2	60	High	Medium	-3.59	68.8	-0.3	-3.5	8.2	Negative	On hold	Medium
Poland	-2.1	121	High	Medium	-4.86	83.1	-6.3	-0.8	5.7	Negative	On hold	High
Romania	-1.7	92	High	Medium	-3.54	101.0	-6.2	-6.6	6.0	Neutral	On hold	High
South Africa	-2.3	NA	Medium	Low	-6.39	96.0	-5.6	-1.2	5.6	Neutral	On hold	Low
Sri Lanka	-4.5	35	Medium	High	-0.63	4.4	-6.5	0.5	4.5	Negative	On hold	High
Taiwan	-5.1	40	Medium	High	-2.25	4.0	-0.2	13.1	15.9	Negative	On hold	Medium
Thailand	-7.8	60	High	High	-4.58	33.5	-2.5	1.3	8.9	Negative	On hold	Medium
Tunisia	-4.0	60	High	High	-3.07	N/A	-6.0	-2.0	4.9	Negative	On hold	High
Türkiye	-3.6	94	High	Medium	-0.64	233	-0.7	-1.3	2.7	Neutral	Hike	Medium
Vietnam	-2.5	15	Medium	High	-0.89	7.7	-2.3	2.4	2.3	Negative	On hold	Medium

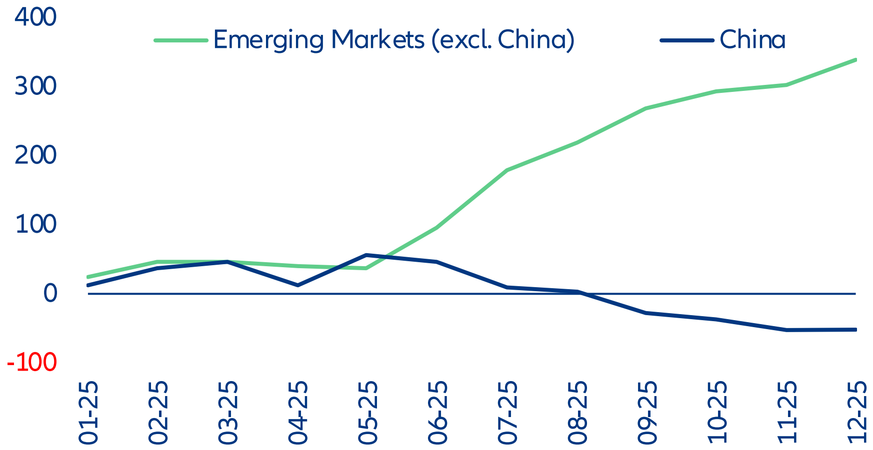
Sources: LSEG, Allianz Research. Note: Data as of 13/03/2026. In green central banks that were cutting before the conflict in the Middle East escalated.

Türkiye and India in particular sit at the intersection of competing blocs – geographically indispensable, politically non-aligned, economically interdependent – and risk paying a price. From a market perspective, both countries continue to post twin deficits (fiscal and current account balance). Last year, Türkiye returned to a positive primary balance for the first time since 2022, while India has had positive primary balance for some time. In the case of Türkiye, recent history shows that when energy prices were at their highest (2022, but also in the 2012-2014 cycle), Ankara has maintained a positive primary balance – at the expense of inflation. The current account will tend to be more negative this year, potentially hitting 2-2.5% of GDP if energy prices remain elevated, with effects on the lira and domestic inflation, but there may be room for subsidies. Fiscal discipline remains in place, but with debt/GDP at 25%, the value of gold reserves increasing further and the ever-present debate about bringing forward the presidential elections before their natural schedule in May 2028, the temptation to mitigate the effects of price increases in some way could also increase. India is also positioned with better footing than the 2022 scenario, with the fiscal deficit improved from the 2022 period (-0.8% in 2025 vs -2% in 2022) and the external balance at -0.2% of GDP. While an energy shortage could harm the country, India could shift oil and natural gas imports to domestic coal consumption, mitigating part of the shock. FX reserves remain very high at 11 months of imports.

Energy price premium ahead for emerging markets?

The new escalation in the Middle East could reverse the record gains of emerging markets. 2025 began with pronounced outflows for emerging markets (excluding China), reaching a cumulative low of -USD10.8bn around the implementation of President Trump's "Liberation Day" tariffs. But that marked a turning point as market sentiment and dollar depreciation turned in EMs' favor, bringing back flows. As a result, 2025 ended with record portfolio flows (USD410bn), almost double from the previous year, much above the historical average and a turnaround from the capital outflows experienced after the pandemic. EM bond funds concluded 2025 with their first annual net inflows since 2021, totaling +USD31.8bn.

Figure 3: Capital flows towards emerging markets, USDbn



Sources: LSEG Workspace, Allianz Research

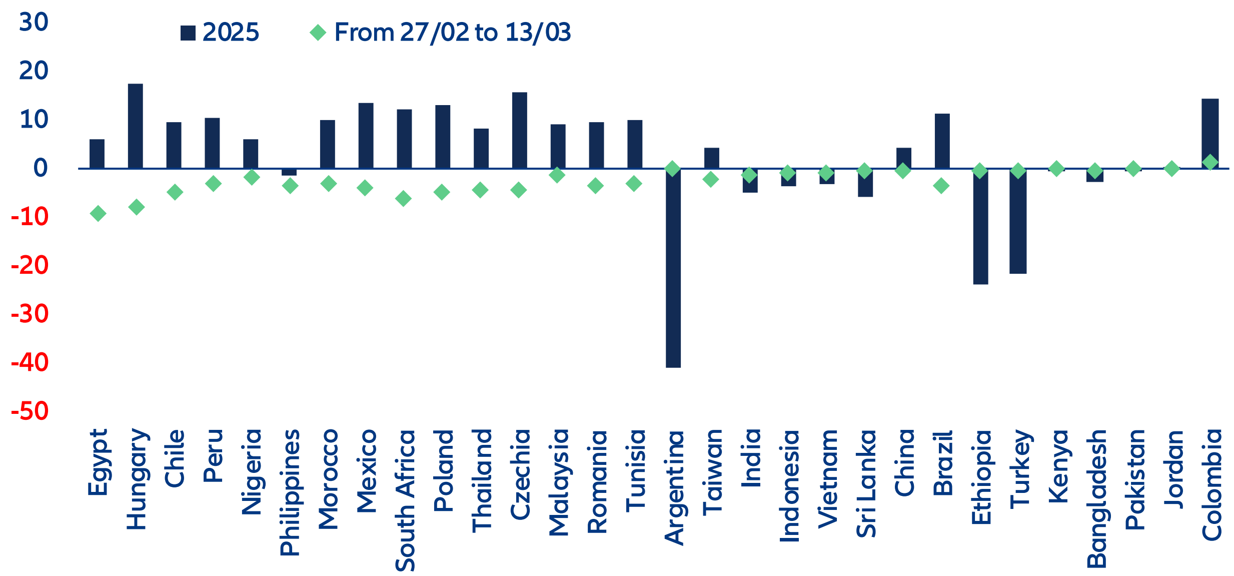
Looking to 2026, while the EM asset class is structurally stronger, selectivity will be an essential aspect of the path ahead. Direct benchmark exposure to the most vulnerable countries (Egypt, India, the Philippines, Thailand, Türkiye) remains limited, as they represent less than 3% of the MSCI Emerging Markets ex-China index. The most vulnerable European economies (Poland, Romania) would indeed feel the pinch of higher inflation but the risk of supply shortages would probably be offset by European mechanisms likely to be reinstated should the conflict last for longer.

Foreign exchange markets reacted quickly to the escalation in the Middle East conflict, with most emerging market currencies weakening over the week. Between 27 February and 13 March, several currencies recorded sharp declines as higher oil prices and a stronger US dollar triggered a broad risk-off move (Figure 4). The Egyptian pound saw the largest depreciation (-9.2%), reflecting its vulnerability as a large net energy importer with significant fiscal and external deficits. Central European currencies also weakened notably, with the Hungarian forint (-8%), Polish zloty (-4.9%) and Czech koruna (-4.5%) under pressure as the region's strong dependence on imported energy combined with the unwinding of investor positions. In Latin America, the Chilean peso (-4.9%) stood out among the underperformers given its negative energy balance, while in Asia the Philippine peso (-3.6%) and Thai baht (-4.6%) also weakened in line with the region's heavy reliance on Middle Eastern oil supplies. Part of the sell-off reflects the unwinding of earlier investor positioning: several currencies that had performed strongly earlier in the year were among the largest underperformers during the recent risk-off move.

As market conditions stabilize, currency movements are likely to more clearly reflect differences in countries' energy exposure. Higher oil prices tend to support currencies of energy exporters while weighing on importers, suggesting that the currencies of some Latin American exporters, such as the Brazilian real and Colombian peso, could stabilize once the initial risk-off adjustment fades. By contrast, currencies such as the Hungarian forint, Korean won, South African rand and Chilean peso appear to have already incorporated part of the energy shock and may remain sensitive to how the conflict and oil prices evolve.

Some currencies have remained relatively resilient despite their exposure to higher oil prices, notably the Indian rupee and Turkish lira, where active central bank management has helped limit volatility. However, the longer energy prices remain elevated, the harder it may become to maintain this stability. Looking ahead, the duration of the disruption in the Strait of Hormuz will remain the key factor determining further FX pressure: a prolonged disruption would keep energy prices elevated and sustain US dollar strength, while a faster resolution would likely allow part of the recent currency weakness to unwind.

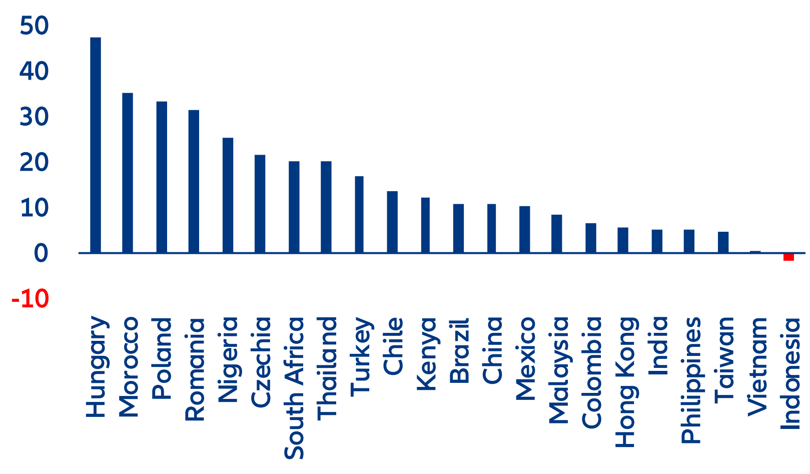
Figure 4: FX moves since the start of the Middle East conflict



Sources: LSEG Refinitiv, Allianz Research

Even with the ongoing shock, we continue to see EM fundamentals as strong, partially mitigating part of the war impact. FX reserves have remained elevated as many EMs used 2025 conditions to rebuild them, despite the challenges of US tariffs. India, South Korea, Taiwan and several Southeast Asian central banks recouped about USD132bn in FX reserves in late 2025 and early 2026, more than half of what they lost during earlier defensive currency interventions, aided by a weaker dollar and inflows. The Indian central bank has been particularly aggressive in rebuilding stockpiles to better defend the rupee on dips.

Figure 5: FX reserves have grown across most EMs over the last year (y/y growth)

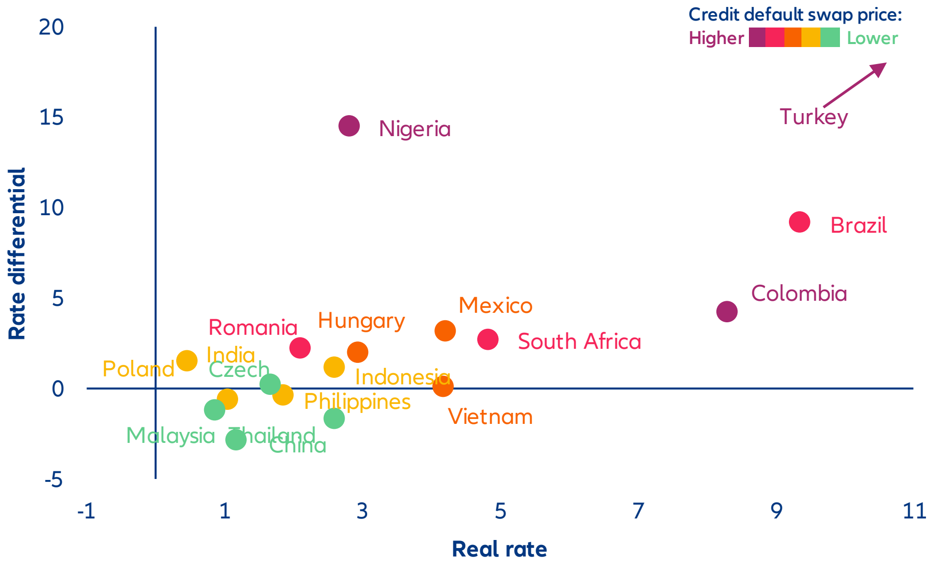


Note: International reserves including gold

Sources: LSEG, Allianz Research

In this context, the Middle East escalation arrives at a positive moment for rate differentials as the US benchmark Treasury saw a broad fall, defined by a structural repricing of the US fiscal risk premium during 2025. During 2025, real rates improved as inflation across EMs decelerated faster than nominal rates, allowing central banks to preserve a meaningful real rate buffer without undermining currency stability or triggering disruptive capital outflows. This parallel configuration revealed a broad improvement the EM universe while highlighting clear groupings within the universe. Türkiye, Brazil, Colombia, Nigeria, stand out with the widest nominal differentials to UST, though the investment case differs substantially across these markets. Türkiye, Brazil and Colombia also offer elevated real rates, creating a rare alignment of internal and external value. South Africa, together with Mexico sit in an intermediate position, with an elevated differential and a high real rate, though less extreme than Brazil and Colombia. Meanwhile, China, Thailand and Malaysia offer small rate differentials to the US, suggesting these markets are priced primarily for domestic investors.

Figure 6: Real rate, rate differential and credit default swaps in selected EMs



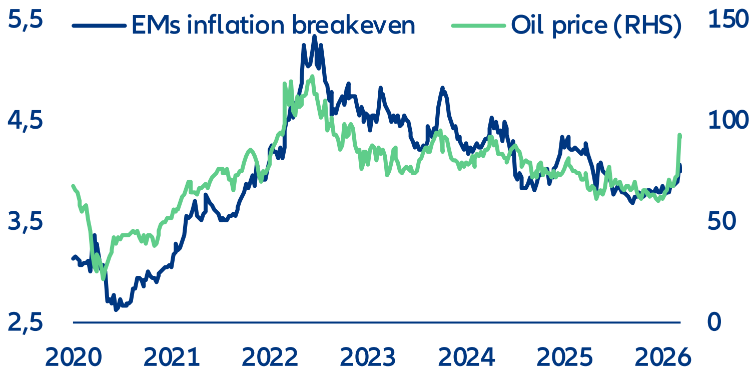
Sources: LSEG Workspace, Allianz Research

However, this new geopolitical shock renews stagflationary fears, with central banks on hold due to higher inflationary pressures. Inflation expectations have surged given the quick increase in prices during the one week of closure of the Strait of Hormuz – a dynamic already visible in US Treasury markets, where yields have risen sharply rather than fallen, as bond markets price the conflict primarily as an inflation shock rather than a growth one. This puts central banks in an acutely uncomfortable position: slowing growth would ordinarily call for rate cuts, but persistent energy-driven inflation prevents them from acting. The Fed, which had been on a gradual easing path, could now change path and keep on hold, or even hike through the rest of 2026. For emerging market central banks – many of which, like Egypt and Türkiye, are already navigating fragile disinflation paths – the constraint is even more binding, as currency depreciation pressures compound imported inflation, leaving little room to support growth without risking a renewed inflation spiral.

Emerging-market sovereign spreads have widened modestly, reflecting the tone of cautious sentiment. The move has been more pronounced across Africa, the Middle East and CEE Europe. The sharpest adjustments have occurred in the most directly exposed countries, such as Bahrain or Pakistan, where acute energy-import vulnerabilities have intensified market sensitivity. In contrast, the rest of the Gulf has remained relatively tight, supported by strong oil-related revenues that partially offset elevated geopolitical risks. This resilience should nonetheless be interpreted carefully. Historically, spread markets have tended to reprice geopolitical shocks with a delay, responding more forcefully once hard economic data begins to reveal the underlying impact. Should the Strait of Hormuz remain closed beyond roughly four weeks, and supply disruptions start feeding into macro indicators, a broader and more disorderly widening across EM credit, particularly among energy-import-dependent sovereigns, cannot be ruled out.

The critical question is whether local bond markets are beginning to price the inflation risk. Breakeven inflation (BEI) offers a real-time measure of market-implied inflation expectations. Since oil prices are an important driver of inflation compensation (Figure 7), a rise in oil should normally push EM breakevens higher. However, before the strikes, inflation protection across EM looked unusually cheap. For a representative basket of economies, the average 10-year breakeven stood at just 3.89%, while the historical sensitivity to a 10% rise in oil prices averaged only +8bp in BEI – well below what our simulations suggest on actual inflation rate. In other words, inflation expectations were still responding to oil, but much less than in past stress episodes. This matters because the sensitivity of BEI to oil tends to rise when markets start to view the shock as more persistent and structural. If the conflict proves prolonged, inflation expectations across EM curves are therefore likely to adjust more forcefully.

Figure 7: EM 10Y inflation breakeven (excl. Türkiye) and oil price

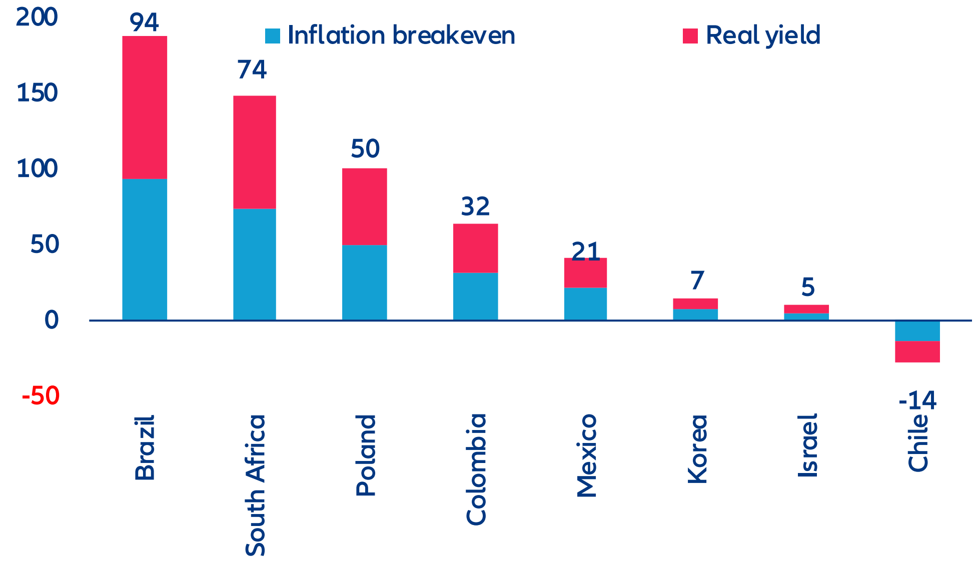


Sources: Bloomberg, LSEG Datastream, Allianz Research. Notes: Inflation breakeven is derived only for countries that have an active inflation-linked bond market. Countries considered are Brazil, South Africa, Poland, Mexico, South Korea, Türkiye, Chile, Israel and Colombia.

Has that adjustment already begun? Early evidence suggests that repricing is underway, but not uniformly. In the first week after the strikes (27 February to 9March), South Africa recorded the largest breakeven widening, at +57bps, from 4.32% to 4.89%, consistent with its current-account vulnerability and energy import dependence. Poland's breakeven rose by +34bps and Mexico's by +28bps – meaningful moves in markets where pre-shock inflation expectations were running close to, or even below, central bank targets. However, the adjustment has not been universal. Colombia's breakeven fell by 45bps, as markets priced in the terms-of-trade gains of a net oil exporter, while Israel's rose only +10bps.

A look at nominal 10-year yield moves helps clarify the forces behind the repricing and the extent to which it reflects inflation versus broader risk aversion (Figure 8). Chile now stands out as the clearest case of inflation-led repricing: its +18bps rise in nominal yields was more than explained by a +33bps widening in breakevens, while real yields fell by 15bps, indicating that the move was entirely driven by higher inflation expectations. Turkey (+233bps nominal, including +181bps from breakevens) and Korea (+26bps, of which +20bps came from breakevens) also saw most of their repricing driven by inflation compensation. Mexico, which initially looked almost entirely inflation-driven, has since experienced a broader sell-off: nominal yields are now up +49bps, with +28bps attributable to breakevens and +21bps to real yields. This still points to an inflation-led move, but with a more visible risk-premium component than before. Brazil presents a much sharper contrast. Nominal yields rose by +88bps even as breakevens fell by 6bps, implying a +94bps increase in real yields and pointing clearly to higher credit and liquidity premia rather than inflation. South Africa has also moved in that direction. With nominal yields now up +96bps, breakeven widening has largely reversed to just +22bps, leaving +74bps accounted for by higher real yields, suggesting that the move is now dominated by risk aversion rather than the inflation repricing seen initially. Colombia (-13bps) remains the outlier, with nominal yields still rallying as markets continue to price terms-of-trade gains for a net oil exporter, while Israel (+14bps) has shifted from a modest rally to a mild sell-off. In CEE, Türkiye, Romania, and Hungary have recorded the sharpest sell-offs in local rates markets, with Poland not far behind. Türkiye's move (+233bps) – the most pronounced across all EM – reflects a confluence of acute oil import dependency, geographic proximity to the conflict, and a market that was already pricing fragile disinflation dynamics. Romania (+101bps) has overtaken Hungary (+94bps) as the second-largest mover in the region: both had benefitted from a strong rates rally going into the conflict, and the sell-off is in part a technical correction as investors take profit on stretched long positions, compounded by the region's energy vulnerability and broad CEE risk-off repricing. Poland (+83bps) and the Czech Republic (+59bps) round out a picture of broad-based CEE weakness.

Figure 8: EM 10Y local yield moves after the conflict: inflation breakeven vs real yield contribution, bps



Source: Bloomberg, LSEG Datastream, Allianz Research. Notes: Changes in yield from 27/02/2026 to 13/03/2026.

The outlook for EM capital flows in 2026 is now firmly scenario-dependent. In a short-lived conflict – contained within four to five weeks – EM inflows should remain broadly positive, with markets normalizing relatively quickly toward pre-Liberation Day conditions. Investors would nevertheless become more selective, differentiating more sharply between countries on the basis of energy exposure and inflation vulnerability rather than treating EMs as a homogeneous asset class. A prolonged conflict would alter that picture materially. Flows would rotate toward safer assets, while the most exposed economies – Egypt and Pakistan foremost among them – would face the sharpest outflow pressure as the combination of currency stress and imported inflation reshapes their risk profile. In this environment, the repricing is likely to take the form of a selective energy risk premium, differentiating sharply between exposed and insulated sovereign issuers rather than triggering a broad-based EM sell-off.

This scenario dependency comes against the backdrop of an exceptionally strong year for EM debt issuance. On the hard-currency side, sovereign issuance rebounded from roughly USD72bn in 2022 to around USD236bn in 2025 – the highest level on record – reflecting genuinely renewed access to international funding markets. Saudi Arabia and Türkiye were the primary drivers, with Türkiye's issuance alone rising from USD12bn in 2022 to roughly USD31bn in 2025. Poland also contributed significantly, with issuance climbing to around USD30bn in 2024, while the UAE and Mexico provided additional support. Local-currency issuance expanded in parallel, led by Brazil, Poland and India, with the Czech Republic emerging as a particularly notable issuer at USD195bn – the largest absolute increase over the period – further supported by Thailand and Mexico. The scale of this recovery underscores how much is at stake: a prolonged conflict risks interrupting a multi-year rebuilding of EM market access at precisely the moment it had regained its strongest footing in years.

Slightly more than two weeks into the conflict, we observe modest underperformance among the more exposed sovereign issuers. While the Philippines continues to trade broadly in line with the wider market, Egypt – and in particular Türkiye and India – have experienced more pronounced spread widening. If the Ukraine conflict is taken as a reference point, spreads could widen by roughly one-third to one-half of pre-crisis levels. In practical terms, this would imply an estimated widening of around 80bps for Türkiye, 35bps for the Philippines, 180bps for Egypt and 35bps for India. It is worth noting, however, that Türkiye's macro-financial position is materially stronger today than in early 2022, and that energy-related risk premia during the Ukraine episode ultimately proved short-lived.

These assessments are, as always, subject to the disclaimer provided below.

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Warsh's double dilemma: when the Middle East rewrites the Fed's playbook

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In Summary

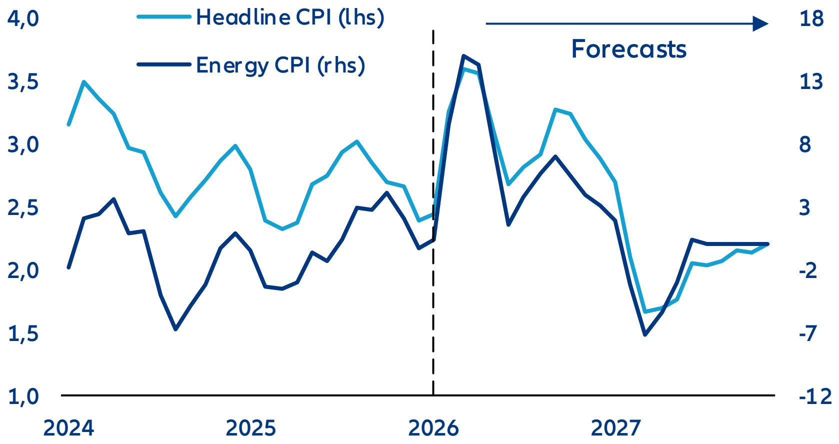
- **The energy price shock will delay the Fed's single rate cut in 2026.** Rising energy prices should push US inflation towards +3.6% y/y in April-May, up from +2.8% expected before the Middle East conflict, under the assumption that oil prices remain around 90 USD/bbl on average in Q2. Despite a still weak labor market, the Fed will have to keep rates on hold at least through the summer amid risks of de-anchoring inflation expectations. We continue to expect only one 25bps rate cut this year, but pushed out to September.
- **Fed chair nominee Kevin Warsh combines a dovish view on interest rates with a hawkish approach to the Fed's balance sheet, which could drain liquidity from a highly leveraged financial system. Increasing frictions in the US money market would have a global impact on financial markets mainly though the channels of FX and interest-rate volatility.** Warsh opposed additional rounds of quantitative easing (QE) in the 2010s, warning that prolonged asset purchases fueled financial excess and risked inflation. With hindsight, his concerns about loose financial conditions and asset valuations appear partly validated. But reducing the Fed balance sheet drastically or even returning to a scarce-reserves regime would ignore the intermediation mechanisms on which the US funding market has been founded since the GFC. The critical point is not the amount of reserves but the dealers' balance sheets available to warehouse Treasuries and intermediate repo flows. Dealers face regulatory limits that are capping the size of their balance sheets. They cannot do both: absorb heavy Treasury issuance and provide smooth funding for the private sector. It would drive up prices for hedging instruments (swap spreads and cross-currency basis) but in a worst-case scenario such mismatch could trigger a deleveraging spiral affecting valuations of risky assets and increasing default risk for highly leveraged entities (e.g. hedge funds). The Treasury and the Fed have options to mitigate this, but it would require a massive easing of leverage regulation and/or a substantial decrease in Treasury issuance, the first being more likely than the latter.
- **We expect the Fed to move cautiously towards a modestly smaller balance sheet from Q4 2026 at the earliest, keeping Treasury holdings steady while continuing the run-down of the USD2trn MBS holdings. This would limit money-market volatility but still risk higher mortgage rates. Wars typically involve monetizing deficits, i.e. restarting QE and departing from Warsh's pledges for rapid quantitative tightening (QT).** The Fed will likely adopt a hybrid model: a leaner balance sheet combined with standing repo facilities as permanent liquidity backstops, preserving rate control while mitigating systemic funding risks. Regulatory moves on bank liquidity regulations could also support the transition toward lower bank's reserves. We would expect banks' reserves balances to drop gently from 9.5% of GDP currently to reach the level of the 2019 money-market meltdown (below 7% GDP) only by Q4 2028, though uncertainties are large and episodes of money market volatility will likely become more frequent. The MBS reduction would however blunt the administration's attempts to lower mortgage rates. The conflict in the Middle East could also drastically change the course of monetary policy in the US. In a downside scenario, quantitative easing may be needed to help finance the effort of the war.

The energy price shock will delay the Fed's single rate cut in 2026

The Fed is likely to keep interest rates on hold at its next meeting on 18 March amid upward pressure on inflation, which we expect to peak at +3.6% by April/May. The conflict in the Middle East poses a difficult dilemma for the Fed, with inflation set to pick up amid higher energy prices while the labor market still looks fragile. Non-farm payrolls declined by -92K in February, and though part of the weakness was due to strikes in the healthcare sector, the private sector excluding healthcare continued to shed jobs. But with inflation now stuck above the 2% target for five years, the Fed clearly does not have the luxury to focus on labor market risks. Inflation expectations are especially sensitive to energy (and food) prices and could risk de-anchoring if the Fed is perceived as too complacent with inflation risks. We expect the Strait of Hormuz to open again by the end of April. In the meantime, partial re-retouring, additional production (Russia, US) and tapping strategic reserves should cover over 50% of the gap in oil and gas, capping oil prices at 90 USD/bbl on average in Q2 2026. We estimate that US energy CPI will rise from +0.4% year-on year in February to peak at around +15% in April. Headline inflation would rise from +2.4% to a peak of +3.6% in April-May (Figure 1).

We continue to expect one 25bps rate cut this year, but pushed out to September. We have long argued that the Fed will be limited in its ability to cut interest rates in 2026, and we were expecting only rate cut in June prior to the conflict in the Middle East – a much more hawkish view than what is currently priced in financial markets. We now expect the Fed to wait until September, when inflation should start to decline though still remain above target. The labor market is likely to remain weak, or even weaker further, through the summer as higher inflation eats into households' incomes and some corporates' profit margins. If the Fed stays on hold in the near term, the risks of second-round effects on prices, which would compound the initial energy price shock, may be smaller than in 2022.

Figure 1: US CPI inflation forecast (left)



Sources: LSEG Datastream, Allianz Research

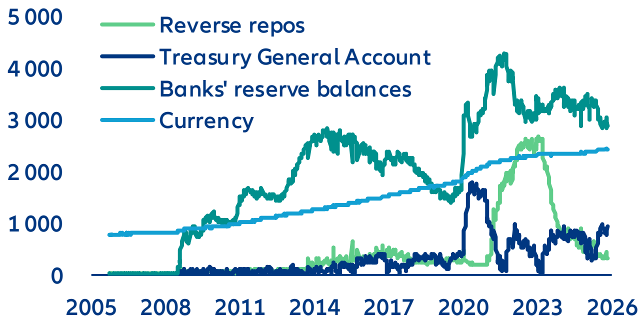
Kevin Warsh' s push for a lower Fed's balance sheet could gain traction...

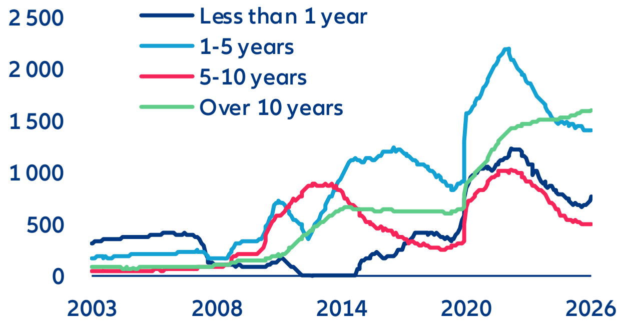
Kevin Warsh, President Trump's nominee for the next Fed Chair, has assuaged financial markets, thanks to his strong credentials and perceived prudent approach on monetary policy. Warsh, who is expected to take over in May, served on the Fed Board from 2006 to 2011 during the height of the Global Financial Crisis (GFC). At the time, he supported an aggressive monetary response to stabilize markets. However, once the recovery gained traction, he opposed the second and third rounds of quantitative easing (QE) and stepped down in 2011, reportedly amid policy disagreements over continued asset purchases. Importantly, his concerns about QE were centered on the blurring of monetary and fiscal policy. He has warned about financial stability risks, asset bubbles and inflation pressures stemming from an expanded balance sheet. With the benefit of hindsight, many of Warsh's views proved to be well founded: Financial conditions have been very loose since the aftermath of the GFC, equity markets have been buoyant, financial risks have built up and inflation picked up – though at a much later stage. Other critics of the Fed's balance sheet policy include Treasury Secretary Bessent, whose line of attack is on the net operating losses that the Fed makes by paying interest on banks' reserves – although Bessent is not pushing for a sharp shrinkage of the Fed's balance sheet. More recently, Warsh argued in op-ed in 2025 for a significant reduction in the Fed's holdings.

However, Warsh's hawkish views on the balance sheet sit alongside more dovish views on policy rates. Warsh argues that money is too tight for small corporates and calls for lower policy rates. He believes strong GDP growth is being driven by a positive supply shock – notably AI – which should generate disinflationary forces, which will in turn be reinforced by the Administration's deregulatory agenda. Besides, in 2025, Warsh wrote that the Fed "should abandon the dogma that inflation is caused when the economy grows too much, and workers get paid too much," signaling limited concern about demand-driven inflation pressures. He also suggested that balance sheet "largesse can be redeployed in the form of lower interest rates to support households and small and medium-size businesses." While the operational implications remain unclear, this could imply a preference for shifting from asset purchases toward more direct credit-support mechanisms.

Warsh's criticism of the Fed's bloated balance sheet could find support among FOMC members. The Fed has already shrunk its balance sheet significantly from its 2022 high of 35% of GDP to less than 20% through a reduction of asset holdings (Treasuries and MBS). All Treasuries maturities were reduced except long-term ones (Figure 2, left). The pre-GFC average of the Fed's balance were close to 5% of GDP, when the Fed operated under a ''scare'' reserve system rather than an ''excess'' reserves one: banks' reserve balances were essentially zero prior the GFC (Figure 1, right). In end-2025, following tensions in the repo market, the Fed decided to launch Reserve Management Purchases (RMPs), i.e. Treasury purchases at short maturities (T-bills) to increase reserve balances again. The T-bills purchases are partly funded by the reinvestments of MBS and longer-dated Treasuries paydowns. However, there was some disagreement among FOMC members around the extent to which the Fed should increase its purchases, with the minutes noting "a couple of participants added that effective standing repo operations may allow for a smaller balance sheet ".

Figure 2: Fed's balance sheet main liabilities items (left) ; Fed's Treasury holdings by maturity (right) – in USD bn



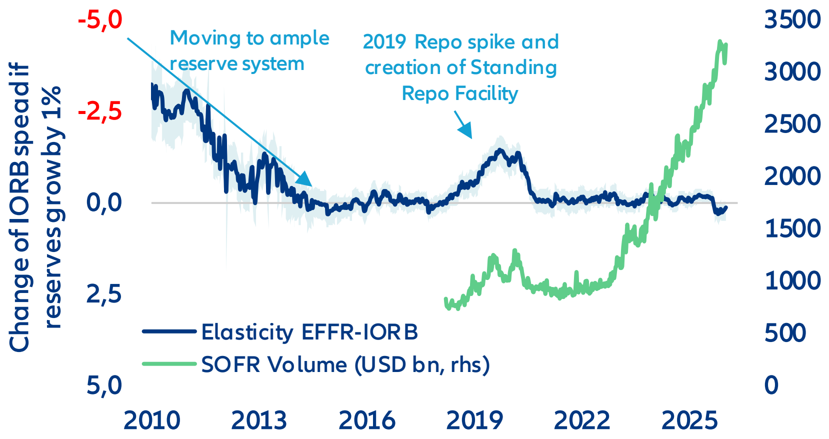


Sources: LSEG Datastream, Allianz Research

...but it would be a risky strategy for little return

The transition towards a reduced balance sheet would entail risks for financial stability as the monetary policy system has shifted from reserve fine-tuning to private-sector intermediation. The current ample reserve balance sheet policy is the result of the structural breaks of the GFC. Before 2007, the Fed ran a scarce-reserves corridor system in a funding market structured by large scale unsecured lending, small dealer balance sheets, loose leverage constraints (pre-Basel III), off balance-sheet repo and a limited role of non-banks (money market funds, hedge funds etc.) in cash supply. After the liquidity collapse during the GFC, the Fed expanded its balance sheet by large-scale asset purchases – creating such excess reserves that the corridor system could no longer be handled with fine-tuned open-market operations – and endorsed the role of market maker of last resort. The US funding market structure shifted from one where banks arbitraged small reserve shortages via interbank lending to one where administered rates and central-bank facilities anchor pricing and balance-sheet-constrained dealers intermediate between non-banks and the Fed. Consequently, the critical point of the system shifted away from the amount of reserves to the amount of dealer balance sheet available to warehouse Treasuries and intermediate repo flows. If balance-sheet capacity is constrained, market liquidity can be scarce despite ample reserves. Going back to a pre-GFC scarce reserve system would not be "doing what worked for decades" but simply ignoring the intermediation nature of the USD funding market. The current ample-reserves floor system with Interest on Reserve Balance (IORB) and standing facilities (Overnight Reverse Repo ON RRP, Standing repo facility SRF) has been a crucial pillar of central banks' volatility compression in the post-GFC era. It has immunized the monetary policy system against shifts in the reserve demand. Changes in reserves today only have a minimal impact on movements in money-market spreads. Under the previous "scarce reserves" regime, a 1% change in reserves could trigger a response of up to 3–4 bps in the money-market spreads (Figure 3). A return to scarce reserves with daily open market operations would create volatility and money-market rate dislocations as rather than simple management of reserves, intermediation capacity is the key point of friction.

Figure 3: Reserve demand elasticity: EFFR* - IORB spread reaction to 1% change in reserves



**EFFR = Effective Fed Funds Rate*

Reserve Demand Elasticity is the responsiveness of money-market spreads (Effective Fed Funds-IORB, SOFR-IORB) to changes in reserve supply, i.e. the slope of the reserve demand curve. When frictions are low, this slope is near zero and spreads hardly move. When transmission is impaired the curve steepens and small reserve shocks lead to large changes in money market spreads. In 2009, a 1% expansion of reserves led to 3-4 bps contraction of money market spreads, versus only 0.15bp today. SOFR-IORB elasticity is more volatile as it captures the transmission between unsecured and collateralized money market rates.

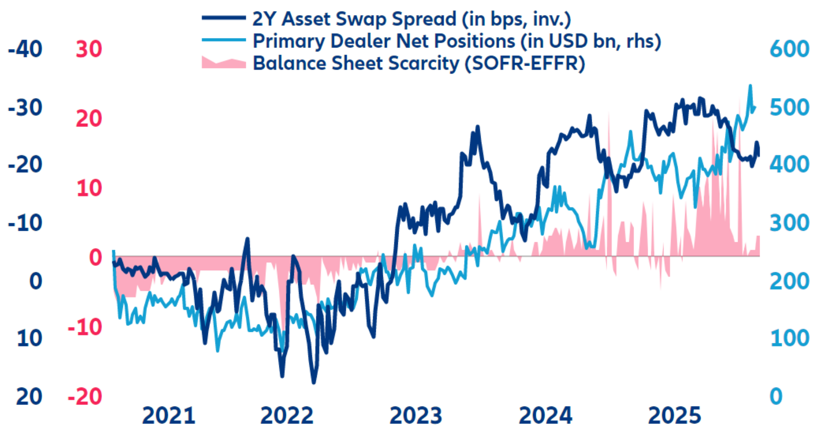
Sources: LSEG Datastream, Allianz Research

A removal of the Interest on Reserve Balance (IORB) would have more negative effects than positive ones – making it unlikely. Alternatively to the Fed balance sheet discussion, there is an ongoing debate on removing the Interest On Reserve Balances. (IORB). The IORB is intended to steer banks' behavior by guaranteeing an administered return on their reserves. The costs and benefits of operating this system are politically challenged. Critics see the IORB as a subsidy for the US banking sector. The Fed sees the IORB as the price to sustain bank credit flow. Eliminating the IORB "would be extraordinarily disruptive" as it would impair the Fed's control of short-term rates, reduce the volume of bank-intermediated transactions, and thereby diminish private-sector liquidity. In addition it would generate massive money market volatility and downward pressure on short-term rates as banks would substitute reserves with interest-bearing alternatives such as newly issued Treasuries or the Fed's reverse repo facility (RRP, which is the lower bound of the policy rate corridor). Replacing reserves with Treasuries would increase banks' liquidity risk and raise Basel III capital charges, boosting banks' demand for central bank liquidity and therefore counteracting efforts to shrink the Fed's balance sheet. Shifting to the RRP would also compress banks' net interest margins, reduce their lending capacity and restraining credit provision to the economy. Given these adverse consequences, the costs of a IORB removal outweigh fiscal benefits.

The Fed is already operating in a framework of scarcity

The Fed is facing a funding paradox where an ample balance sheet coincides with scarce intermediation capacity. While the discussion is focused on the scarcity of reserves, we find that the Fed is already operating in a framework of scarcity but from the intermediation side. Intermediation in funding markets is the willingness and ability of balance-sheet-constrained dealers and banks (subject to regulatory leverage and liquidity limits) to act as a broker between cash-rich non-banks (money market funds, pension funds etc.) and collateral-rich issuers (US Treasury, corporates etc.). During the QE era, balance-sheet constraints were rarely the binding margin. With QT, excess liquidity declined and dealers' balance-sheet capacity decreased as did their debt-absorption ability. This situation prevails today. With Treasury (notably T-Bills) issuance taking over an increasing part of the scarce dealers' balance sheet, funding becomes more expensive for other investors, especially those running leveraged risky strategies (i.e. basis trade, asset swap spread). In Figure 4, we can see the effect on balance sheet scarcity: the more Treasuries issuance increases, the more dealers must increase their holdings on their books. Their cost of carry is reflected in the risk premium, which is the asset swap spread, and in the increased demand for cash reflected by the SOFR-Fed Funds spread (SOFR can be understood as the price for balance-sheet capacity) as well as volatility in short term rates.

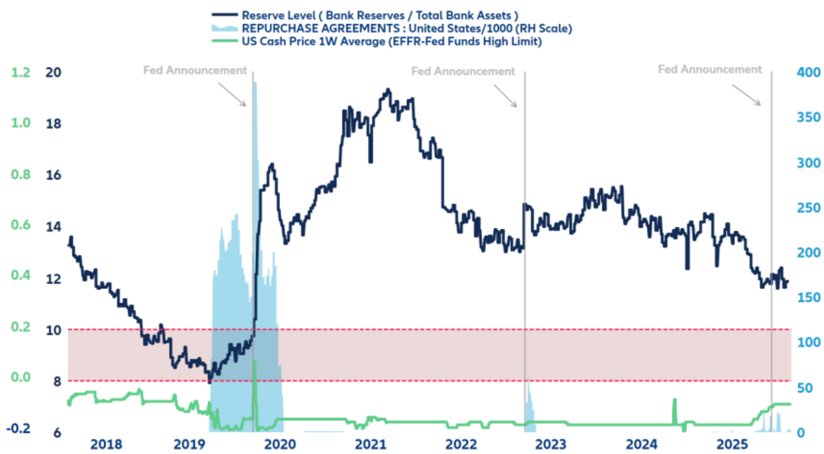
Figure 4: Balance-sheet scarcity illustrated



Sources: LSEG Datastream, Allianz Research

The recent decision of the Fed to resume buying short-term US Treasuries (T-Bills) must be seen as a reaction to this scarcity, acting as cash support to absorb heavy US Treasury issuance. If intermediation scarcity prevails we will see further Fed interventions mostly though the SRF (Temporary Open Market Operations) and further asset purchases. The probability of Fed interventions is linked to the ratio of bank reserves on bank total assets. Above 10% reserves are "abundant", below 10% reserves are "ample" while 8% marks reserves as "scarce". As shown in Figure 5, when the market reaches the level of absorption stress the Fed generally intervenes and restores market liquidity. These interventions are often accompanied by announcements of new facilities (SRF in September 2019) or programs (Bank Term Funding Program following Silicon Valley Bank Crisis, March 2023).

Figure 5: The Fed intervention framework - cash price, reserve levels and standing Repo facility use



Sources: LSEG Datastream, Allianz Research

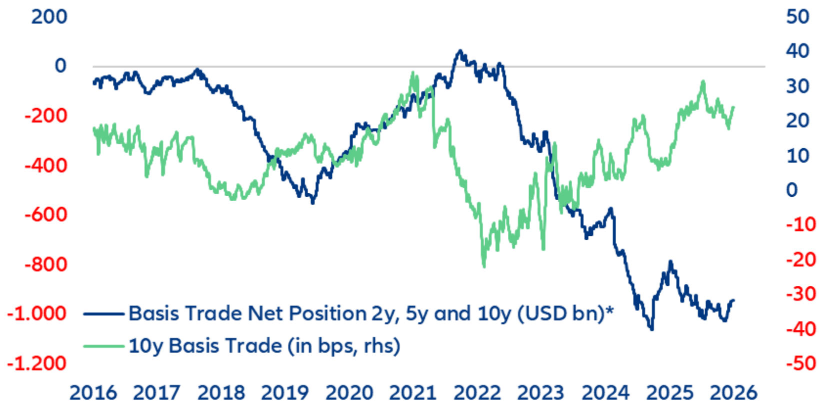
The Fed is moving from being a structural provider of liquidity to a relief valve while the balance sheet is playing a limited role. The problem is that non-banks (esp. leveraged hedge funds) are not "plumbed" to the Fed's valves.

Elevated and volatile repo rates with tighter haircuts can force them to unwind risky leveraged strategies (such as basis trades) and sell Treasuries into less liquid market, amplifying price moves and volatility. Such a destabilizing deleveraging spiral was seen in March 2020, for instance. The Treasury and the Fed have option to mitigate these risks. A substantial easing of leverage regulation (i.e. SLR, ISLR) could alleviate some balance-sheet constrains and free up cash that could ease the liquidity pressure. A substantial slowdown in Treasury issuance could also free balance-sheet capacity, but remains unlikely. Both would anyway only be temporary.

The Fed could lose its dampening grip on global volatility

US money markets are the primary providers of liquidity and leverage across the global financial system. When US money-market spreads widen and volatility rises due to pressure on SOFR, the FX swap market and interest-rate swaps become the main transmission channels. For FX swaps this occurs because the same intermediaries use their constrained balance sheets to price FX derivatives. Therefore intermediaries can pass on the cost of scarce balance sheets through higher risk premia onto the end users, resulting in higher FX volatility and widening the cross-currency basis, appreciating the USD, which tightens global financial conditions. The transmission channel through interest-rate swaps operates via Treasury issuance dynamics. Swap spreads reflect liquidity at specific maturities as well as the banking system's absorption capacity. If newly issued debt is not enough subscribed by the private sector, primary dealers (banks) must by law keep it on their books, affecting their funding capacity of the private sector. Negative swap spreads are a sign that issuance absorption is tight. This reduces the demand for duration, increases bid-ask spreads, tightens trading volumes and steepens the curve. This effect is reinforced by huge positions of hedge funds in bond basis trades (long cash bond, short future) funded in US money markets. Official numbers from the CFTC (Commodity Futures Trading Commission) indicate a total volume of USD1,300bn with a USD1,000bn net short position in the US basis trade (Figure 6).

Figure 6: Hedge funds have a minimum net short position of USD1000bn in US Treasury futures

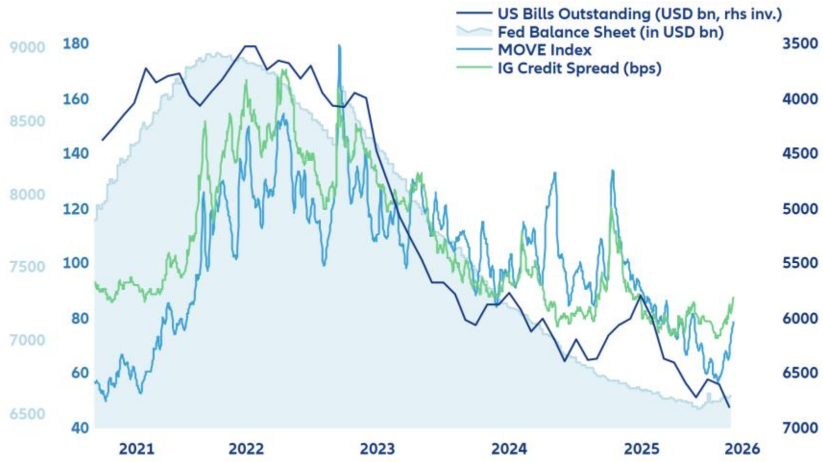


**net open interest in USD for leveraged funds as reported by the CTFC (Commodity Futures Trading Commission)*

Sources: LSEG Datastream, Allianz Research

Scarce money market liquidity therefore creates a risk of fixed income market selloff and increases the risk of a massive deleveraging spiral. The US Treasury market is now more exposed to this risk as hedge funds have become the marginal buyer of US Treasuries since the of the US curve (on a FX hedged basis) has become relatively unattractive for foreigners. But hedge funds are not buy-and-hold investors; they use these US Treasuries as collateral to borrow cash in the repo market to fund leveraged positions such as yield curve arbitrage, swap spreads, credit spreads, securitized products on bonds, crypto or stock indices. This demand from leveraged investors compresses implied volatility on many asset classes and is a reason for the unusually low global volatility. But this mechanism only works as long as there is subdued volatility on money-market rates. This is why any change in the monetary policy system by the Fed would be very risky (Figure 7).

Figure 7: Volatility compression is linked to monetary and fiscal policy



Sources: LSEG Datastream, Allianz Research

How would a global volatility shock unfold?

The transmission of rising money-market rates volatility runs via higher risk premia that would hurt leverage, affect risky-asset valuations and increase default risk within the banking and non-bank system. It would unfold the following way: First, a margin call due to rates volatility would trigger emergency refunding demand and so a selloff in leveraged assets. Fringe assets would be the first in line to correct (leveraged crypto, highly leveraged ETFs). Second, default risk among leveraged investors (hedge funds) would rise. Since hedge funds usually fund themselves on the sponsored repo market[1], the sponsoring banks would finally have to step in and bail out the clearing house if hedge funds fail. This procyclicality is the result of the "Active Treasury Issuance" (ATI) strategy. Its objective is to massively issue T-bills to release excess liquidity created by QE and parked into the RRP (reverse repo program) to counter restrictive effects from Quantitative Tightening (QT). But this aggressive supply of T-Bills also increased the potential for leveraged trades for non-banks (hedge funds). ATI therefore contributed to fuel asset prices and compresses asset volatility. So far we do not see stress on sponsored repo, but outstanding volumes indicate a slight deleveraging trend since the beginning of the year. The deleverage risk from the non-bank sector is reinforced by additional layers of leverage: collateral transformation and collateral re-use. Collateral transformation refers to the practice where counterparties swap lower-quality assets, such as investment-grade corporate bonds, for high-quality assets like US Treasuries in uncleared bilateral markets. These transactions occur off-balance sheet, their outstanding amount is hard to quantify. Collateral re-use (rehypothecation) occurs when dealers re-use received collateral forming collateral chains in which the same asset runs through multiple transactions. This is a widely used practice as around 85% of collateral received by the biggest primary dealers is said to be re-used. In a massive sell-off scenario, collateral re-use might trigger a chain reaction across markets that the Fed liquidity back stop could only partially contain as it is occurs outside its sphere of influence in the non-bank and/or uncleared segment of USD money markets (Eurodollar system).

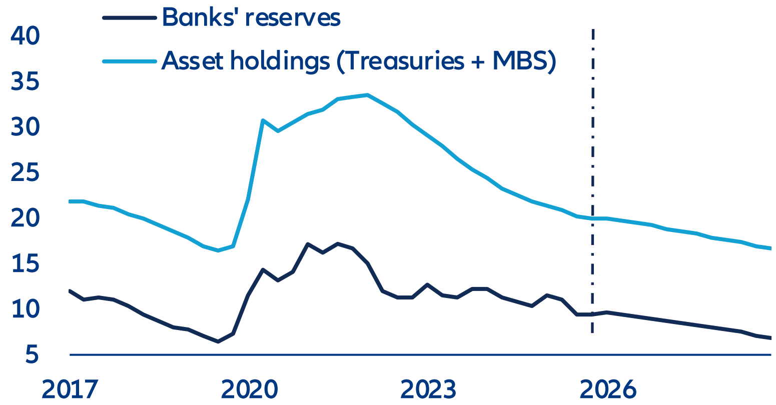
The Fed will reduce its balance sheet in slow-motion

Given these market fragilities, we expect the Fed to embark cautiously on a balance-sheet reduction path, from Q4 2026 at the earliest. Treasury holdings could be maintained at flat levels, while the Fed could continue the run-down of the USD2trn MBS holdings. The Fed would then instead be relying more heavily on standing repo facilities as permanent liquidity backstops, preserving rate control while mitigating systemic funding risks while reserves decline. Regulatory moves on bank liquidity regulations, such as a lowering of liquidity requirements for banks, which include banks' reserves, could support the transition towards a smaller balance sheet and lower reserves. This would maintain financial market stability but risk higher mortgage rates. We would expect it would

[1] *Sponsored repo market refers to cleared repo transactions in which a dealer (bank) sponsors non-dealer clients (such as hedge funds or money market funds) into the Fixed Income Clearing Corporation's central clearing platform, allowing those clients indirect access to centrally cleared repo while the dealer guarantees their obligations.*

take a bit of time for Warsh to convince a majority of the FOMC to start reducing the Fed's balance sheet. Most likely, a compromise could be found on flattening total Treasuries holdings, rather than shrinking them, but to let the MBS run-down continue at the current pace of around -USD17bn per month. In this baseline scenario, we would expect the Fed to start shrinking its balance sheet in Q4 2026. Banks' reserves would drop from 9.5% GDP currently to less than 7% in Q4-2028 – a level close to the 2019 money-market meltdown. Put otherwise, under prudent balance-sheet reduction, it would take a bit of time before the Fed potentially faces money-market volatility on the back of scarcer reserves. However, forecasting is fraught with large uncertainties. In particular, should nominal GDP grow faster than we expect, we could see a faster reduction of reserves as a share of GDP.

Figure 8: Banks' reserves at the Fed & Fed's asset holdings (Treasuries + MBS), % GDP



Sources: LSEG Datastream, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed
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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

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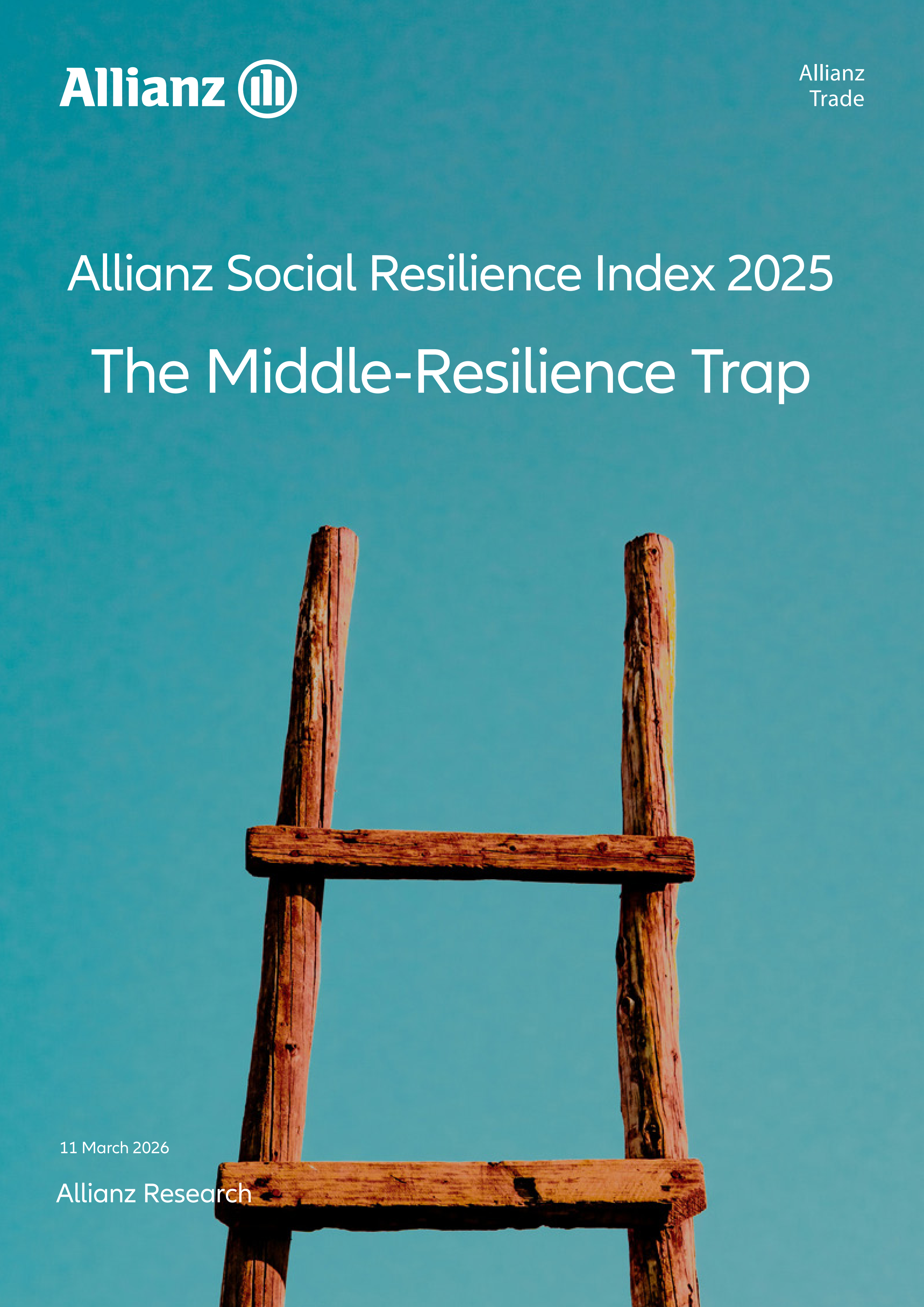

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Allianz Social Resilience Index 2025

The Middle-Resilience Trap

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Content

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Executive Summary


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- **In 2025, the Allianz Social Resilience Index (SRI) records a third consecutive – albeit modest – improvement, but underlying divergence remains pronounced.** The global average rose from 47.4 in 2024 to 47.9 (on a 0-100 scale) across 171 economies, driven by lower imported inflation pressures and greater currency stability in Emerging Asia and several advanced economies. Firmer governance was another driving force, including institutional stabilization in parts of Emerging Asia and Central Europe. At the same time, weaker social cohesion in the Middle East, Emerging Europe and high-income countries continued to weigh on social resilience. Northern European economies lead the ranking, with Finland (1st, 84.3), Denmark (2nd, 83.8), and Iceland (3rd, 81.4) at the top of our index. Germany (8th, 78.5), France (13th, 74.7), and the UK (18th, 72.9) remain strong, while Italy (28th, 66.8) and the US (32nd, 65.3) ranked lower but stayed in the top third. China (59th, 52.5), India (81st, 46.8), Mexico (105th, 41.3), and Brazil (112th, 40.0) ranked significantly lower, with conflict-affected states like Lebanon (169th, 17.7) and South Sudan (171st, 11.8) at the bottom.
- **The conflict in the Middle East and the resulting energy inflation will test the strength of social fabrics, particularly in countries such as Vietnam, Thailand, Morocco, Tunisia, and Malaysia.** The SRI helps measure vulnerabilities to such exogenous shocks: countries with weaker resilience and limited fiscal buffers could be particularly affected by higher- for- longer energy prices. Indeed, sustained energy- price increases raise inflation (including food inflation), weaken growth and heighten political tensions, particularly where societies have limited capacity to respond. Coping mechanisms will be the pivotal, including crisis- response mechanisms, economic stabilizers, and optionality. Two clusters stand out: Low-to-mid resilience economies with high exposure – including Vietnam, Thailand, Morocco, Tunisia and Malaysia – could see price spikes more easily translate into social pressures. Meanwhile moderate- resilience countries with medium exposure to the energy price shock – including Chile, Egypt, Serbia and South Korea – should be in a better position to cushion the impact of a temporary energy shock on social resilience. Europe is likely to be more affected than the US given its greater reliance on imported energy, though social spending may cushion the impact.
- **Taking the long view, four trajectories of global progress in social resilience emerge: low-resilience countries catching up (e.g., Romania and Saudi Arabia); countries weakening further (e.g., Nigeria and Brazil); high-resilience countries consolidating strength (e.g., Germany and Finland); and countries experiencing slippage (e.g., Canada and Sweden). Some countries seem stuck in a middle-social-resilience trap among these diverging paths.** Rankings show limited change since 2024, though Sri Lanka (+12.7pts; +39 ranks) and India (+7.8pts; +27 ranks) improved notably, while Brazil (-8.7pts; -41 ranks) and the Czech Republic (-6.9pts; -15 ranks) saw

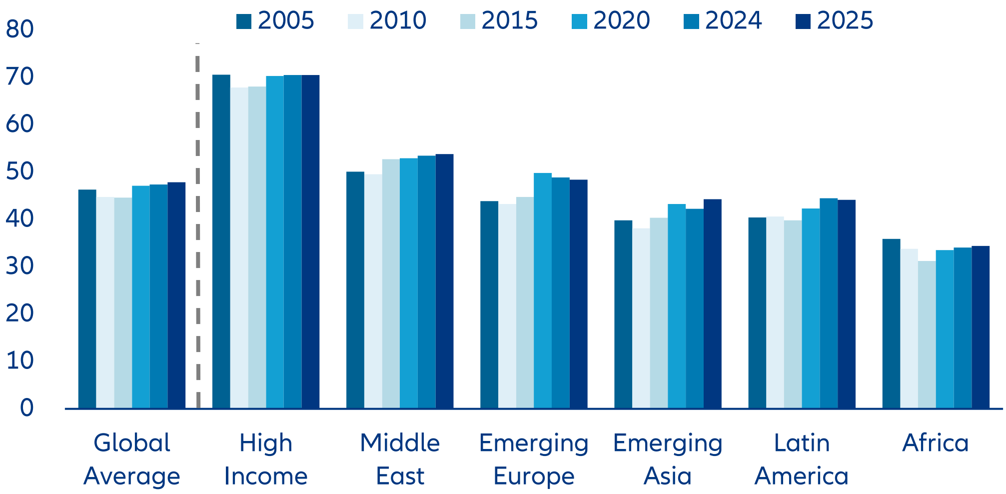
steep declines. A cluster of advanced economies persistently scoring 65-70 – including Czechia, Hungary, Italy, the US and Japan – appears caught in a middle-SRI trap (in reference to the middle-income trap phenomenon coined by Indermit Gill and Homi Kharas in 2007). Material prosperity increasingly coexists with political polarization and policy volatility. Prolonged stagnation in this range risks greater instability, weaker reform capacity and declining policy predictability, with implications for long-term growth and sovereign risk. These tensions are increasingly visible: between 2020 and 2025, mid-resilience countries – including India, Indonesia, South Korea, the UK and the US - accounted for up to 70% of global strikes, riots and civil commotion (SRCC) events.

- **Social resilience matters for macroeconomic performance, capital-market development and sovereign risk.** For investors and policymakers alike, the SRI functions an both early-warning system and reform compass. Higher resilience is associated with stronger real GDP per capita growth. In high-income economies, higher SRI levels also correlate with deeper capital markets, suggesting stronger market stability and investor confidence. Sovereign exposure, contract frustration and operational disruption are distinct in their mechanics but interconnected in their root cause: low and deteriorating social resilience. Sharp deteriorations in the SRI have preceded episodes of sovereign stress by 12-24 months. Recoveries tend to be asymmetric, with resilience gains often preceding spread compression as markets reprice structural repair with a lag. Social resilience is not a one-way street: declines can be reversed where institutional credibility is strengthened, polarization reduced and opportunity broadened, whereas reliance on short-term fixes risks gradual erosion. For investors, tracking both the level and momentum of the SRI sharpens sovereign risk assessment and helps anticipate structural turning points; for policymakers, sustained reform rather than reactive crisis management remains central to rebuilding long-term resilience.

Uneven and polarized:
resilience consolidates, slowly

Global social resilience gaps remain wide – and structurally entrenched. Social resilience is not merely a social or political concern; it is increasingly a determinant of economic performance and financial stability. Countries with stronger institutions, cohesive societies and lower polarization tend to deliver more predictable policies, more stable growth and lower macro-financial volatility, while weaker resilience is associated with policy uncertainty, reform paralysis and higher risk premia. Against this background, our Social Resilience Index (SRI) assesses 171 jurisdictions on a scale from 0 (lowest resilience) to 100 (highest), aggregating 12 indicators spanning economic fundamentals, social cohesion and institutional and external strength. This year, the global headline SRI score increased only marginally to 47.9 from 47.4 in last edition of the SRI ranking in 2024. This 0.5pt improvement masks persistent divergence across regions (Figure 1). High-income economies remain the clear leaders, with an average score of 70.5 in 2025 – unchanged from 2024 and virtually identical to their 2005 level (70.6). The gap between high-income economies (70.5) and Africa (34.4) exceeds 36 points, underscoring the structural nature of global resilience inequality.

Figure 1: Global evolution of the Social Resilience Index by region 2005-2025



Sources: Allianz Research

Short-term resilience dynamics remain modest in aggregate but diverge across regions, revealing renewed momentum in parts of Asia and stagnation or reversal elsewhere. High-income economies in Europe and North America remained stable at the top of the ranking with a SRI score of 70.5, unchanged compared with 2024. Emerging Asia recorded the most visible improvement, rising from 42.4 in 2024 to 44.3 in 2025, supported by broad-based growth and institutional stabilization. The Middle East edged up from 53.5 to 53.8, maintaining a mid-tier position. By contrast, Emerging Europe declined slightly from 48.9 to 48.4, while Latin America softened from 44.5 to 44.2 despite the earlier post-pandemic recovery. Africa improved only marginally from 34.1 to 34.4 and remains below its 2005 level of 35.9, highlighting persistent structural fragility. While year-on-year changes are modest, the dispersion across regions remains substantial and persistent over time.

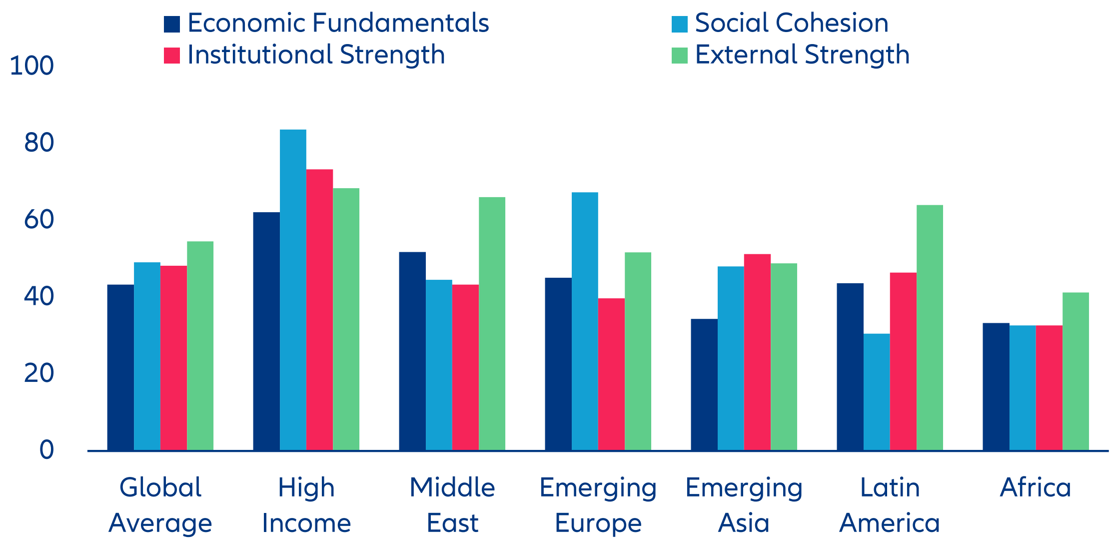
High-income countries recorded the highest levels of social resilience in 2025, led by Northern Europe[1]. Finland ranks first with a score of 84.3, Denmark second (83.8) and Iceland third (81.4). Among advanced economies, Germany (8th, 78.5), France (13th, 74.7) and the UK (18th, 72.9) are in the leading group of countries in the ranking, while Spain (27th, 66.9), Italy (28th, 66.8) the US (32nd, 65.3) and Poland (34th, 64.9) sit in the upper-mid tier. Among emerging economies, the United Arab Emirates (5th, 79.3) records one of the highest social resilience levels globally, while Saudi Arabia (38th, 63.2) ranks in the upper-mid of the distribution. By contrast, large emerging markets such as China (59th, 52.5), India (81st, 46.8), Mexico (105th, 41.3) and Brazil (112th, 40.0) are placed markedly lower in the ranking. At the bottom of the ranking are conflict-affected states, such as Lebanon (169th, 17.7), Democratic Republic of Congo (170th, 16.7) and South Sudan (171st, 11.8).

While the ranking has remained broadly stable for most countries since the last edition of the SRI in 2024, some short-term shifts are particularly noteworthy. Sri Lanka posted the strongest improvement (+12.7pts), climbing 39 ranks from 153rd to 114th, followed by India (+7.8pts), rising 27 positions from 108th to 81st, and China (+4.0pts), which advanced 15 places from 74th to 59th. By contrast, Brazil recorded the steepest deterioration (-8.7pts), falling 41 ranks from 71st to 112th; the Czech Republic dropped 15 places from 20th to 35th (-6.9pts), and Chile slid 30 ranks from 61st to 91st (-5.9pts).

Economic fundamentals are currently – and perhaps surprisingly – the weakest pillar of global resilience, with a global average score of 43.3, indicating that the recent slowdown in global growth and softer labor markets have weighed on the SRI (Figure 2). High-income economies perform considerably better (62.2), supported by stronger labor market participation and fiscal capacity, while structurally weaker regions such as Africa (33.4) and Emerging Asia (34.4) remain constrained by weaker growth dynamics and limited fiscal buffers. At the country level, strong economic fundamentals underpin resilience in Germany and the Nordic economies, while persistent macroeconomic instability and fiscal pressures weigh on South Africa, Nigeria, Egypt, Brazil and Türkiye. Meanwhile, social cohesion – with a global average of 49.2 – remains the primary differentiator between advanced and emerging economies. High-income countries record the highest cohesion levels globally (83.7), reflecting dense welfare systems and stronger institutional trust, as illustrated by Finland, Denmark and Germany. Emerging Europe also maintains relatively strong cohesion (67.4), particularly in Poland, Romania and Serbia, where employment gains and social transfers have supported stability. By contrast, Latin America posts the lowest cohesion score globally (30.5), reflecting structural inequality, informality, crime and limited trust in state institutions; countries such as Brazil, Guatemala and Colombia rank among the weakest in this dimension, and repeated protest cycles in Chile and Peru have tested institutional credibility. Institutional strength has shown limited upward momentum in recent years (48.2 globally), while external strength has become the broadest pillar of resilience (54.6 globally). This partly reflects short-term effects of a fragmenting global economy shifting from highly efficient global supply chains toward more regional and politically aligned ones. The medium-term effects of this shift from efficiency to resilience warrant caution as higher costs and weaker productivity could ultimately weigh on social resilience. In several countries, resilience has shifted from growth-driven to buffer-based stabilization – reducing exposure to sudden external shocks, but not necessarily strengthening long-term structural capacity.

1 The full SRI country ranking for 171 countries is provided in Appendix Table 1.

Figure 2: Drivers of SRI scores by region in 2025

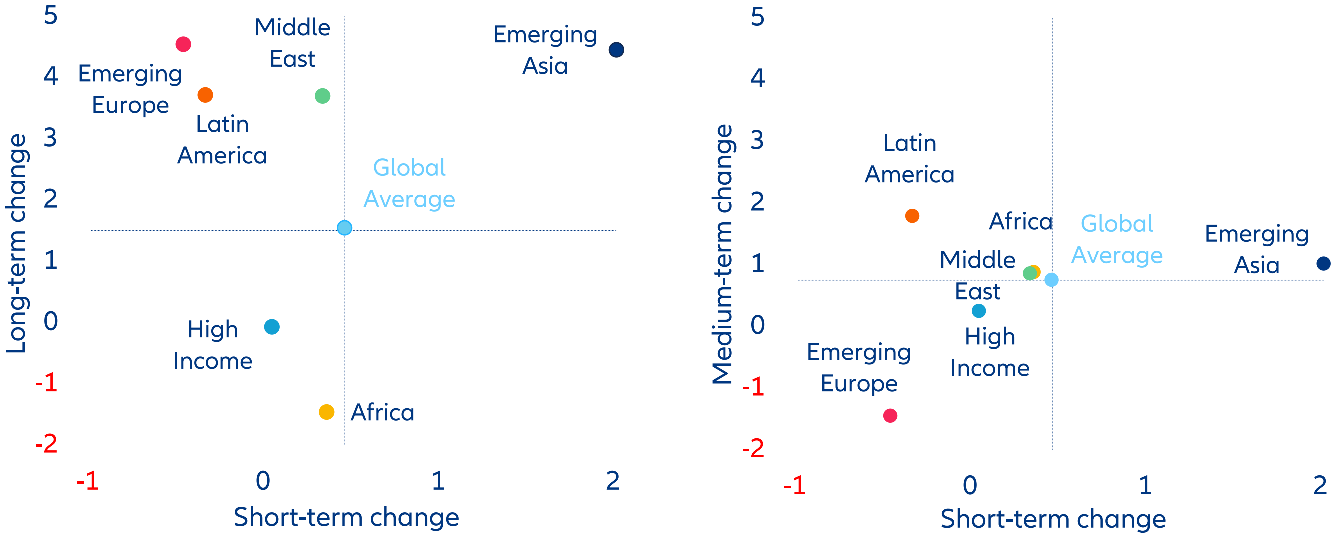


Sources: Allianz Research

Zooming out to a longer time horizon reveals substantial divergences among geographical areas. For the first time, we have reconstructed the SRI back to 2005, allowing us to identify not only current resilience levels but also secular trends and turning points over two decades. The global average has increased only modestly, from 46.3 in 2005 to 47.9 in 2025. High-income economies have remained broadly flat over the period (70.6 vs 70.5), confirming their structural plateau at a high level. The Middle East improved from 50.1 to 53.8, while Emerging Asia rose from 39.8 to 44.3, marking the strongest long-term gain among regions. Latin America recorded a more moderate increase from 40.4 to 44.2, though with visible cyclical setbacks along the way. Emerging Europe advanced from 43.9 to 48.4 before recent moderation. Africa, by contrast, declined overall from 35.9 to 34.4, remaining structurally at the lower end of the distribution. Despite major global shocks – including the financial crisis, the pandemic and the energy shock – the relative distance between regions has narrowed only marginally. Social resilience therefore evolves cumulatively and unevenly, with structural gaps persisting across regions.

Switching to a medium-term perspective, the period from the eve of the Covid-19 pandemic until today shows gradual adjustment amid longer-term scarring. Between 2020 and 2025, the global average increased from 47.1 to 47.9, reflecting gradual recovery from the pandemic shock. High-income economies rose only marginally from 70.3 to 70.5, underscoring limited additional strengthening at the top. The Middle East improved from 53.0 to 53.8, while Emerging Asia recorded a more pronounced increase from 43.3 to 44.3, recently moving back above its pre-pandemic level. Latin America rebounded from 42.4 in 2020 to 44.2 in 2025, recovering from its 2019-20 trough linked to social unrest and institutional stress. In contrast, Emerging Europe declined from 49.9 in 2020 to 48.4, reflecting geopolitical strain and slower post-pandemic momentum. Africa improved only modestly from 33.6 to 34.4, remaining below its 2005 level. Overall, convergence since the pandemic has been gradual and uneven, with some regions regaining lost ground but few achieving a decisive structural break.

Figure 3: Drivers of SRI scores by region in 2025



Sources: Allianz Research

Relating long-term (since 2005) and medium-term (since 2020) changes in the SRI to the most recent movements of the index (2025 vs. 2024) reveals clear regional differentiation (Figure 3). Emerging Asia stands out as the only region improving consistently across all three horizons, outpacing other regions through stronger economic performance and gradual gains in social welfare and institutional capacity. Emerging Europe and the Middle East both show positive long-term trajectories, but their medium-term momentum has weakened: resilience has stagnated in the Middle East and declined in Emerging Europe, where institutional pressures and geopolitical strains have weighed on recent performance. Latin America posts gains over the medium- and long-term, while remaining broadly stable in the short-term. By contrast, high-income economies show limited movement across all horizons, consolidating their structural lead at the top of the global resilience ranking.

Energy price shock triggered by conflict in the Middle East will put resilience gains to the test

The conflict in the Middle East and the resulting surge in energy prices will test the strength of social fabrics across countries[2]. As oil and gas prices surge and supply chains face disruptions – particularly through shipping routes such as the Strait of Hormuz – sustained increases in energy prices could feed into higher inflation, weaker growth and rising political tensions. The SRI helps identify such vulnerabilities by capturing the structural ability of economies and institutions to absorb exogenous shocks. Countries with weaker resilience and limited fiscal buffers are particularly vulnerable, as higher energy costs quickly translate into higher living costs and food inflation where societies have limited capacity to respond. In this context, coping mechanisms – including crisis response capacity, automatic stabilizers and policy optionality – will be pivotal. Among advanced economies, Europe is likely to be more affected than the US given its greater reliance on imported energy, although stronger social spending may cushion part of the impact.

Figure 4 provides a forward-looking perspective by mapping countries' SRI levels against their exposure to food and fuel import shocks. The horizontal axis shows SRI levels in 2025, while the vertical axis measures resilience in food and fuel supply: higher values indicate greater resilience and therefore lower exposure to price shocks. Figure 3 Panel A highlights two clusters among advanced economies. A first green cluster combines upper-middle to high SRI levels with relatively low exposure, reflecting either greater energy independence – as in the US – or strong institutional resilience despite higher import dependence, as in Denmark and Finland. A second yellow cluster groups countries with medium SRI and medium-to-high exposure, including Latvia, Estonia, Slovenia, Hungary and Portugal, where resilience remains relatively solid but exposure to energy price shocks is more pronounced. Among lower- and mid-resilience economies, exposure to food and fuel imports creates a clearer gradient of vulnerability. Figure 3 Panel B shows three clusters, largely composed of emerging markets. A first yellow cluster – similar to the mid-resilience group in advanced economies – includes Poland and Czechia, which combine SRI levels around 65 with moderate exposure and therefore appear relatively well positioned to absorb a temporary shock. A second orange cluster – including Chile, Egypt, Serbia and South Korea – combines moderate resilience with medium exposure to energy price shocks. These countries should be better positioned to cushion the impact of a temporary energy shock on social resilience. The third red cluster represents the most vulnerable group: low-to-mid resilience economies with high exposure, including Vietnam, Thailand, Morocco, Tunisia and Malaysia, where energy price spikes could more easily translate into social pressures.

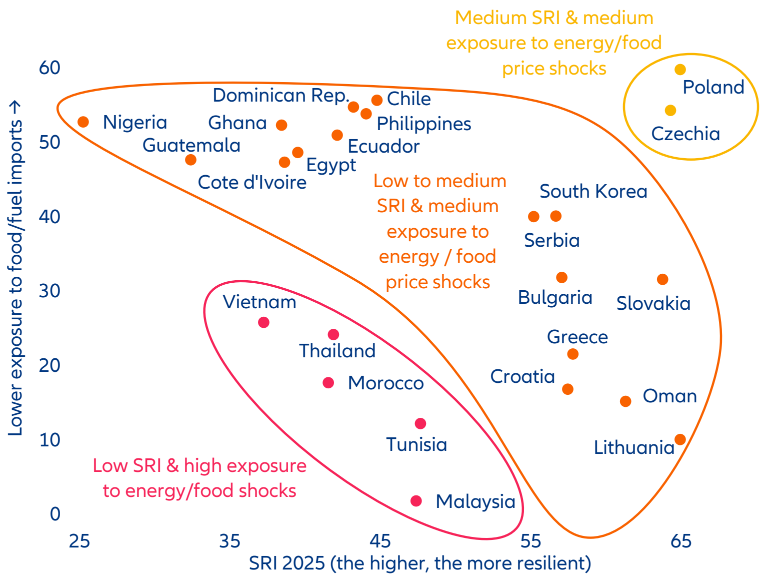
Ultimately, the broader macroeconomic and political effects of an energy price shock could amplify existing social vulnerabilities and increase the risk of instability. Higher energy prices may translate into weaker growth, persistent inflation and higher-for-longer interest rates, weighing on economic fundamentals and potentially aggravating existing distributional conflicts. In countries heading into major elections, these pressures could become a focal point for political contestation and social

2 The conflict that began on 28 February 2026 following the US and Israel's attack on Iran is not reflected in the 2025 SRI results presented in this report.

unrest. Even in the case of the US, for example, a 10-cent increase in the price of gasoline historically leads to a 0.6% drop in the president's approval rating[3]. In parts of the Middle East, disruptions to tourism-driven prosperity could also strain social contracts. Overall, the analysis suggests that energy price shocks tend to amplify structural vulnerabilities already captured by the SRI rather than create new ones, making underlying resilience the key determinant

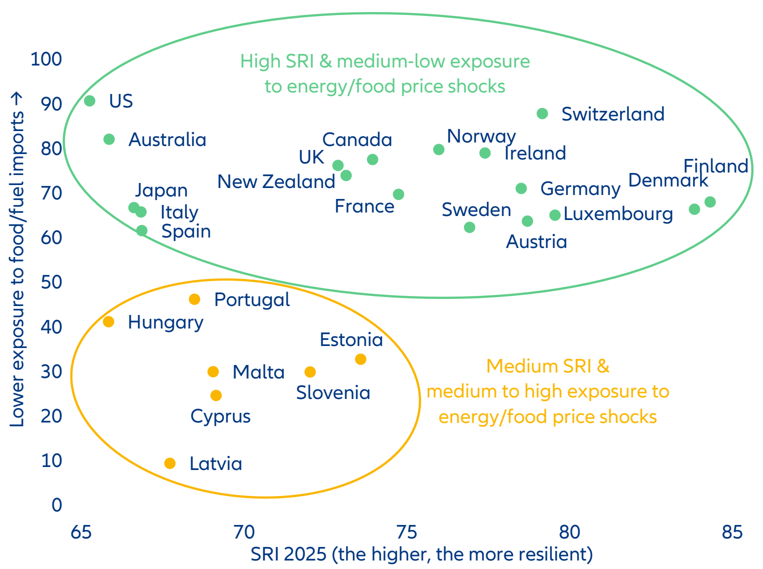
Figure 4: Social resilience and exposure to energy price shocks

Panel A. High-SRI countries



Sources: Allianz Research

Panel B. Low-to-medium SRI countries



Sources: Allianz Research

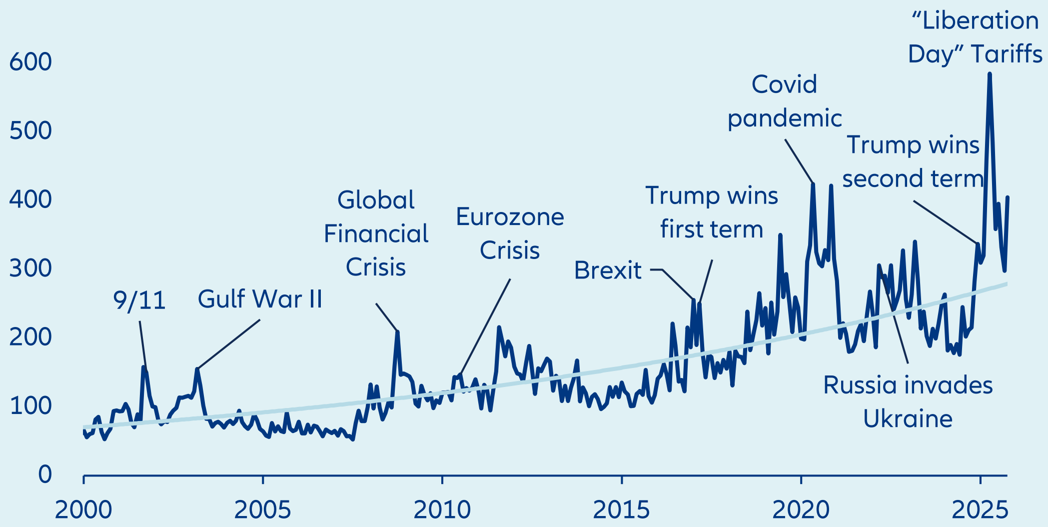
3 Sources: Gupta, R., C. Pierdzioch, A.K. Tiwari (2025), "Gasoline Prices and Presidential Approval Ratings of the United States", American Politics Research, September 2025 https://doi.org/10.1177/1532673X251325458; Harbridge, L., J.A. Krosnick, J.M. Wooldridge (2016), "Presidential Approval and Gas Prices: Sociotropic or Pocketbook Influence?", Political Psychology.

Structural drivers: policy uncertainty rises while liberal democracies decline

As this year's report takes a longer historical perspective on social resilience, it is useful to consider two structural forces shaping the global environment in which resilience evolves: rising economic policy uncertainty and the rise of authoritarian regimes worldwide. These trends are distinct but increasingly interconnected. Higher global uncertainty can trigger diverse domestic responses, including the emergence of charismatic or strongman leadership styles promising stability or protection. In turn, political systems characterized by concentrated power or personalized leadership can reinforce polarization and institutional fragmentation, shaping the trajectory of social resilience across countries.

Figure 5 illustrates the long-term evolution of global economic policy uncertainty (EPU), based on the widely used index[4]. The index measures policy uncertainty by tracking the frequency of newspaper articles containing terms related to the economy, policy and uncertainty across major economies, combined with information on tax-code expirations and economic forecaster disagreement. Over the past two decades, the index shows a clear upward trend. While spikes were previously associated with major shocks such as the 9/11 attacks or the Global Financial Crisis, today's baseline level of policy uncertainty is structurally higher than those earlier peaks. Recent surges have reflected geopolitical tensions, trade conflicts and political polarization, with one of the latest spikes occurring around the announcement of the "Liberation Day" tariffs in April 2025.

Figure 5: Economic policy uncertain worldwide, 2000-2025



Source: Baker et al. (2016), LSEG Workspace, Allianz Research

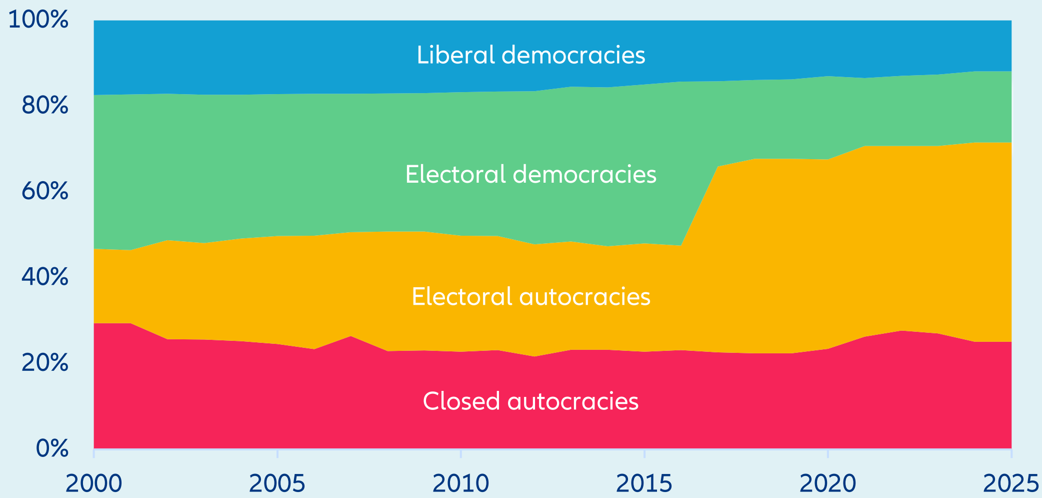
At the same time, the global distribution of political regimes has shifted markedly. Data from the Varieties of Democracy (V-Dem) project, shown in Figure 6, indicate a gradual decline in the number of liberal democracies since the early 2000s, alongside a rise in various forms of autocratic governance[5]. V-Dem classifies regimes based on indicators covering electoral integrity, civil liberties, rule of law and constraints on executive power. A particularly notable category is that of electoral autocracies – systems where elections are formally held but key democratic safeguards, such as media freedom, judicial independence or opposition rights, are significantly constrained. The visible shift around 2017 is largely driven by V-Dem's reclassification of India – the world's largest democracy by population – as an electoral autocracy, reflecting sustained declines in indicators such as media freedom, civil society autonomy and constraints on executive power.

4 Source: Baker, S., N. Bloom and S. Davis (2016), "Measuring Economic Policy Uncertainty," The Quarterly Journal of Economics, 131(4), 1593-1636, https://doi.org/10.1093/qje/qjw024.

5 Source: Coppedge, M. et al. (2024), Varieties of Democracy (V-Dem) Dataset v14, Varieties of Democracy Institute, University of Gothenburg, https://www.v-dem.net/data/the-v-dem-dataset/.

Taken together, these trends point to a more uncertain and politically fragmented global environment. Part of this shift reflects the broader transition from a highly integrated global economy toward more regionalized and geopolitically aligned economic structures as supply chains adapt to geopolitical tensions. In such contexts, domestic political dynamics may amplify underlying social pressures: when economic opportunity, social mobility or institutional performance are perceived to weaken, societies may become more receptive to nationalist or populist leadership narratives. These dynamics do not determine social resilience directly, but they help shape the broader structural backdrop against which resilience evolves across countries..

Figure 6: Share of the world population living in democracies and autocracies, 2000-2025



Source: Baker et al. (2016), LSEG Workspace, Allianz Research

Four clusters of social resilience

Over the long-term, there has been a gradual catch-up and resilience improved globally from 46.3 in 2005 to 47.9 in 2025 – but trajectories across regions and countries are heterogeneous along four structurally differentiated paths. Plotting 2005 SRI levels against long-term changes in SRI reveals four distinct clusters (Figure 7). The analysis shows that resilience is path-dependent but dynamic, and not income-determined. Both convergence and erosion occur among both high- and low-income countries, and institutional momentum – positive or negative – shapes long-term resilience.

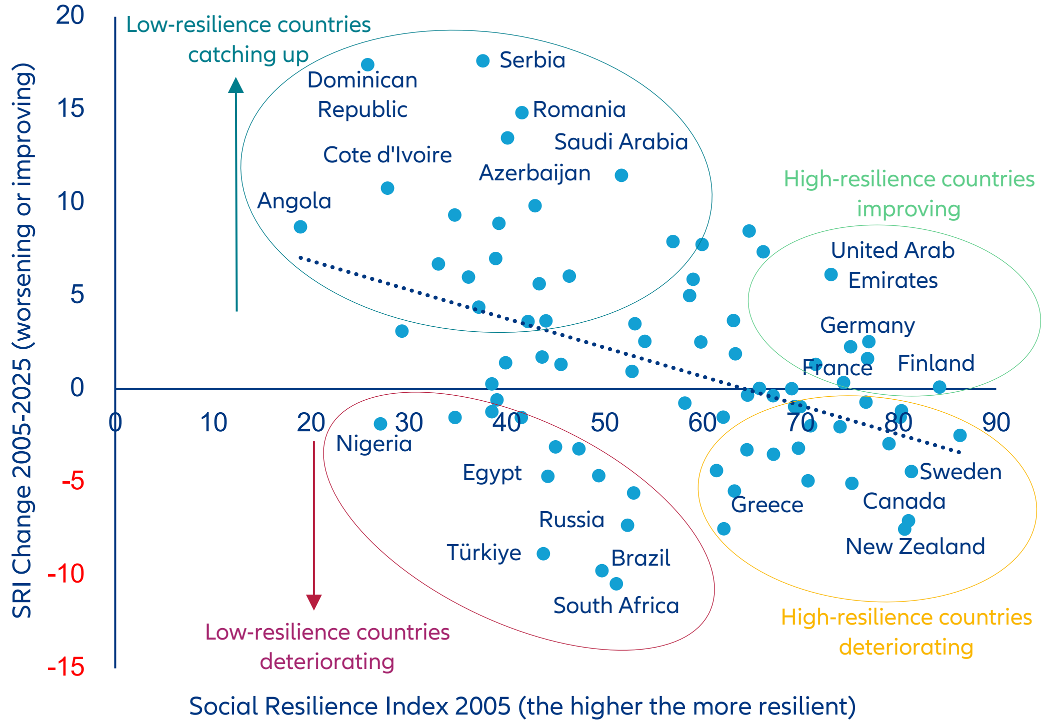
1. Low-resilience countries catching up. Starting from weak institutional and economic baselines, these countries record sustained cumulative gains. Romania and Serbia benefited from EU alignment, FDI-led industrial upgrading and improvements in public administration. Saudi Arabia diversified beyond hydrocarbons and modernized regulatory environments, raising institutional and economic resilience even as social reforms remain uneven. Azerbaijan, Côte d'Ivoire and the Dominican Republic combined stronger growth, improved macroeconomic management and infrastructure investment with institutional consolidation and policy predictability. These cases illustrate how reform momentum, external integration and institutional upgrading can gradually lift structural resilience.

2. Low-resilience countries deteriorating. These countries combine fragile starting positions with ongoing institutional or macroeconomic erosion. Nigeria remains vulnerable due to underinvestment, governance constraints and high poverty, leaving resilience exposed to oil-price volatility and security risks. Egypt faces persistent inflation, repeated IMF programs and geopolitical tensions, while South Africa struggles with weak growth, high unemployment and governance strain. Brazil's institutions are relatively strong, but low productivity growth, polarization and security concerns limit resilience. Türkiye faces sticky inflation, limited fiscal space and tight monetary conditions. Russia stabilized briefly through capital controls and energy revenues after 2014 and 2022, but sanctions, isolation and war-related fiscal and demographic pressures are now weakening resilience. Across these economies, macroeconomic volatility and institutional rigidities reinforce each other, constraining resilience.

3. High-resilience countries improving. These countries already had strong systems and have made further progress on social cohesion, institutional strength and policy adaptability. Germany and Finland sustained fiscal credibility, social cohesion and governance effectiveness through successive shocks, while the United Arab Emirates combined institutional modernization with a strengthening of its external position. These cases demonstrate that resilience can be reinforced even from high starting levels when institutional depth and reform capacity are preserved.

4. High-resilience countries deteriorating. These advanced economies show signs of structural erosion despite solid macro frameworks. Canada, Sweden, Greece and New Zealand have seen declines in institutional trust or social cohesion, reflecting polarization, reform fatigue and distributional tensions. Among the 19 countries clustered around an SRI score of 70, 16 declined this year and seven remain below pre-pandemic levels despite cyclical recovery. This suggests a "hidden scarring" from the 2020–2022 shock: while output, employment and corporate balance sheets have largely recovered, institutional trust and social cohesion have weakened. In several economies, the pandemic and cost-of-living crisis eroded confidence in expert authority, heightened perceptions of unfairness and widened generational and regional divides. The slippage of even top-tier economies highlights a key finding of this report: resilience is not permanent and can deteriorate under compound shocks, with implications for political stability, policy predictability and long-term investment risk.

Figure 7: SRI level in 2005 vs. long-term changes 2005-2025, selected countries



Sources: Allianz Research

While the long-term picture reveals clear secular trends across countries that can be grouped into structural clusters, the medium-term dynamics since 2020 are far less uniform and considerably more fragmented. Figure 8 plots medium-term movements in social resilience since the pandemic in 2020 against the short-term change since 2024, showing dispersed results across countries. Within each region, trajectories have diverged markedly: some economies consolidated post-pandemic stabilization, while others slipped back amid political volatility, inflation pressures or reform fatigue. Latin America illustrates this fragmentation: although the region shows the strongest medium-term rebound since 2020 (+1.8pts), several large economies remain structurally constrained and vulnerable to renewed stress. Emerging Asia has returned above its pre-pandemic level at the aggregate level (+1.0pt since 2020), but underlying institutional and social cohesion paths vary significantly across countries. Europe, despite high starting levels, underperformed in the short term (-1.5pts in Emerging Europe since 2020), reflecting geopolitical strain and uneven institutional adjustment.

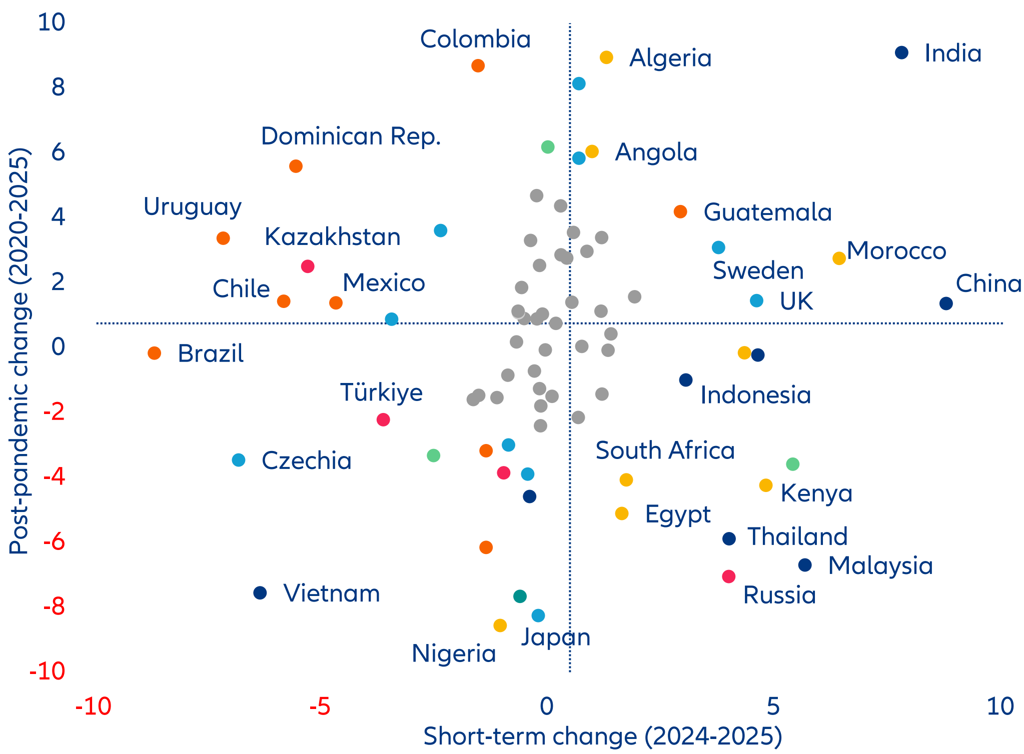
The post-pandemic period therefore reinforces a central conclusion of this report: resilience adjusts slowly in aggregate but can diverge rapidly at the country level. Medium-term movements are shaped less by structural convergence and more by domestic policy choices, geopolitical exposure and institutional credibility – factors that can alter sovereign risk profiles within relatively short horizons.

The long timeframes required to improve resilience also apply to several high-income economies: several markets are now facing a "middle-SRI trap", which is accompanied by greater polarization and marked political alternation – in reference to the middle-income trap phenomenon coined by Indermit Gill and Homi Kharas in 2007[6]. High-income countries that remain around an SRI of 65-70 for too long may turn toward nationalist or populist leaders, reflecting underlying dissatisfaction with economic opportunity, social mobility or institutional performance (see Figure 9). This typically coincides with rising political volatility, weaker reform capacity and declining policy predictability, with negative

6 Source: Kharas, H. and I. Gill (2007), An East Asian Renaissance: Ideas for Economic Growth, World Bank, http://hdl.handle.net/10986/6798.

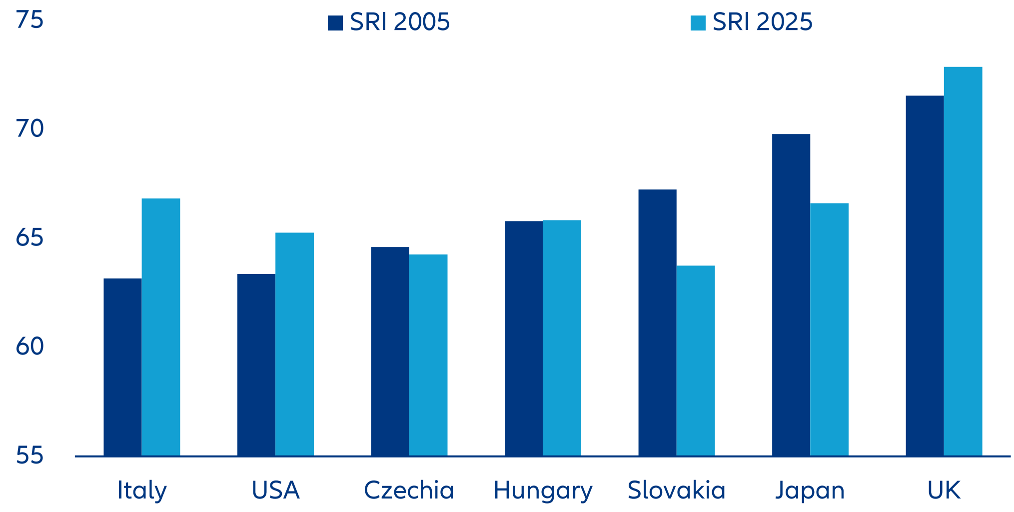
ramifications for economic growth and sovereign risk. In countries such as Czechia and Hungary, relatively weaker scores on institutional checks and balances, media freedom and minority protections have coincided with the rise and consolidation of governments that emphasize national sovereignty, cultural conservatism and skepticism toward supranational constraints (e.g. the EU). Italy and the US have seen repeated electoral successes for parties or leaders promising to protect "insiders" from globalization, migration and perceived elite capture, mirroring the frustration of segments of the population who feel left behind despite high average income levels. Even in Japan, where political turnover is more contained and social cohesion relatively strong, low growth, demographic pressures and persistent concerns over stagnation have made nationalist narratives and calls for a more assertive security posture more salient. These cases suggest that when structurally high-income societies cannot translate wealth into broad-based opportunity and trust, political demand for disruptive, populist or nationalist alternatives grows and protests become more frequent and lasting.

Figure 8: SRI medium-term (post-pandemic) and short-term evolution, selected countries



Sources: Allianz Research

Figure 9: Selected countries with middle-SRI scores



Sources: Allianz Research

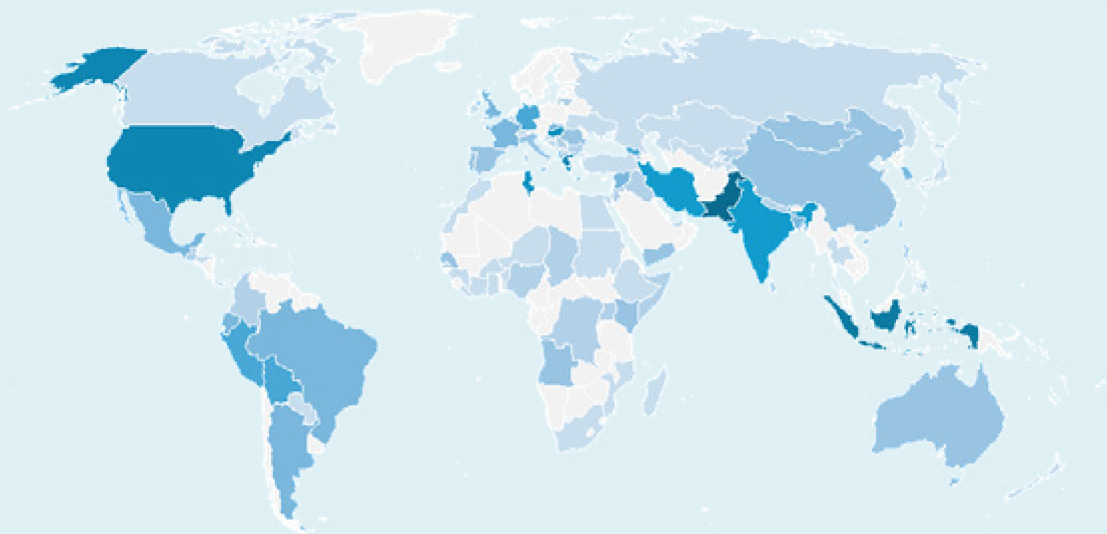
Social unrest: Just the tip of the iceberg

Historically, mass protests have emerged as one of the most visible indicators of social and political stress across both high income and emerging markets. From the streets of Berlin and Paris to the squares of Islamabad and Dhaka, citizens have increasingly taken to public spaces to voice grievances ranging from economic inequality and democratic backsliding to climate policy and government corruption. It has been estimated that a nationwide protest capable of blocking economic activities for more than 24 hours can shave off 0.2pp from GDP[7] .

However, protest activity represents only the visible tip of a broader iceberg of potential business and operational disruption. The additional risk dimensions assessed later in this report – encompassing factors such as supply-chain exposure, political risk and institutional volatility – must be read alongside the protest data to form a complete picture. Taken together, the overall framework allows us to flag countries where social unrest is a meaningful but manageable risk factor, particularly among high and upper mid-SRI states where institutional buffers remain operative. The calculus shifts materially, however, in contexts characterized by very low resilience and weak voice and accountability: here, protest events are less likely to be contained within established channels and more likely to translate into sustained, highly disruptive episodes with direct consequences for business continuity. Countries that saw at least four major strikes, riots and civil commotion (SRCC) events involving more than 1,000 people and lasting more than one day between 2020 and 2025 have been grouped into three SRI tiers for comparative purposes.

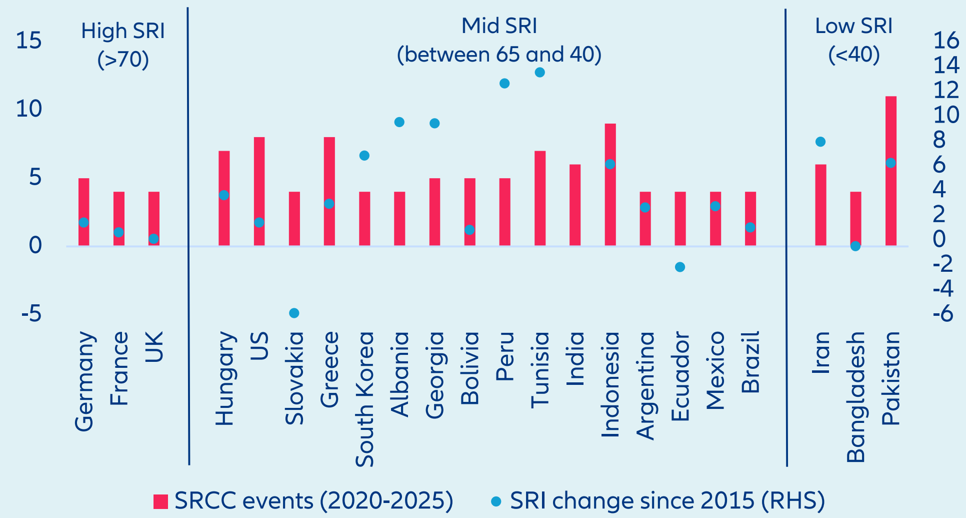
7 Source: Baker, S., N. Bloom and S. Davis (2016), “Measuring Economic Policy Uncertainty,” The Quarterly Journal of Economics, 131(4), 1593-1636, https://doi.org/10.1093/qje/qjw024.

Figure 10: Map of reported events in 2015-2025 with active participation exceeding 1,000 people and lasting more than one day, the darker the higher



Source: Carnegie Endowment, Allianz Research

Figure 11: SRCC events and SRI changes, selected countries sorted by SRI level



Source: Carnegie Endowment, Allianz Research

High-SRI countries: Institutional stability despite protest pressure

Germany, France and the UK – all scoring above 70 on the SRI – recorded a modest four to five significant protest events each over the review period. This relatively contained protest frequency, combined with stable or marginally improving SRI scores, suggests that in high-resilience democracies, public mobilization tends to operate within established channels and does not fundamentally destabilize institutional order. Germany's SRI of 78.5 – the highest in the sample – reflects strong rule of law, deep social trust networks and a political culture accustomed to managed dissent. The minor uptick of 1.4pts since 2015 points to continued, if gradual, consolidation of social resilience. France and the UK show similarly stable, if marginally lower, SRI trajectories, with the latter's near-flat change of +0.1pts consistent with the post-Brexit turbulence that has weighed on institutional confidence.

Mid-SRI countries: A heterogeneous and contested middle ground

The mid-SRI tier is the most internally diverse and analytically interesting grouping, being the main contributor to protest figures globally, with around 70% SRCC events localized within this group. It spans democracies experiencing different forms of stress (Brazil, Hungary, Mexico, Slovakia and the US), newly mobilized societies in East Asia and the Caucasus (South Korea, Georgia, Albania) and economically volatile states in Latin America and Southeast Asia (Bolivia, Peru, Argentina, Ecuador, Indonesia). The United States, with eight protests recorded – the joint-highest within this tier alongside Indonesia – shows in parallel only a modest SRI increase over the past decade. India stands out with six recorded protest events and an SRI of 46.8; it has nonetheless registered the largest positive SRI shift in the entire sample, at +16.6pts since 2015, distancing itself from neighboring Bangladesh and Pakistan, which remain in the low-SRI segment. The SRI performance reflects measurable gains in state capacity and economic integration metrics, though civil liberties indicators remain contested. A similarly sharp upward trajectory appears in Tunisia (+13.5pts) and Peru (+12.6pts), though both countries continue to face acute governance fragility and population is much smaller. At the other end of the mid-tier, Slovakia and Ecuador have seen SRI declines of 5.9pts and 2.2pts respectively, pointing to deteriorating institutional environments that merit monitoring. Indonesia leads the mid-tier in protest frequency at nine events, yet maintains an SRI of 45.9 with a positive trajectory (+6.1pts). This combination may reflect a democratizing society in which civic mobilization is itself a sign of growing political engagement rather than systemic instability.

Low-SRI countries: High protest intensity, fragile state structures

The three low-SRI countries – Iran (38.1), Bangladesh (30.8) and Pakistan (27.0) – present the starkest case for the relationship between state fragility and protest intensity. Pakistan, with 11 recorded protest events, records both the highest protest frequency and the lowest SRI in the sample. Its modest positive change of +6.2pts since 2015 does not yet bring it into the mid-SRI band and should be interpreted cautiously in light of continued political volatility, civil-military tensions and macroeconomic stress. Iran's SRI of 38.1, combined with six protest events, reflects a pattern of recurrent, high-stakes mobilization against a backdrop of authoritarian control. The positive SRI change of +7.9pts appears paradoxical given documented crackdowns on civil liberties, and may be driven by improvements in economic or health-related sub-indices that partially offset governance deficits. Bangladesh, with four protests and an SRI of 30.8, has seen a marginal SRI decline of 0.5pts. Its position near the boundary between the low and mid tiers warrants attention, particularly in light of recent political transitions and their potential downstream effects on institutional stability.

Skating on thin ice

The data reveal a broadly consistent, if nuanced, relationship between social resilience and the frequency and character of protest activity. High-SRI countries tend to experience fewer large-scale mobilizations, and when they do occur, institutional frameworks appear better equipped to accommodate and channel dissent without systemic rupture. Low-SRI environments, by contrast, are characterized by higher protest frequency alongside weaker state capacity to manage social conflict, amplifying both the probability and the potential consequences of instability.

The SRI change variable introduces a critical temporal dimension. Countries that have improved resilience over the past decade – even from a low base – may be developing the institutional buffers necessary to better manage future social pressures. Conversely, states with declining SRI scores, regardless of their current tier, face a compounding risk: weakening institutions coinciding with heightened mobilization creates conditions in which protest can more readily escalate beyond manageable thresholds. Decision makers should treat protest frequency not merely as a symptom to be managed, but as a diagnostic signal. Where mobilization coincides with declining resilience, early investment in institutional capacity, social trust and inclusive economic policy may prove far less costly than reactive crisis management

Macro-financial and country-risk relevance of social resilience

Relating social resilience to macroeconomic and financial outcomes is conceptually challenging because these variables co-evolve and may influence each other in both directions. The relationships presented in this section should therefore be interpreted as correlations rather than causal effects. Stronger economic growth and stable financial conditions can improve social outcomes and institutional trust, thereby strengthening social resilience. At the same time, more resilient societies – characterized by stronger institutions, social cohesion and policy credibility – may support macroeconomic stability, reduce risk aversion and encourage investment and capital-market development. In other words, reverse causality is possible and likely, as resilience and macro-financial conditions evolve together over time. Moreover, the SRI captures structural social and institutional conditions that typically change slowly, whereas many macroeconomic and financial indicators fluctuate at a much higher frequency.

To account for these differences, the analysis focuses on within-country correlations over longer horizons, relating changes in resilience to structural macro-financial variables rather than short-term cyclical movements. While the results should not be interpreted as evidence of causality, they nevertheless provide useful insights into the co-movement and long-term interaction between social resilience, macroeconomic performance and country risk across economies.

A broad academic and policy literature suggests that social cohesion, institutional stability and political polarization can have meaningful macro-financial consequences. Higher political fragmentation and policy uncertainty tend to increase economic uncertainty, raising risk premia and discouraging investment and hiring[8]. In such environments, firms and households often postpone spending decisions and financial markets demand higher compensation for risk[9]. Empirical studies also link institutional quality and political stability to stronger growth performance and improved investment dynamics. For example, cross-country research finds that institutional quality is positively associated with long-term economic growth, while instability and governance weaknesses tend to weigh on investment and capital accumulation[10]. Taken together, these findings suggest that diminished social cohesion, institutional fragility and polarization are not only political phenomena but also macro-financial risk factors that can influence growth trajectories, capital formation and sovereign risk.

Across world regions and over time, improvements in social resilience are positively associated with stronger long-term real GDP per capita growth, although the relationship remains modest and far from mechanical. This link is not circular: growth enters the SRI only as a short-term deviation from a country's long-run growth trend, whereas the growth variable used here reflects secular long-term growth performance. Overall,

8 Source: Baker, S., N. Bloom, and S. Davis (2016), "Measuring Economic Policy Uncertainty," Quarterly Journal of Economics 131(4), 1593–1636; M. Azzimonti (2018), "Partisan Conflict and Private Investment," Journal of Monetary Economics 93, 114–131.

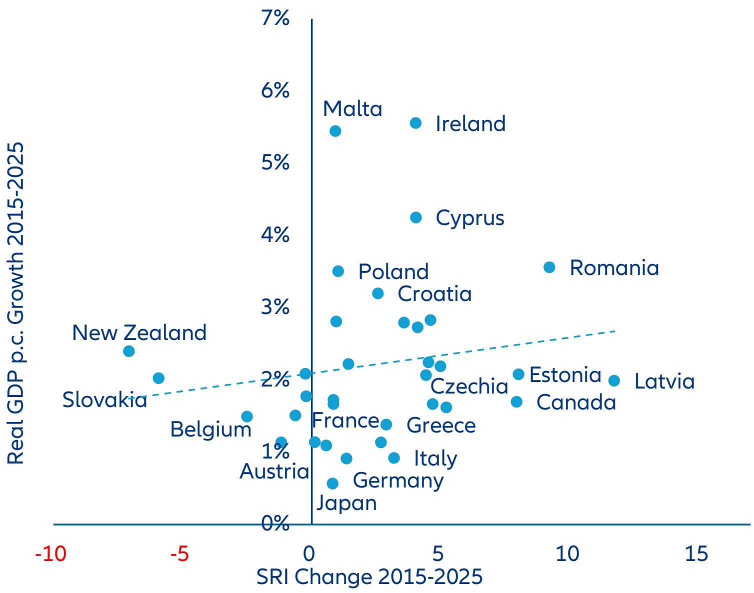
9 Source: Bloom, N. (2009), "The Impact of Uncertainty Shocks," Econometrica 77(3), 623–685; Kempf, E. and M. Tsoutsoura (2024), "Political Polarization and Finance," Annual Review of Financial Economics, 16, 121–148.

10 Source: Alesina, A., S. Özler, N. Roubini, and P. Swagel (1996), "Political Instability and Economic Growth," Journal of Economic Growth 1(2), 189–211; J. Duca and J. Saving (2016), "Income Inequality and Political Polarization: Time Series Evidence over Nine Decades," Review of Income and Wealth 62(3), 445–466.

the evidence suggests that resilience improvements tend to coincide with stronger long-term growth outcomes, particularly in emerging economies, though structural factors such as institutional quality, economic fundamentals and external vulnerabilities remain important drivers. The relationship is strongest in emerging markets, where a 1pt (10pts) increase in the SRI corresponds to roughly +0.1% (+1%) higher long-term GDP per capita growth per year. In high-income economies, the association is smaller, with a 1pt (10pts) increase corresponding to about +0.05% (+0.5%) higher growth (R^2 = 0.0257). Regionally, the relationship appears somewhat stronger in Emerging Asia, where a 10pts increase in resilience corresponds to roughly +0.6% higher growth (R^2 = 0.0858). India stands out as a positive outlier, combining robust long-term growth (+5.7%) with large resilience gains, while Vietnam represents a divergence case, posting strong growth despite declining resilience. The relationship becomes more pronounced in Latin America (R^2 = 0.1767), where a 10pts increase in resilience corresponds to about +0.9% higher growth, illustrated by the Dominican Republic, and is strongest in Africa, where a 10pts increase in SRI is associated with roughly +1.5% higher GDP per capita growth.

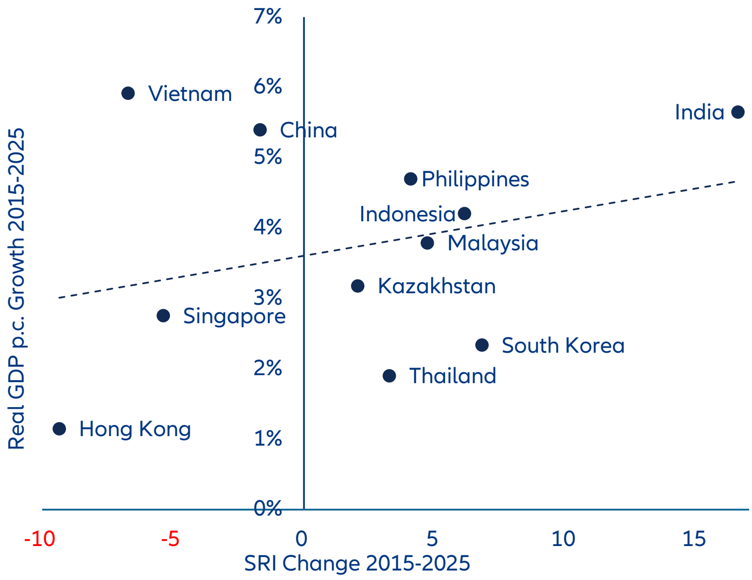
Figure 12: Correlation of SRI changes 2015-2025 and average real GDP p.c. 2015-2025, by region

Panel A: High-income economies



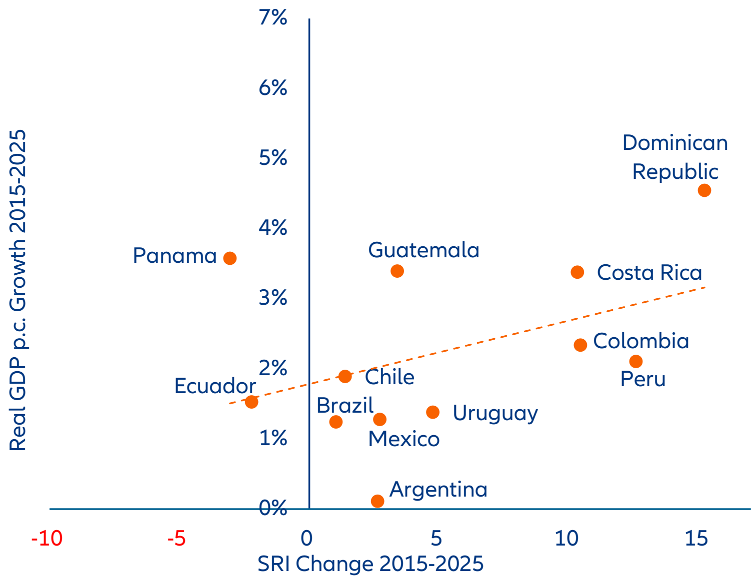
Sources: LSEG Workspace, Allianz Research

Panel B: Emerging Asia



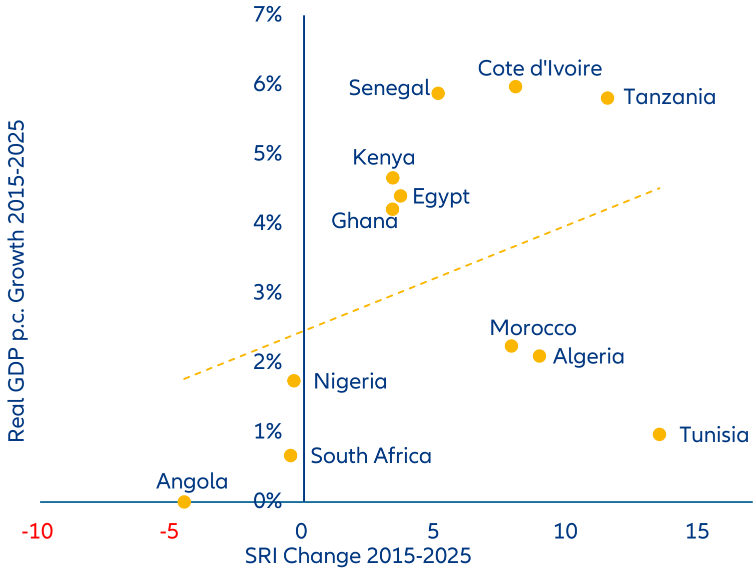
Sources: LSEG Workspace, Allianz Research

Panel C: Latin America



Sources: LSEG Workspace, Allianz Research

Panel D: Africa



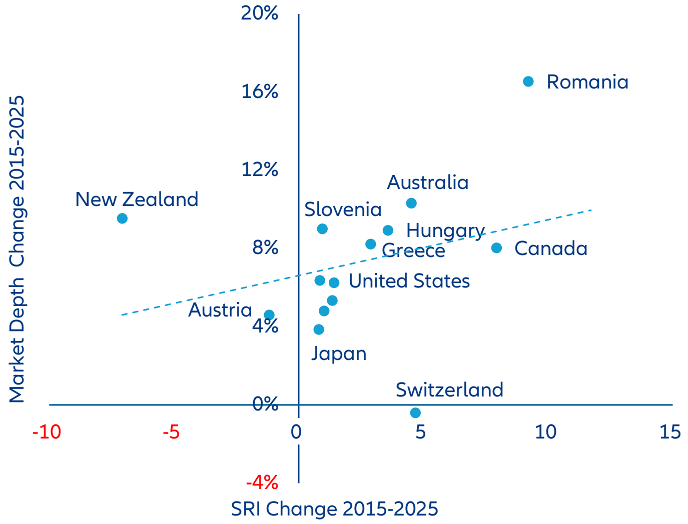
Sources: LSEG Workspace, Allianz Research

Social resilience can be associated with greater financial market depth, measured at the country level as the sum of outstanding government bonds and stock market capitalization relative to GDP. This metric is an approximation of long-term development of financial market depth. We focus on public traded assets (bonds and equity) and do not consider the financial intermediation by the banking sector. In many advanced economies, deeper and more sophisticated financial markets co-evolve with broader social and institutional development, reflecting stronger legal frameworks, better contract enforcement and more effective risk-sharing. However, social resilience should not be interpreted as a sufficient condition for deep financial markets as much as deep financial markets are not necessarily sign of social resilience. Many specific factors can come into play. In economies where small and medium-sized enterprises (SMEs) dominate production, bank-based lending often remains the primary funding channel. Legal and regulatory barriers, along with cultural attitudes toward risk, further shape the structure of the financial system. In more risk-averse societies, agents may preferentially allocate savings into bank deposits and insurance products rather than traded assets, which slows financial market deepening without necessarily implying weaker underlying economic potential. But well-developed financial intermediation can enhance resilience as it increases the capacity of households, firms and governments to absorb and recover from shocks by mobilizing savings, providing access to liquidity and credit. The link between financial development and social resilience is mutually reinforcing and primarily correlational rather than strictly causal. Consequently, relating changes in the SRI to financial market depth should be interpreted as suggestive and illustrative rather than as providing a fully analytical causal assessment.

Across advanced economies, improvements in resilience correlate positively with changes in market depth. The estimated slope suggests that a 10pts increase in SRI corresponds to roughly a +2.8% annual increase in capital-market depth over the long-run, while the explanatory power remains moderate ($R^2 = 0.083$). Country examples illustrate both the alignment and limits of this relationship. Romania combines one of the strongest resilience improvements with a marked increase in financial market depth, reflecting the rapid development of its capital market and growing participation on the Bucharest Stock Exchange[11]. By contrast, Switzerland experienced resilience gains despite a slight decline in market depth, while New Zealand recorded falling resilience even as financial depth increased modestly. These cases highlight that financial development and resilience are related but not mechanically linked, as regulatory frameworks and domestic financial dynamics also play an important role.

11 Source: OECD (2022), "Capital Market Review of Romania: Towards a National Strategy", OECD Capital Market Series, OECD Publishing, Paris, https://doi.org/10.1787/9bfc0339-en.

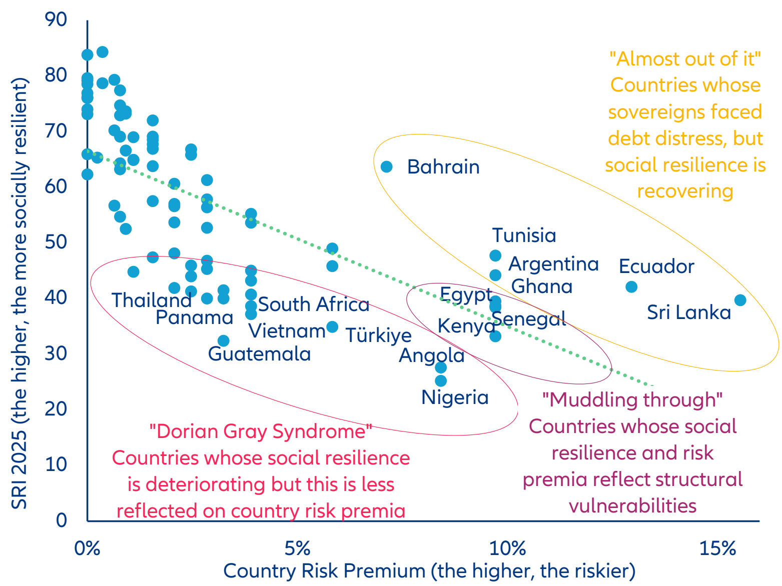
Figure 13: Correlation of SRI changes 2015-2025 and capital market depth changes 2015-2025, high-income economies



Sources: LSEG Workspace, Allianz Research

Beyond macroeconomic and financial market outcomes, the SRI may also serve as an early warning indicator that foreshadows systemic crises, while revealing recovery asymmetries in which societies rebound faster than their sovereigns. While protests and social unrest represent the most visible manifestations of compromised social resilience, they constitute only one tip of the iceberg. Capturing latent risks and emerging signals across diverse geographies may help identify key threats before they materialize into crisis events. For example, marked deterioration in SRI scores or persistently low levels can foreshadow broader systemic crises and spikes in country risk (see purple-circled countries in Figure 14). These deteriorating trends often precede sovereign defaults, political instability or economic collapse by 12-24 months, providing a critical window for risk mitigation. Notably, improvements in social resilience tend to outpace the normalization of sovereign risk spreads following defaults or crises (see yellow-circled countries in Figure 14). This divergence reveals that societies possess fundamental adaptive capacities that enable them to recover more rapidly than their governments' creditworthiness – a crucial insight for investors with longer time horizons.

Figure 14: SRI level and country risk premium trends, selected countries



Sources: Damodaran, Allianz Research

A three-dimensional risk framework for investors. This future outlook is particularly relevant for trade partnerships and infrastructure-related capital that are exposed to political violence, arbitrary measures (confiscation, expropriation and nationalization) or strikes, riots and civil commotion (SRCC) events leading to outright business interruption. From a foreign investor perspective, low social resilience goes in parallel with a multi-layered risk matrix:

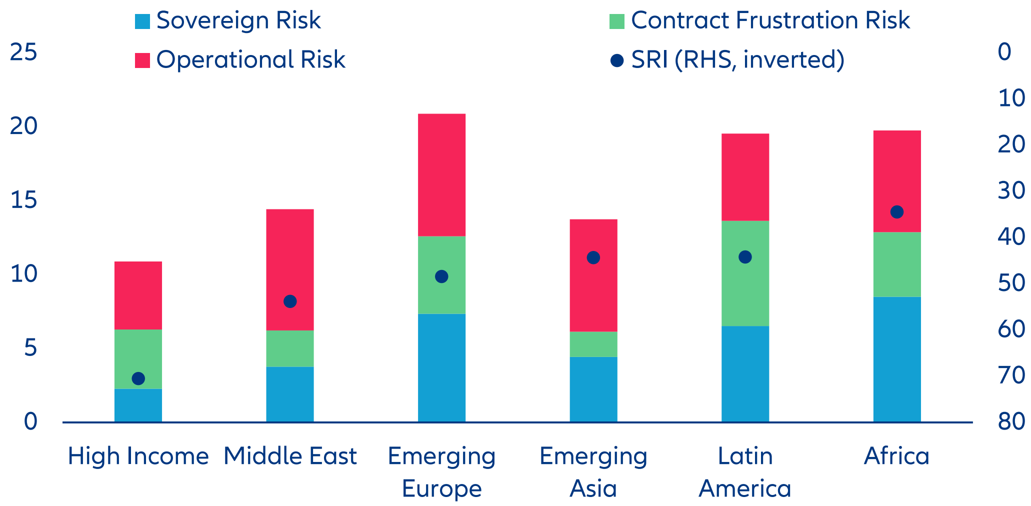
1. Sovereign Risk: Low SRI scores frequently correlate with elevated sovereign default risk, currency instability and capital controls, which are particularly frequent in Africa, South-Eastern Europe and Central Asia.

2. Contract Frustration Risk: Weak social resilience can increase the likelihood of investor-state disputes, particularly around large or politically sensitive projects. Governments under social pressure may seek to revise, delay or unilaterally terminate contracts – especially those perceived domestically as unfair, opaque or misaligned with social and environmental priorities. This creates a higher probability of expropriation (direct or creeping), contract renegotiation and arbitration. This risk is particularly acute in extractive industries, infrastructure concessions and public-private partnerships, with Latin America as a notable example.

3. Operational Risk: Deteriorating social resilience is associated with more frequent strikes, protests, road blockades and localized violence, disrupting logistics, restricting site access, endangering staff and causing production and revenue losses. Firms may face repeated shutdowns and higher costs for security, insurance and contingency planning, while strikes, riots and civil commotion (SRCC) can have lasting effects on institutional stability and sovereign credit. Countries with limited space for voice and accountability are particularly prone to generational, anti-establishment or cost-of-living protests, as seen in China, Ecuador, Guatemala, Iran, Kenya, Mexico, Morocco, Russia, Serbia, Sri Lanka and Tunisia. Since 2020, protest activity has also risen in Emerging Asia (notably India, Indonesia, Pakistan and South Korea) and high-income economies such as Germany, the UK and the US, indicating that operational risks from social unrest are no longer confined to EMs.

The three risk dimensions outlined above – sovereign exposure, contract frustration and operational disruption – are distinct in their mechanics but interconnected in their root cause: low and deteriorating social resilience. For investors, the SRI's most actionable feature is its lead time, having historically anticipated sovereign stress by 12 to 24 months, offering a window to reposition before spreads widen or ratings agencies move. For companies holding infrastructure concessions, extractive licenses or public-private partnerships, that same deterioration signals a shifting political economy in which governments under social pressure will prioritize domestic legitimacy over contractual commitments – making political risk insurance and early legal safeguards essential rather than optional. For exporters and logistics-dependent businesses, the critical shift is geographic: SRCC-driven disruption is no longer an emerging-market footnote but a scenario to be stress-tested across the full operational footprint, including high-income countries where protest frequency has surged since 2020. Across all three dimensions, the SRI reframes social unrest from a reputational or contextual concern into a structured, forward-looking input to risk pricing, contract design and contingency planning.

Figure 14: Risk dimensions by region, index: 0 = lowest, 10 = highest



Sources: Damodaran, ICSID, Carnegie Endowment, World Bank, Allianz Research

Long-term strategies vs. unforeseen disruptions

Social resilience trends are not irreversible, as countries can recover through targeted policy interventions and institutional reforms that address structural vulnerabilities. Improvements in SRI scores often stem from equitable wealth distribution, reduced polarization, restored trust between citizens and institutions, and enhanced institutional adaptability to shocks. Recovery paths, however, vary based on initial conditions, political will, and external pressures. Some countries remain stuck in a middle-SRI equilibrium, marked by stagnation, while others succeed in establishing virtuous cycles of resilience-building. Case studies, such as the Baltic region's progress versus Latin America's stagnation, highlight effective strategies for escaping the mid-SRI trap and fostering resilience. These examples underscore that while challenges are universal, successful pathways require context-specific strategies tailored to local political economies and social structures.

The Baltic economies, particularly Estonia, Latvia, and Lithuania, have strengthened their social resilience during periods of transition, leveraging geopolitical challenges and geoeconomic fragmentation as catalysts for well-being consolidation. Estonia, in particular, has led with transformative reforms. In 2023, it introduced adaptive unemployment benefits, extending payment periods based on labor market conditions to prevent poverty during downturns. The 2024 Social Welfare Act amendments prioritized aging in place by funding home-care services, reducing reliance on institutional facilities. To address healthcare disparities, the Primary Health Care Reform expanded nurses' roles and incentivized general practitioners to serve rural areas, ensuring equitable access to care. Estonia also advanced its resilience through digital innovation and security. The 2017 launch of the world’s first Data Embassy in Luxembourg safeguarded government continuity against cyber and physical threats. The 2023 National Security Concept institutionalized civil preparedness by integrating private sector IT expertise and local government responsibilities into a unified defense framework. Socially, the Adaptive Unemployment Insurance system provided an automated safety net during economic shocks, while the 2024-2025 Civil Protection Law mandated crisis-readiness training and emergency supply standards for households. These initiatives, combined with digital governance, energy independence through the REPowerEU initiative, and community-level crisis management, have created a deeply interconnected and resilient society.

Conversely, in Latin America, social resilience in many of the major countries is stagnant or declining – signaling the need for comprehensive reform and greater sustainability in the literal sense of the term, i.e., from the perspective of future generations. Economic growth rates consistently lower than the global average, the erosion of savings and pension funds to maintain purchasing power in a context of persistently high interest rates, declining real wages and increasing insecurity have weighed heavily on the SRI particularly in recent years, with several countries falling significantly in our ranking. These structural issues are deeply entrenched and difficult to resolve in the short term. The feeling of a lost decade accompanies the ongoing electoral cycles in the region, compounded by the external interference that many countries are subject to in the current geopolitical context, which further undermines institutional stability and public trust. Unlike the Baltic experience, where geopolitical pressure catalyzed institutional innovation and social cohesion, Latin American countries have struggled to translate external challenges into opportunities for systemic reform. The fragmentation of social protection systems, inadequate investment in digital infrastructure and weak coordination between national and local governments have left populations vulnerable to both economic shocks and environmental crises. Moreover, the region faces a critical demographic transition without having built adequate institutional capacity: pension systems remain underfunded and fragmented, healthcare coverage is uneven and often excludes informal workers who constitute a large share of the labor force and educational outcomes have stagnated or declined, limiting intergenerational mobility. The lack of a unified strategic vision for resilience-building, combined with political volatility and fiscal constraints, has prevented the emergence of comprehensive frameworks comparable to those developed in the Baltic states. To reverse these trends, Latin American countries would need to prioritize long-term institutional strengthening over short-term political gains, invest in digital governance and civic preparedness and develop adaptive social protection mechanisms that can respond dynamically to economic and environmental shocks while rebuilding the social contract between governments and citizens.

In a time of increasing geoeconomic fragmentation and rapid social change, proactively addressing social resilience dynamics is crucial for long-term success. This requires shifting from reactive crisis management to proactive monitoring, recognizing that social cohesion and institutional capacity are dynamic and demand continuous evaluation. Strengthening portfolio resilience involves diversifying exposure across countries with varying SRI profiles, reducing concentration risks and integrating SRI metrics into political risk insurance and contract structuring, such as force majeure clauses and dispute resolution mechanisms. Organizations should adopt scenario-based stress tests to model business impacts under different SRI deterioration scenarios and engage proactively with stakeholders in low-SRI environments to build trust and prevent tensions. Beyond mitigating risks, identifying opportunities in countries with reforms or institutional innovation can drive long-term investment, particularly in resilience-building sectors like digital infrastructure, healthcare, education and renewable energy. Persistent declines in social resilience should prompt caution and engagement to assess whether declines are cyclical or structural and explore constructive roles for external actors. By embedding social resilience analysis into core strategies, organizations can better navigate volatility, support sustainable development and seize opportunities overlooked by those focused solely on traditional economic metrics.

Appendix: Country ranking and methodology

Appendix 1: Social Resilience Index 2025

Country	SRI 2025	Rank 2025	SRI Change 2024-2025	Rank Change 2024-2025	SRI Change 2005-2025	Rank Change 2005-2025
Finland	84,3	1	▼ -0,7	▬ 0	▬ 0,1	▲ 2
Denmark	83,8	2	▼ -0,7	▬ 0	▼ -2,5	▼ -1
Iceland	81,4	3	▲ 3,1	▲ 5	▼ -3,7	▼ -1
Luxembourg	79,6	4	▼ -0,2	▬ 0	▲ 2,5	▲ 6
United Arab Emirates	79,3	5	▼ 0,0	▬ 0	▲ 6,1	▲ 12
Switzerland	79,2	6	▼ -0,9	▼ -3	▼ -1,2	▲ 1
Austria	78,7	7	▼ -0,2	▼ -1	▼ -1,5	▲ 1
Germany	78,5	8	▼ -0,3	▼ -1	▲ 1,6	▲ 3
Ireland	77,4	9	▲ 0,8	▬ 0	▲ 2,3	▲ 5
Sweden	76,9	10	▲ 3,7	▲ 7	▼ -4,4	▼ -6
Netherlands	76,1	11	▲ 0,5	▬ 0	▼ -2,9	▼ -2
Norway	76,0	12	▲ 0,2	▼ -2	▼ -0,7	▬ 0
France	74,7	13	▼ -0,3	▼ -1	▲ 0,3	▲ 2
Canada	74,0	14	▼ -0,2	▬ 0	▼ -7,1	▼ -9
Estonia	73,6	15	▼ -0,2	▬ 0	▲ 7,4	▲ 14
Kuwait	73,2	16	▼ -0,3	▬ 0	▲ 8,5	▲ 16
New Zealand	73,1	17	▼ -1,6	▼ -4	▼ -7,5	▼ -11
United Kingdom	72,9	18	▲ 4,6	▲ 7	▲ 1,3	▬ 0
Slovenia	72,0	19	▲ 1,4	▲ 2	▼ -2,0	▼ -3
Qatar	70,2	20	▼ -2,6	▼ -2	▼ -5,1	▼ -7
Cyprus	69,1	21	▲ 0,2	▲ 3	▲ 0,0	▲ 3
Belgium	69,1	22	▼ -1,2	▬ 0	▼ -2,0	▼ -3
Malta	69,0	23	▼ -3,5	▼ -4	▼ -0,9	▼ -2
Portugal	68,5	24	▼ -0,5	▼ -1	▼ -0,9	▼ -1
Bahamas	68,4	25	▲ 0,8	▲ 2	▲ 3,6	▲ 6
Latvia	67,7	26	▲ 0,6	▲ 2	▲ 7,8	▲ 15
Spain	66,9	27	▼ -0,1	▲ 2	▼ -0,4	▲ 1
Italy	66,8	28	▲ 1,1	▲ 3	▲ 3,7	▲ 9
Japan	66,6	29	▼ -0,2	▲ 1	▼ -3,2	▼ -7
Australia	65,9	30	▼ -2,4	▼ -4	▼ -4,9	▼ -10
Hungary	65,8	31	▲ 0,6	▲ 2	▲ 0,0	▼ -1
United States	65,3	32	▼ -0,2	▬ 0	▲ 1,9	▲ 3
Lithuania	64,9	33	▲ 1,9	▲ 6	▲ 5,9	▲ 10
Poland	64,9	34	▲ 1,3	▲ 4	▲ 7,9	▲ 12
Czech Republic	64,3	35	▼ -6,9	▼ -15	▼ -0,3	▼ -2
Slovakia	63,8	36	▲ 0,1	▲ 1	▼ -3,5	▼ -9
Bahrain	63,7	37	▼ -0,4	▼ -3	▲ 5,0	▲ 7
Saudi Arabia	63,2	38	▲ 0,5	▲ 2	▲ 11,5	▲ 16
Singapore	62,3	39	▼ -1,7	▼ -4	▲ 2,5	▲ 3
Barbados	61,3	40	▼ -0,4	▲ 2	▼ -6,7	▼ -14
Oman	61,3	41	▼ -0,7	▬ 0	▼ -3,3	▼ -7
St. Lucia	61,2	42	▲ 0,5	▲ 1	▲ 7,1	▲ 6
Israel	60,6	43	▲ 5,4	▲ 10	▼ -1,5	▼ -4
Greece	57,8	44	▲ 1,2	▲ 4	▼ -5,5	▼ -8
Croatia	57,5	45	▼ -0,9	▬ 0	▼ -0,8	▬ 0

Country	SRI 2025	Rank 2025	SRI Change 2024-2025	Rank Change 2024-2025	SRI Change 2005-2025	Rank Change 2005-2025
Bhutan	57,1	46	6,6	16	12,2	30
Bulgaria	57,0	47	0,7	2	-4,4	-7
South Korea	56,7	48	-0,4	-1	2,6	-1
Uruguay	56,6	49	-7,2	-13	3,5	0
Romania	56,4	50	0,2	0	14,8	48
Brunei	56,3	51	-1,4	-5	-12,2	-26
Serbia	55,2	52	-0,7	-1	17,6	62
Hong Kong	54,7	53	-0,4	1	-7,5	-15
Mongolia	54,3	54	8,4	33	12,2	38
Kazakhstan	53,7	55	-5,3	-11	0,9	-4
Costa Rica	53,6	56	0,6	3	13,5	49
Timor-Leste	52,9	57	-1,1	-2	4,4	5
Azerbaijan	52,7	58	-1,0	-2	9,8	32
China	52,5	59	4,0	15	6,1	12
Albania	52,3	60	1,1	0	11,5	42
Maldives	51,7	61	-1,4	-3	15,4	55
Vanuatu	51,5	62	6,7	28	1,0	-5
Tonga	51,0	63	0,6	0	20,0	71
Montenegro	50,8	64	1,6	5	4,0	4
Moldova	50,6	65	0,7	0	4,4	7
Georgia	50,6	66	-2,6	-9	9,1	33
Bolivia	49,5	67	0,1	1	19,3	71
Belarus	49,3	68	-6,3	-16	-2,4	-13
El Salvador	49,3	69	2,1	13	4,7	10
Trinidad and Tobago	49,0	70	0,9	5	-1,0	-12
Algeria	49,0	71	1,3	7	5,7	18
Jordan	48,8	72	0,8	4	2,4	-2
North Macedonia	48,6	73	0,7	4	2,9	0
Jamaica	48,3	74	-1,9	-10	3,9	6
Peru	48,1	75	0,4	4	8,9	33
Kyrgyzstan	48,0	76	1,2	9	12,1	43
Botswana	47,9	77	-0,7	-5	0,8	-11
Tunisia	47,7	78	2,8	10	3,7	6
Armenia	47,7	79	-0,5	-5	14,8	49
Malaysia	47,4	80	5,6	18	-5,6	-30
India	46,8	81	7,8	27	1,3	-7
Cambodia	46,8	82	-0,8	-2	6,8	24
Cabo Verde	46,8	83	-1,8	-10	2,5	-1
Samoa	46,4	84	-1,0	-3	6,1	20
Indonesia	45,9	85	3,0	10	7,0	25
Tanzania	45,8	86	4,3	14	3,6	7
Fiji	45,7	87	-1,3	-4	-2,7	-24
Mauritius	45,4	88	-4,4	-22	-6,6	-35
Colombia	45,3	89	-1,6	-5	1,7	-2
Russia	45,0	90	4,0	13	-7,3	-38

Country	SRI 2025	Rank 2025	SRI Change 2024-2025	Rank Change 2024-2025	SRI Change 2005-2025	Rank Change 2005-2025
Chile	44,8	91	-5,9	-30	-4,6	-31
Belize	44,7	92	3,5	10	-0,2	-15
Solomon Islands	44,6	93	-4,9	-26	-4,1	-32
Argentina	44,2	94	-0,6	-5	-3,2	-29
Philippines	44,0	95	4,6	11	9,3	28
Bosnia and Herzegovina	43,3	96	1,1	1	-1,2	-15
Dominican Republic	43,2	97	-5,6	-27	17,4	57
Uganda	43,2	98	5,1	18	2,3	3
Benin	42,7	99	0,6	-3	-0,9	-13
Uzbekistan	42,5	100	-1,8	-9	11,5	35
Ecuador	42,1	101	-1,4	-7	6,0	16
Thailand	41,9	102	4,0	15	-3,1	-27
Morocco	41,5	103	6,4	26	4,4	12
Guyana	41,4	104	0,5	0	5,5	14
Mexico	41,3	105	-4,7	-19	1,4	2
Sao Tome and Principe	40,8	106	3,3	16	-2,1	-15
South Africa	40,7	107	1,7	2	-10,4	-51
Turkmenistan	40,5	108	5,5	22	18,5	55
Rwanda	40,5	109	1,5	2	9,1	23
Nicaragua	40,3	110	1,1	-3	12,2	36
Niger	40,1	111	-1,4	-12	-0,6	-8
Brazil	40,0	112	-8,7	-41	-9,8	-53
Panama	40,0	113	-1,4	-12	-1,5	-13
Sri Lanka	39,7	114	12,7	39	6,7	13
Egypt	39,5	115	1,6	3	-4,7	-32
Senegal	38,7	116	0,1	-4	0,3	-4
Cote d'Ivoire	38,6	117	1,1	3	10,8	32
Namibia	38,5	118	0,0	-4	-9,3	-54
Ghana	38,4	119	-0,6	-9	-0,6	-10
Libya	38,4	120	-1,2	-15	-6,4	-42
Iran	38,1	121	1,6	2	10,2	27
Mozambique	38,0	122	0,6	-1	-5,3	-34
Paraguay	37,8	123	-0,1	-4	7,2	14
Gabon	37,5	124	-0,8	-9	-4,4	-29
Vietnam	37,2	125	-6,4	-32	-1,2	-14
Iraq	36,9	126	0,7	-1	16,1	40
Congo, Rep.	36,2	127	0,7	0	4,9	6
Burkina Faso	35,8	128	-0,5	-4	-6,2	-34
Madagascar	34,9	129	5,8	19	6,2	14
Turkiye	34,9	130	-3,7	-17	-8,8	-45
Ethiopia	34,7	131	0,3	0	-6,9	-34
Equatorial Guinea	34,3	132	1,2	4	1,9	-1
Nepal	34,2	133	6,8	19	10,0	25
Malawi	34,0	134	-1,4	-6	-1,8	-14
Suriname	34,0	135	1,3	3	-0,2	-11

Country	SRI 2025	Rank 2025	SRI Change 2024-2025	Rank Change 2024-2025	SRI Change 2005-2025	Rank Change 2005-2025
Myanmar	33,8	136	8,2	23	14,1	31
Kenya	33,2	137	4,8	12	-1,5	-15
Yemen, Rep.	32,6	138	-1,4	-5	6,7	15
Guinea-Bissau	32,4	139	-0,6	-2	0,0	-9
Guatemala	32,4	140	2,9	6	3,1	2
Lesotho	32,4	141	-0,7	-6	-14,1	-72
Liberia	32,3	142	1,0	-3	4,2	3
Mali	31,6	143	0,8	-2	-10,1	-47
Sierra Leone	31,5	144	-1,7	-10	6,5	13
Guinea	31,3	145	1,1	-1	10,4	20
Cameroon	31,2	146	1,1	-1	-7,0	-33
Togo	30,9	147	0,1	-7	3,3	3
Bangladesh	30,8	148	6,7	14	3,7	3
Tajikistan	30,8	149	-5,2	-23	5,1	7
Honduras	30,1	150	-0,3	-7	7,9	12
Laos	29,6	151	0,2	-4	-1,2	-15
Djibouti	29,4	152	2,5	3	-4,1	-26
Eswatini	28,9	153	-1,6	-11	-18,1	-86
Gambia	28,2	154	-6,0	-22	4,8	5
Angola	27,6	155	0,9	1	8,7	13
Pakistan	27,0	156	0,0	-2	-6,5	-31
Haiti	27,0	157	9,4	12	5,5	7
Central African Republic	26,4	158	0,0	-1	-3,5	-17
Burundi	26,2	159	-1,7	-9	-8,6	-38
Zimbabwe	26,2	160	-1,5	-9	-1,7	-13
Comoros	25,6	161	0,4	0	8,3	8
Nigeria	25,2	162	-1,1	-4	-1,9	-10
Mauritania	25,2	163	2,2	0	-7,7	-34
Afghanistan	25,0	164	5,5	3	12,2	7
Zambia	24,1	165	-1,5	-5	-5,9	-25
Chad	23,7	166	1,5	-1	-2,0	-11
Papua New Guinea	22,4	167	-0,5	-3	-6,2	-23
Venezuela	18,3	168	6,3	3	-4,9	-8
Lebanon	17,7	169	0,2	-1	-12,4	-30
Congo, Dem. Rep.	16,7	170	-3,8	-4	0,9	0
South Sudan	11,8	171	-4,7	-1	n/a	n/a

Appendix 2: Methodology of the Social Resilience Index

We use 12 indicators for the SRI that are readily available for most countries:

1. Real GDP per capita growth trend: We compare the average annual growth in the last three years to the average growth prior to that since 2000. This approach reflects that the potential for social risk can also rise in high-income EMs (such as Chile or in the GCC) and AEs if the (relatively high) level of economic welfare is deteriorating or being perceived to deteriorate.

2. Labor force participation: The higher the share of the labor force in the working-age population, the lower the potential for discontent. This indicator is better than the unemployment rate, which is measured very inconsistently across countries.

3. Income inequality is measured by the GINI index.

4. Public social spending on education, health and social protection reflects the importance of social policies and networks in a given country.

5. Political stability and absence of violence captures the prevalence of political instability or politically motivated violence.

6. Government effectiveness measures the perceived quality of public services, policy implementation and the credibility of government commitments.

7. Corruption perception reflects the perceived extent of corruption in the public sector and the degree to which public power is used for private gain.

8. Trust in government indicates the share of people that trust their national government.

9. Vulnerable employment is made up of own-account workers and contributing family workers who are less likely to have social security coverage and to benefit from other forms of social protection.

10.Imports of food and fuels as % of GDP reflects a country's exposure to external price shocks in essential goods.

11. Currency depreciation captures exchange rate pressures that can amplify imported inflation, particularly for food and energy.

12. Fiscal revenue as % of GDP captures a government's capability to respond with fiscal stimulus to crises.

To make the data comparable across indicators, each of them was rescaled from 0 to 100, with 0 denoting the lowest resilience/higher risk and 100 the highest resilience/lower risk. Then the SRI was calculated as the average of the sub-indicators, thus also ranging between 0 and 100.

Note that in 2023, we renamed the indicator Social Resilience Index (from Social Risk Index) to reflect the scoring system more appropriately, but the methodology remained unchanged.

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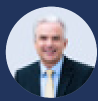

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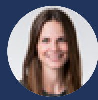

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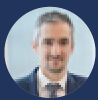

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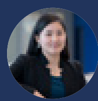

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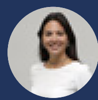

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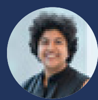

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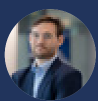

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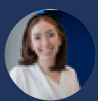

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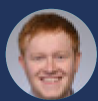

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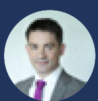

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Allianz Research | 10 March 2026 | Future of Europe series

The second energy shock: Why Europe still isn't energy secure

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In Summary

Strategic autonomy has become Europe's north star in an increasingly fragmented world, almost ten years after President Macron's first Sorbonne speech. But what will it take to achieve it? In our Future of Europe series over the coming weeks, we will examine the necessary reforms, instruments and investments — from trade policy to energy security and the role of the euro — that will enable Europe to address its critical dependencies and vulnerabilities. Today we look at what it will take for Europe to achieve energy autonomy.

- **The escalation of the conflict in the Middle East is a stark reminder of Europe's unfinished business: achieving energy autonomy. Although Europe's reliance on Russian gas has improved since 2022, it is merely replacing one dependency with another.** Recent events have destabilized global energy markets and highlighted the need for greater energy resilience. Around 20 million barrels of oil per day – about a quarter of global seaborne trade – pass through the Strait of Hormuz, making it a key supply chokepoint. Since hostilities began, oil prices have risen, peaking near 120 USD/bbl. Although the Middle East supplies only about 4% of Europe's gas, European benchmark prices have nearly doubled as traders fear Asian buyers could compete for supplies. Europe's industry is most vulnerable to gas disruptions (39% of final energy), transport to oil shocks (with road transport consuming 73% of EU oil), while gas often sets power prices, spreading risks across the economy. The situation recalls the 2022 Ukraine energy crisis and highlights the need to strengthen energy resilience and competitiveness. Russian pipeline gas to the EU fell from 150 bcm in 2021 to 38 bcm in 2025, largely replaced by US LNG (45%), increasing exposure to global market cycles and shipping limits. The EU now operates 33 LNG terminals (215 bcm capacity), with 22 bcm under construction and 78 bcm planned, but without faster electrification and grid expansion it risks replacing pipeline dependence with costly LNG reliance.
- **First, Europe must reactivate its energy emergency toolkit.** The 2022 energy crisis showed that while price volatility cannot be avoided, its economic impact can be limited through coordinated policy responses combining demand reduction, supply stabilization and targeted fiscal support. Policymakers should therefore replay the crisis playbook: demand reduction, encourage efficiency and compensate industry for temporary curtailment. Power generation can also shift away from gas by maximizing nuclear output and temporarily using other available capacity. On the supply side, rebuilding storage and securing LNG imports are essential ahead of winter. Finally, emergency market tools and targeted financial support can stabilize markets while protecting vulnerable households and energy-intensive industries.
- **Secondly, this time around, Europe's energy autonomy agenda should also be a strategic transition program towards more resilience.** The near-term gas pivot worked because governments treated energy security as wartime logistics. But resilience now requires fixing structural weaknesses in market design and delivering a best-in-class power grid. Nuclear power can strengthen Europe's energy resilience by providing reliable, low-carbon baseload electricity that stabilizes power prices, and ensures system reliability, however it is costly and comes with other challenges (e.g. uranium sourcing, waste management etc.). Europe should address the primary bottleneck to the

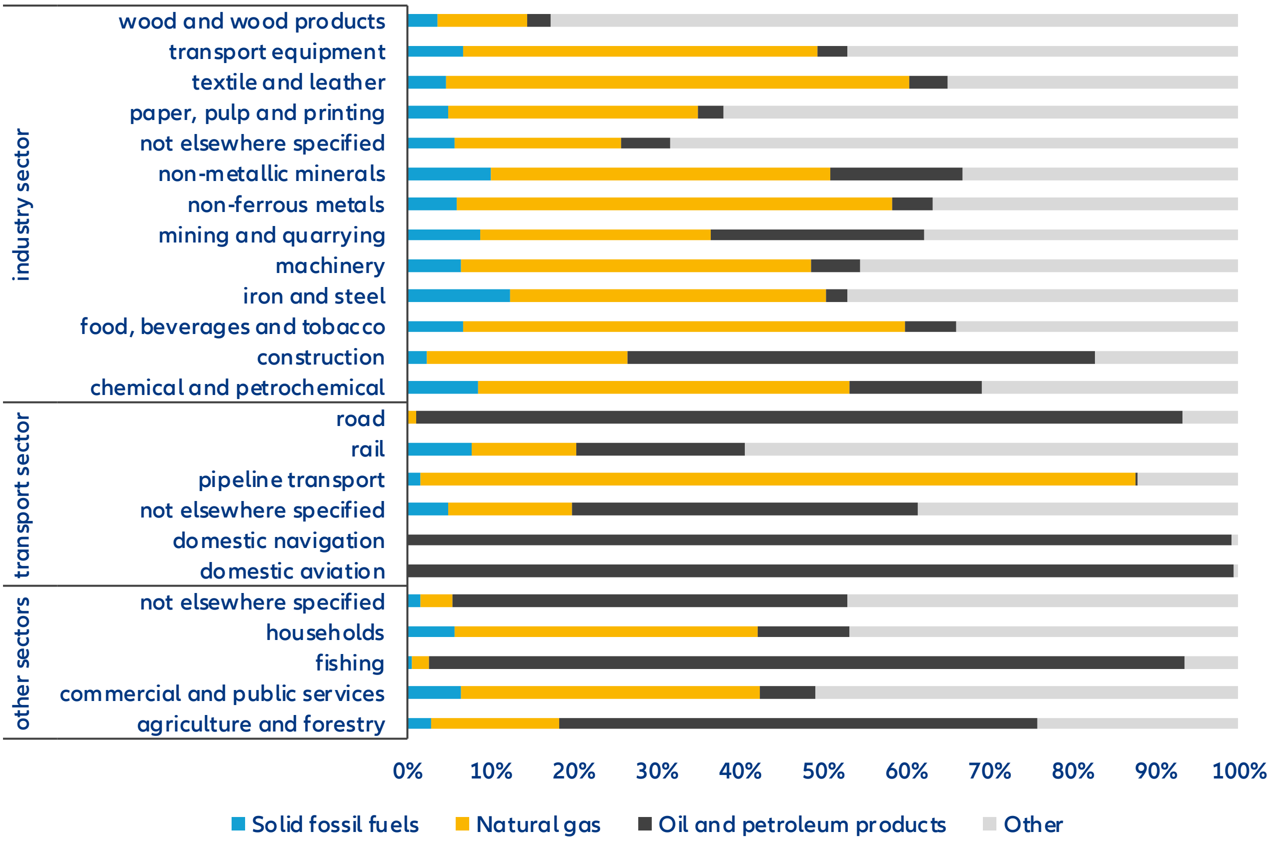
renewable transition : the inflexibility of the power grid, by building and digitalizing networks, expanding cross-border interconnection and making demand price-responsive at scale in order to enable consumers to shift electricity use when power is cheapest. With congestion costs projected to reach EUR12.3bn by 2030 and EUR56.7bn by 2040 without upgrades, Europe needs to act to avoid price shocks that could reach +22% by 2030 and up to +103% by 2040. To solve this and improve its energy system, we estimate Europe will need to invest around EUR2.27trn in the grid by 2050 (i.e. about EUR 91bn per year), plus EUR101bn per year in wind and solar through 2030 to substantially improve networks, interconnectors, and enable consumers to shift electricity use when power is cheapest.

- **The EU ETS is a central tool for reducing Europe's fossil fuel dependence, having already cut power sector emissions by 54% and fossil-based generation by 47% since 2005, with coal imports down 58%.** Yet gaps remain: transport and buildings are largely outside the system, with ETSII only launching in 2028, and over 90% of industrial emissions covered by free allowances. The ongoing ETS review is a critical opportunity to reinforce rather than dilute the instrument, and should prioritize sector expansion, phase-out of free allowances, Carbon Contracts for Difference, better revenue recycling, and a broader Carbon Border Adjustment Mechanism to advance energy autonomy while preserving competitiveness.
- **2040 should be the target for a robust EU strategic autonomy on energy.** Europe will achieve energy autonomy once it can heat homes and power industrial production primarily with renewables, using electricity where possible, green hydrogen where needed and bridging remaining gaps with a diversified portfolio of independent and long-term stable energy sources. A realistic timeline in three stages would be: 2026-2027 delivers "Russian independence on paper", 2028-2035 is the window to reach "operational autonomy" by structurally reducing gas demand and relying less on gas plants during periods of high electricity demand, instead using system flexibility such as storage, demand response and stronger grid links, and 2035-2040 is the more conservative horizon for "robust autonomy" if grid projects and flexibility measures expand more slowly than expected.

The war in Iran is a timely reminder of Europe's energy weaknesses. The ongoing conflict involving Iran is already reverberating across global energy markets, highlighting once again the structural vulnerability of energy systems to geopolitical shocks. Sharp price volatility and renewed concerns over supply security are back, particularly because of the region's central role in global oil and gas markets. This reinforces a key policy target for Europe: energy resilience. Although Europe does not get much oil or gas through the Strait of Hormuz. It is a key route for Qatar's LNG, which might lead its usual Asian buyers to compete with Europe for LNG from other suppliers. Indeed, disruptions affecting Gulf export infrastructure could temporarily remove close to 20% of global LNG supply. Such shocks would translate quickly into electricity price spikes, inflationary pressure and industrial competitiveness risks for the region. The current situation echoes the 2022 energy crisis triggered by Russia's invasion of Ukraine, which exposed Europe's dependence and its vulnerability to gas price volatility. Gas prices averaged over 130 EUR/MWh in 2022, more than seven times the historical average. Europe ultimately stabilized its energy system through rapid LNG diversification, demand reductions exceeding 10% and aggressive storage policies that pushed gas inventories to around 95% capacity ahead of winter. Europe should build from that episode to prepare not only for the potential Iranian shock but more importantly for futures ones, in a world where geopolitical risk remains structurally high. Energy resilience will increasingly define economic security and stability.

Europe's final energy-consuming sectors exhibit markedly different exposure to fossil fuel shocks. Industry relies heavily on natural gas, which accounts for 39% of its final energy consumption, leaving it particularly sensitive to gas supply disruptions (see Figure 1). Transport is largely dependent on oil, with road transport alone consuming 73% of EU oil in 2024, making it the most vulnerable sector to oil price spikes. The power sector's reliance on gas as the primary fossil fuel and marginal price-setter spreads upstream supply risks across the entire economy, amplifying the impact of any gas shock well beyond the energy sector itself. Reducing sectoral vulnerability therefore requires a combination of structural demand reduction, fuel switching toward electricity, and decarbonization of power supply. Electrification of transport through electric vehicles is central to lowering exposure to global oil markets, while broader electrification and grid expansion can mitigate the systemic risks that stem from gas dependency in power generation.

Figure 1: Final energy consumption shares by sub-sector and energy source in 2024 (in %)

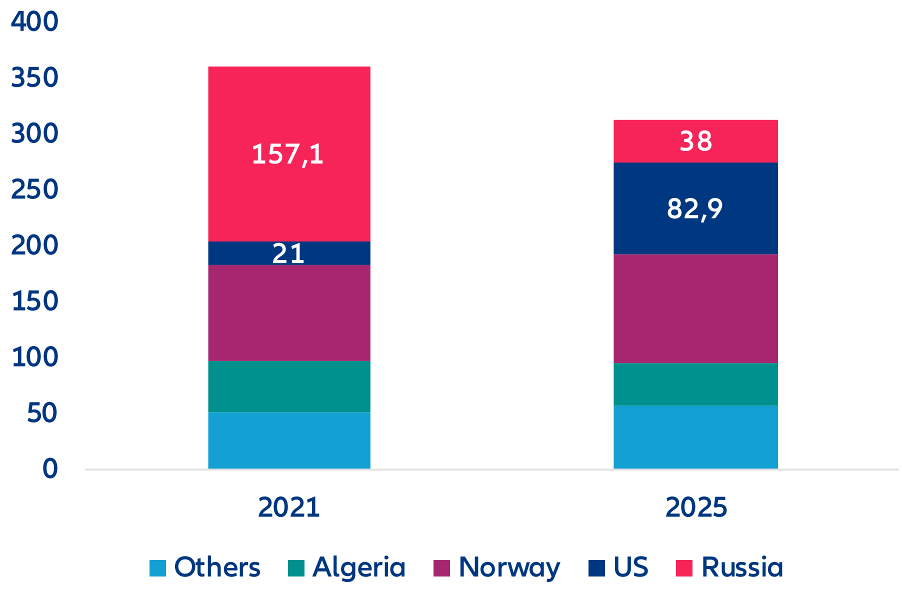


Sources : Allianz Research based on Eurostat; Note: Final energy consumption for electricity & heat was attributed to primary energy sources using gross generation shares.

From single-supplier pipeline trap to single-supplier LNG habit? Russian pipeline gas deliveries to the EU collapsed from 150bn cubic meters (bcm) in 2021 to 38 bcm in 2025. LNG has been the main substitute, and the US the marginal supplier, providing almost 45% of EU LNG in 2024 and roughly 60% in 2025. But this leaves Europe exposed to US commercial cycles, shipping constraints and Washington's shifting political mood (see Figure 2). Resilience will require treating gas as a strategic balancing commodity, not a comfort blanket, by keeping demand structurally lower than the 2021 level (EU consumption in 2024 was down -19% compared to 2021), ringfencing storage and system services, diversifying the "energy portfolio" across Norway, North Africa and domestic biomethane, as well as accelerating the renewables energy transition, rather than locking in a second era of long-term overconcentration.

Europe can only be independent if it lowers the strategic value of imported hydrocarbons in its industrial model. Europe has built enormous LNG optionality: 33 large EU terminals in operation with 215 bcm of annual regasification capacity, plus 22 bcm under construction and 78 bcm planned. Yet these solutions do not remove geopolitical risk, they re-route it. The security challenge is to shift from "supply substitution" to "system substitution": electrify heat and low-temperature industry where it is cost-effective, accelerate efficiency that permanently reduces peak demand and ensure firm low-carbon capacity and flexibility

Figure 2: EU gas imports (bcm)



Sources : Eurostat, Allianz Research

The experience of 2022 showed that while price volatility cannot be eliminated, its economic impact can be contained through a combination of demand reduction, supply stabilization and targeted fiscal support. Europe could be again confronting an energy shock that echoes the crisis of 2022. However, conditions are slightly different: this time the continent is approaching spring rather than entering winter, which means heating demand will soon fall. Yet this seasonal relief is deceptive. Gas storage levels are currently below historical averages and the refilling season begins precisely when global energy markets are turning more volatile. The structural vulnerability also remains intact and policymakers should turn to a familiar crisis playbook:

- **Demand management remains the fastest and most cost-effective lever.** Because gas-fired plants frequently set the marginal electricity price in Europe, reducing power consumption directly lowers gas demand and wholesale electricity prices. Governments should therefore encourage coordinated demand-reduction efforts across households, services and industry. Public campaigns and regulatory guidance can promote lower heating levels and reduced non-essential consumption. Dynamic pricing and demand-response schemes can shift electricity use away from peak periods, while voluntary curtailment programs can compensate large industrial users for temporarily reducing demand. Careful coordination of gas storage withdrawals can also smooth market tensions and limit sudden surges in LNG demand.
- **Reducing gas use in power generation provides another immediate buffer**. Since gas-fired plants often determine electricity prices, even modest reductions in gas-fired output can stabilize markets. Utilities can temporarily increase generation from other available sources. Where capacity remains available, coal plants can be brought back online—as Germany did during the 2022 crisis—while maximizing nuclear output in France could significantly reduce reliance on gas across European power markets. Although these measures carry environmental trade-offs, they can provide a rapid safeguard against supply shortages and price spikes.
- **On the supply side, rebuilding gas storage and securing LNG cargoes remain Europe's primary defenses ahead of next winter.** Competition for shipments could intensify if global markets tighten, particularly if Asian demand strengthens. Maintaining high storage levels is therefore essential to cushion supply disruptions and dampen price volatility.
- **Governments also retain/reactivate emergency market tools.** Strategic oil reserves can be released during periods of acute stress to stabilize global oil markets and indirectly ease energy prices. At the European level, coordination remains crucial. During the 2022 crisis the EU introduced a "market correction mechanism" allowing authorities to cap gas futures prices on the Dutch TTF hub if they exceeded 180 EUR/MWh and diverged significantly from global LNG benchmarks. Although never triggered, the

mechanism helped calm markets by signaling a credible willingness to intervene. Joint gas purchasing, coordinated storage refilling and flexible state-aid rules can similarly strengthen Europe's collective response.

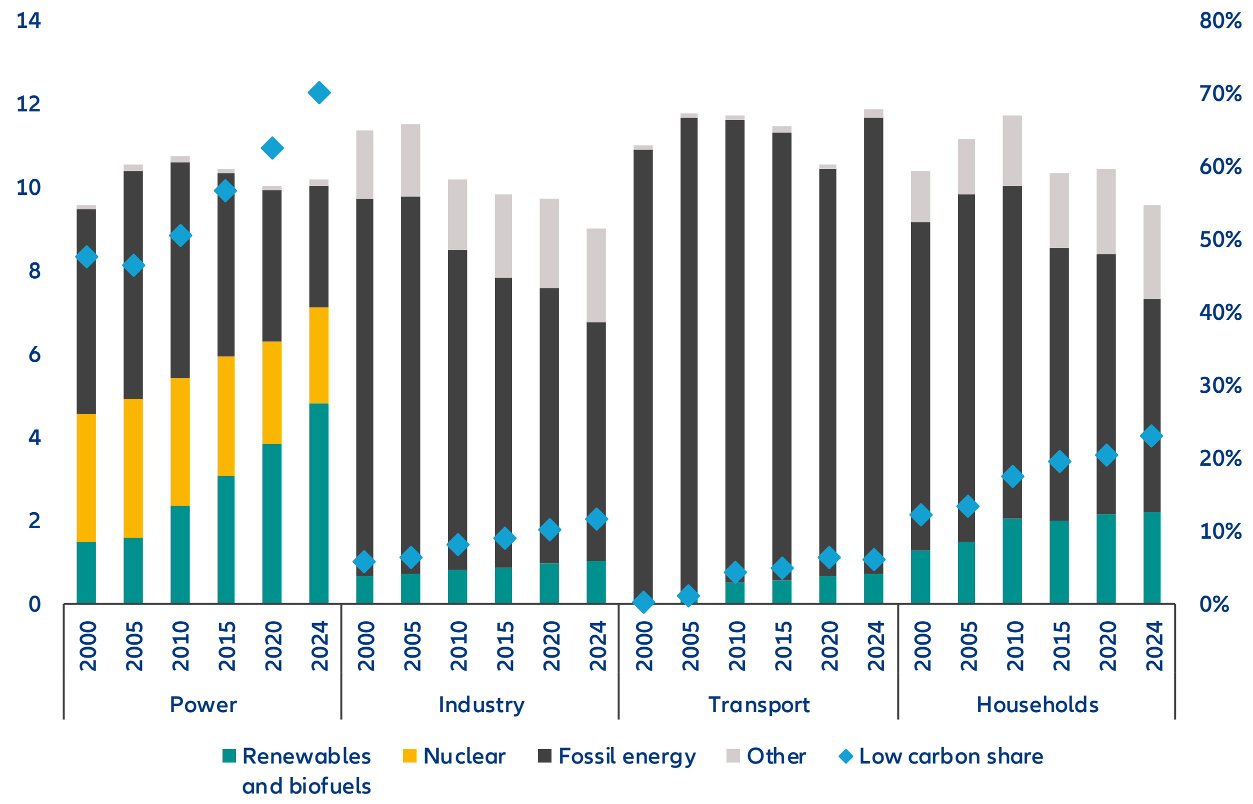
- **Finally, targeted financial support can shield households and firms without blunting incentives to conserve energy.** Lifeline prices (subsidizing a basic level of consumption while charging market prices for additional use) can protect essential energy use while preserving price signals. Income-tested transfers, energy vouchers and temporary protections against disconnection can provide additional relief for vulnerable households. For energy-intensive industries exposed to international competition, governments may need to offer temporary credit lines, guarantees or limited state aid to ease liquidity pressures. Windfall revenue caps on low-cost electricity producers can help finance such measures, recycling extraordinary profits to support households and businesses while limiting additional strain on public finances.

Europe's energy resilience agenda should therefore be framed as a strategic transition program, not another emergency procurement scramble. The near-term gas pivot worked because governments treated energy security as wartime logistics. The next phase requires treating it as market design: faster permitting, stable network returns and EU-level coordination that makes cross-border infrastructure and flexibility remunerative. Underinvestment in the energy transition is the expensive option, because indirect GDP losses dwarf the direct congestion bill – this is a fiscal argument as much as a climate one. On geopolitics, the strategic discomfort of US "energy dominance" is significant even if LNG is less easily weaponized than pipelines and the US is still a European ally. However, Europe should aim at reducing US leverage by ensuring it can run its economy with materially less gas in a cold winter, a dry hydro year or a shipping disruption. Practically, that means a disciplined sequence: lock in the electricity backbone (grids, interconnection, digital control), monetize flexibility (dynamic rates, smart meters, demand response) and then let clean generation and electrification do the strategic work of shrinking import dependence.

Nuclear energy is a short term fix that offers firm, low-carbon baseload capacity that could improves Europe's grid stability, seasonal reliability and fuel security. The EU now has about 98 GW of nuclear installed, and the Commission estimates EUR 241 bn investment needed by 2050 for new builds and life extensions.. US (about 100 GW), China (60 GW), Russia (27 GW) and India (9 GW) are the other key players in nuclear energy. China is the country that is building most of the world's new capacity. Nuclear operates nearly continuously, unaffected by weather, so it supplies reliable winter power and cuts the hours when gas turbines must run. Wind and solar are intermittent and need backup (storage or gas) during lulls. Nuclear's near-zero fuel cost means it typically displaces gas plants in the market order, smoothing prices and cutting volatility. European countries with large nuclear shares (e.g. France, Finland) see far fewer price spikes than others in the region. Demand response and batteries can shift load peaks, but cannot yet substitute tens of GW of capacity. Nevertheless, nuclear deployment faces long lead times (10–15 years) and high costs (capex at EUR 2 000 to 6 500 per kW). Complex permitting, public opposition and waste disposal are also key hurdles. Small Modular Reactors (SMRs) promise repeatable, factory-built construction and can supply industrial heat or hydrogen, but none are built yet – first EU SMRs are unlikely before the early 2030s without major policy support. Extending lifetimes of existing reactors could significantly raise EU nuclear capacity to about 144 GW by 2050 in a favorable scenario. Uranium fuel supply is also another risk factor: the EU imports ore (mainly from Kazakhstan and Canada) and has domestic conversion/enrichment. EU policy should therefore treat nuclear as a strategic resilience asset: accelerate licensing, ensure stable financing, coordinate cross-border grids and fuel planning, support joint SMR development and maximize life-extension of existing plants.

Strengthening and expanding the EU ETS as Europe's primary decarbonization instrument will be central to bolstering the region's energy independence. Since its introduction in 2005, the system has delivered meaningful results in the power sector, reducing emissions by -54% and cutting power generation from fossil sources by -47% (Figure 3). These reductions have significantly reduced Europe's coal dependence, with coal imports falling by -58% between 2005 and 2024. The same shift has not occurred in sectors outside the ETS scope, such as transport and buildings, nor in the industrial sector, where over 90% of emissions remain covered by free allowances. Therefore, with the ongoing ETS review presenting a critical opportunity, reinforcing rather than diluting the instrument is essential to any credible energy autonomy strategy. This requires both a timely expansion to currently excluded sectors and a continued phase-out of free allowances, as well as targeted reforms such as Carbon Contracts for Difference, better revenue recycling, and a broader Carbon Border Adjustment Mechanism to ensure competitiveness is maintained throughout the transition.

Figure 3: Final energy consumption and gross electricity production (EJ, lhs) and low carbon share (in %, rhs)

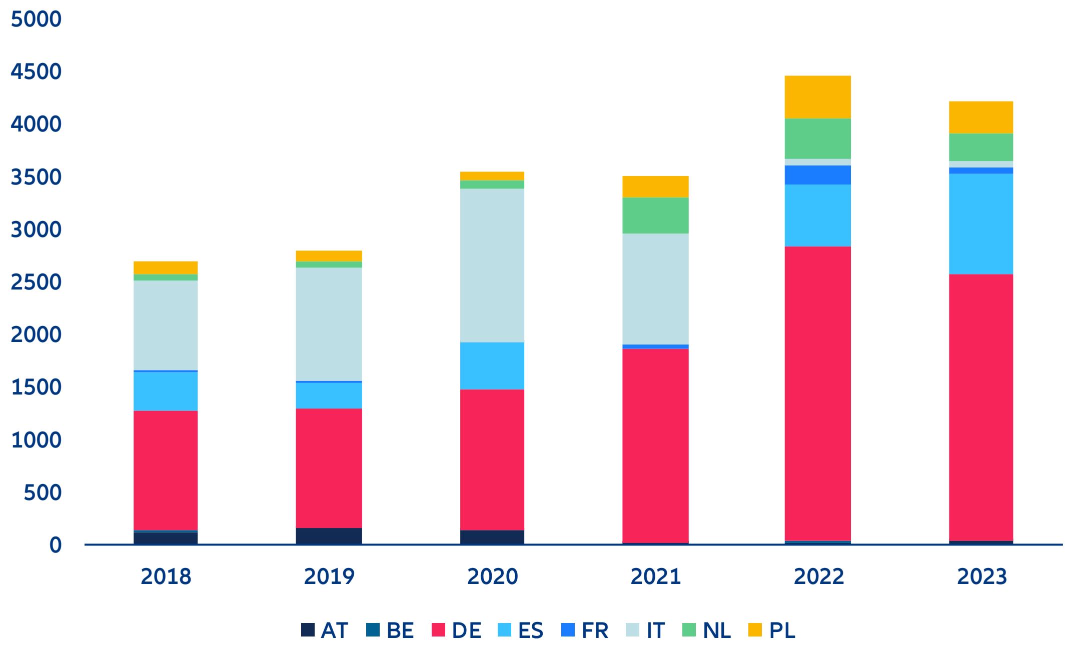


Sources : Eurostat, Allianz Research

The primary bottleneck to a rapid, full renewable transition in Europe is the inflexibility of the power grid. Congestion and limited flexibility amplify intraday volatility (i.e. high prices at peaks, negative prices off-peak) while shifting costs from the network into the bill (see Figure 4). In Germany alone, compensation for renewables reached EUR21bn in 2024. Congestion costs are projected to reach EUR12.3bn by 2030 and EUR56.7bn by 2040 without upgrades, translating into implied price shocks of +22% by 2030 and up to +103% by 2040 in a business-as-usual pathway1. The macroeconomic stakes are also significant: We estimate that cumulative GDP losses from electricity congestion could reach EUR4,700bn across the EU by 2050 under a business-as-usual pathway, even before counting wider strategic damage from de-industrialization and investment flight. The remedy is to build and digitalize networks, expand cross-border interconnection and make demand price-responsive at scale so that renewables displace imported gas in practice, not just on annual averages. On investment, we put the order of magnitude at about EUR2,270bn of grid infrastructure by 2050 (about EUR390bn over 2025–2030 and EUR1,880bn over 2030–2050), averaging EUR91bn per year, with distribution networks absorbing 56% of the total. The region also needs to increase wind and solar generation capacity through EUR101bn of investment per year to 2030. This would build an energy-security asset that reduces reliance on LNG during stress events, when cargoes and politics tighten simultaneously.

[1] *See our previous report : Plug, baby, plug: Unlocking Europe's electricity market*

Figure 4: Congestion management costs in Europe (EUR mn)



Sources: ACER monitoring reports, Allianz Research

2040 should be the target for a robust EU strategic autonomy on energy. Europe will achieve energy autonomy once it can heat homes and power industrial production primarily with renewables, using electricity where possible, green hydrogen where needed and bridging remaining gaps with a diversified portfolio of independent and long-term stable energy sources. On the gas side, the EU's own legislative trajectory implies the formal endpoint is close: The Council and Parliament struck a deal in December 2025 to phase out Russian gas imports, with a ban on Russian LNG by end-2026 and Russian pipeline gas by autumn 2027. In practice, however, the strategic endpoint is later, because it requires enough electricity-system flexibility to keep gas from setting marginal prices and emergency procurement. A realistic timeline in three stages would be: 2026–2027 delivers "Russian independence on paper", 2028–2035 is the window to reach "operational autonomy" by cutting gas demand structurally and replacing gas peaking with flexibility and 2035–2040 is the more conservative horizon for "robust autonomy" if permitting, interconnection and demand response scale only gradually.

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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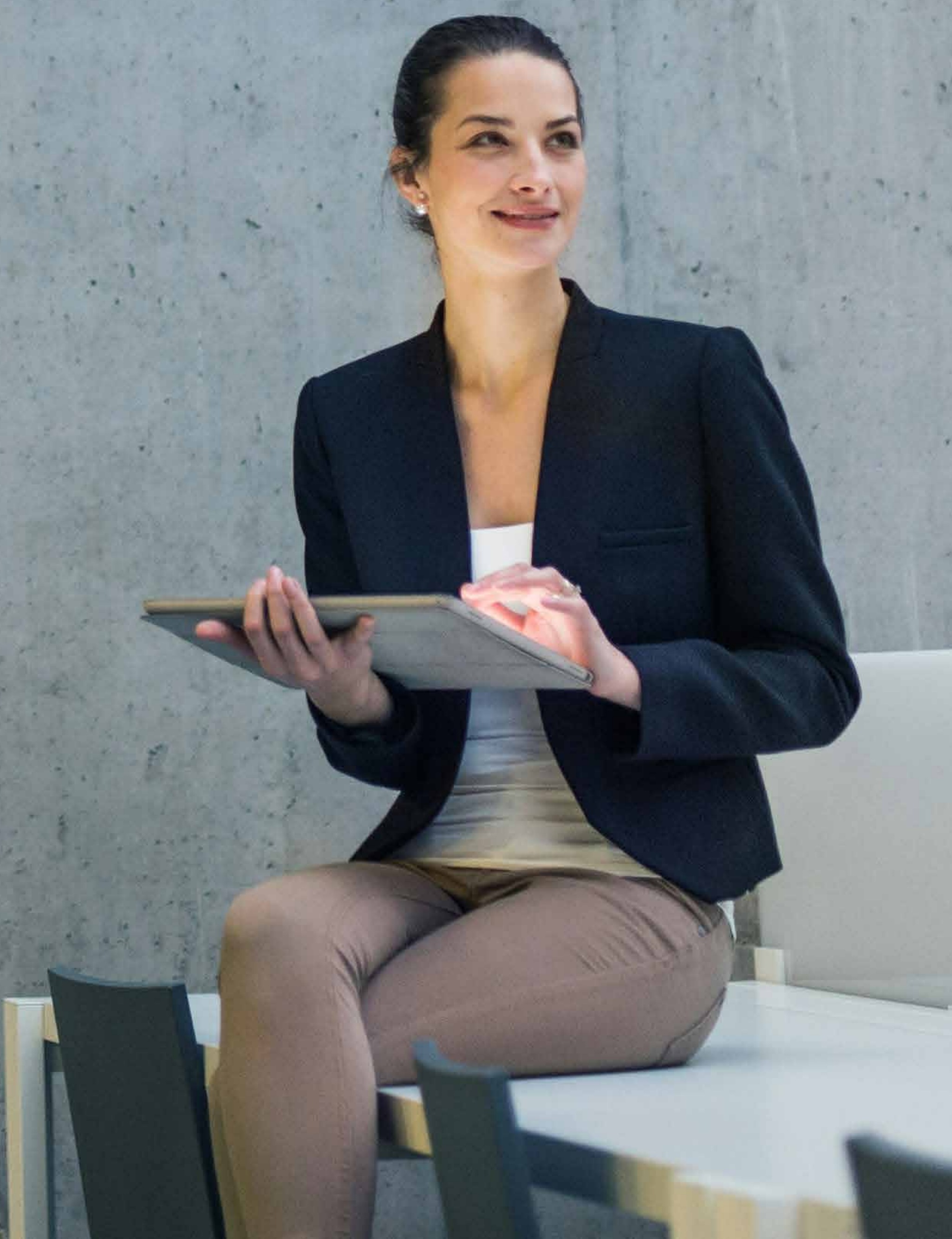
Closing the gender income gap: from paycheck to pension

05 March 2026



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Content

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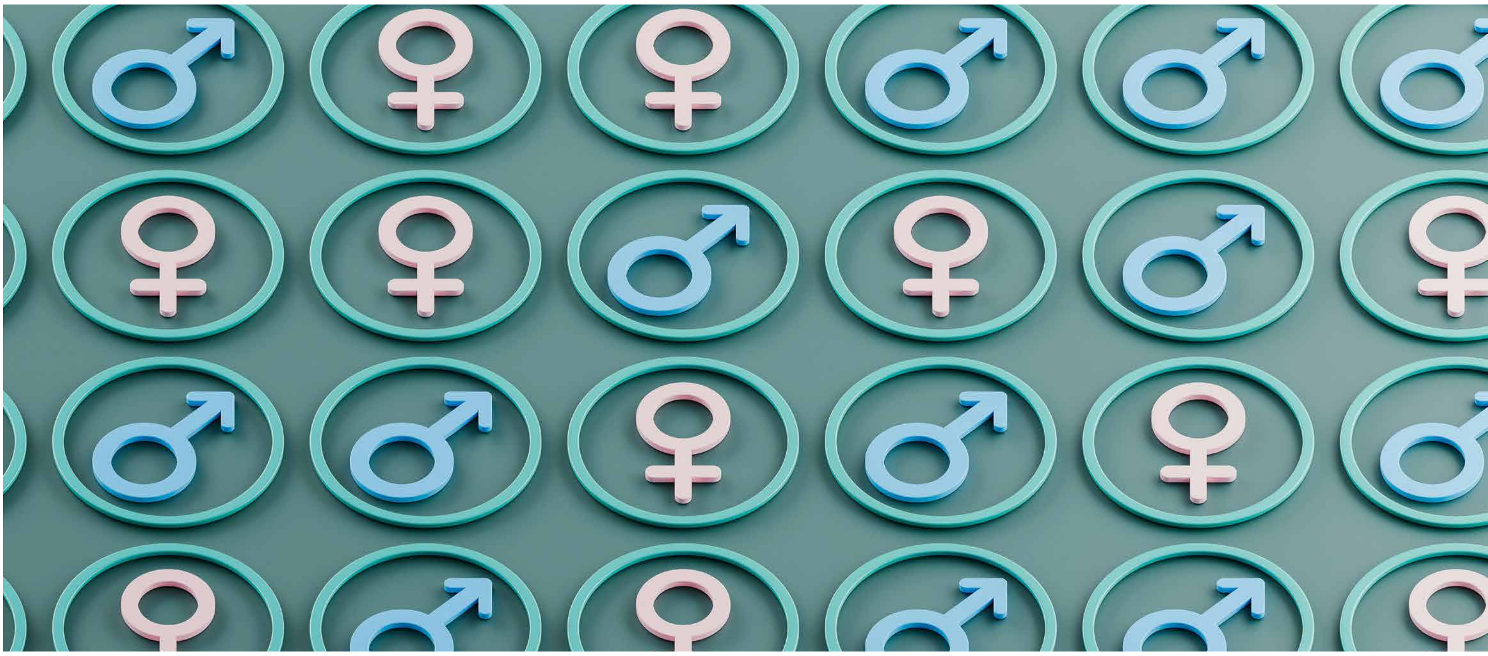
- **Women have made measurable progress over recent decades in narrowing gender pay gaps and increasing labor-force participation – but structural gaps persist.** Across the OECD, the unadjusted gender pay gap – the percentage difference between the average earnings of all men and all women without accounting for differences in job type, hours worked, experience or seniority – has declined from 21% in the early 2000s to 13.7% in 2024, and female labor-force participation has risen steadily to 71%, compared with 81% of men. However, single metrics do not capture the full economic impact of these gender disparities as important structural differences remain: women continue to work fewer paid hours, experience more career interruptions and receive on average 23.7% lower pension income than men.

- **It's about more than equal pay: What ultimately matters is total income over the lifecycle.** Lower earnings during working years reduce savings capacity, investment returns and pension entitlements. To measure this cumulative effect, we develop an integrated lifecycle model that combines labor income, capital income and pensions into a unified lifetime income measure, tracing women and men born in 1975, 2000 and 2025 across major OECD countries. Our results show that lifetime income gaps across the 14 analyzed countries decline markedly across cohorts: from roughly 33% for those born in 1975 to 16% for the 2000 cohort. This lifetime income gender gap is driven primarily by labor income (79%), followed by pensions (16%) and capital income (5%). Assuming a continuation of current structural trends, progress looks set to stall, with the lifetime income gap still at 16% for those born in 2025.

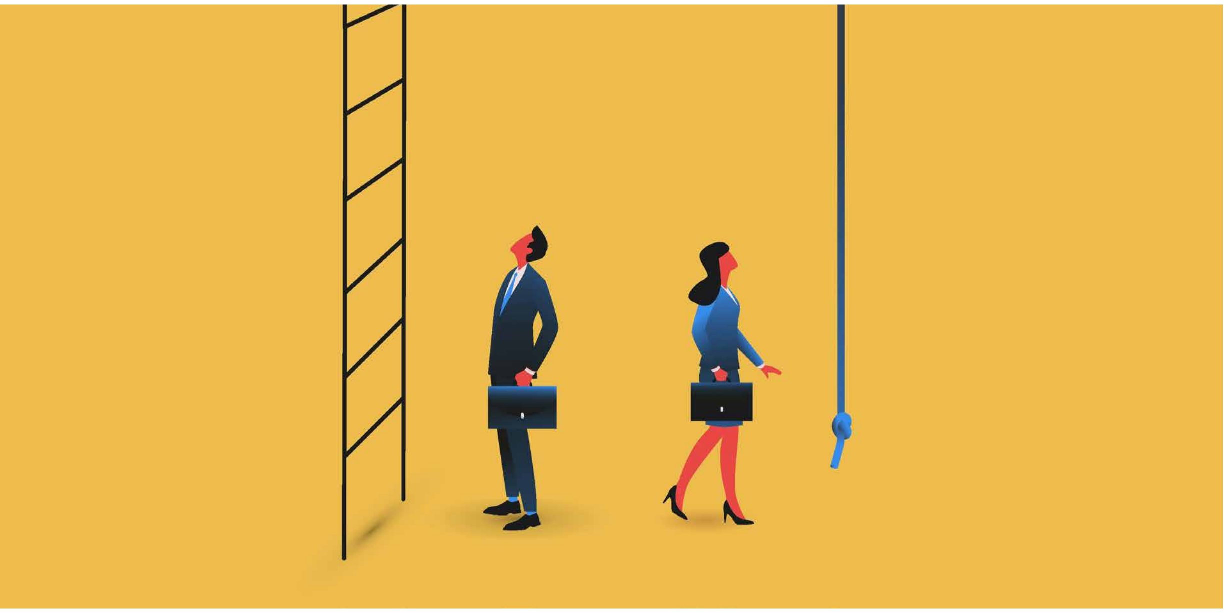
- **Country level results show sharply diverging trajectories:** Sweden is projected to reach near parity (-2%), followed by the US and Spain (around 7%), while France, Czech Republic and Belgium approach low double-digit levels. By contrast, Italy, Germany and Switzerland continue to display sizeable gaps above 20% over the lifecycle of women and men born in 2025, indicating the need for continued reform efforts to create equal conditions for men and women to foster the convergence of lifetime incomes.

- **Under current structural trends, progress slows – and the key driver is hours worked, not hourly wages.** Moving the focus from the generational view to annual labor income dynamics clearly shows this diverging dynamic: A leading group – Sweden, the US, Spain, France, Czech Republic, Belgium, Poland and the Netherlands – is projected to see annual labor income gaps fall below 10% by the end of the century. In lagging countries – Germany, Italy, the UK, Denmark, Austria, and Switzerland – sizeable gaps persist as structural differences in part-time employment outweigh gains in participation and hourly pay. The future of gender income convergence will thus be determined less by wage equality per hour and more by the distribution of working hours across the lifecycle.

- **Closing gender income gaps faster requires structural change.** To close the labor income gap, public policy must take the lead. Expanding affordable childcare, reducing disincentives for second earners and supporting continuous full-time employment would address disparities at their source – and represent sound fiscal policies that can pay for themselves over time.[1] At the same time, women need to close emerging gender gaps in AI adoption - currently at 16%[2] across the EU - to ensure full participation in future productivity gains and technological transformation. With regard to the pension gender gap, public pension design can help cushion the cumulative effects of career interruptions and lower lifetime earnings. However, in times of increasingly strained public finances, private pension provision will become more important. Women should therefore focus on strengthening financial literacy, which can increase annual investment returns by up to 1.5pps[3], while also starting to save and invest as early as possible to fully benefit from the power of compounding.



[1] Source: Hendren, N. and B. Sprung-Keyser (2020), "A Unified Welfare Analysis of Government Policies", The Quarterly Journal of Economics, 135(3), 1209-1318, https://doi.org/10.1093/qje/qjaa006.
[2] Source: Eurostat (2025), Individuals - Use of Generative AI Tools, https://doi.org/10.2908/ISOC_AI_IAIU.
[3] Source: Allianz Research (2023), "Playing the Squared Ball: the Financial Literacy Gender Gap", https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/financial-literacy.html



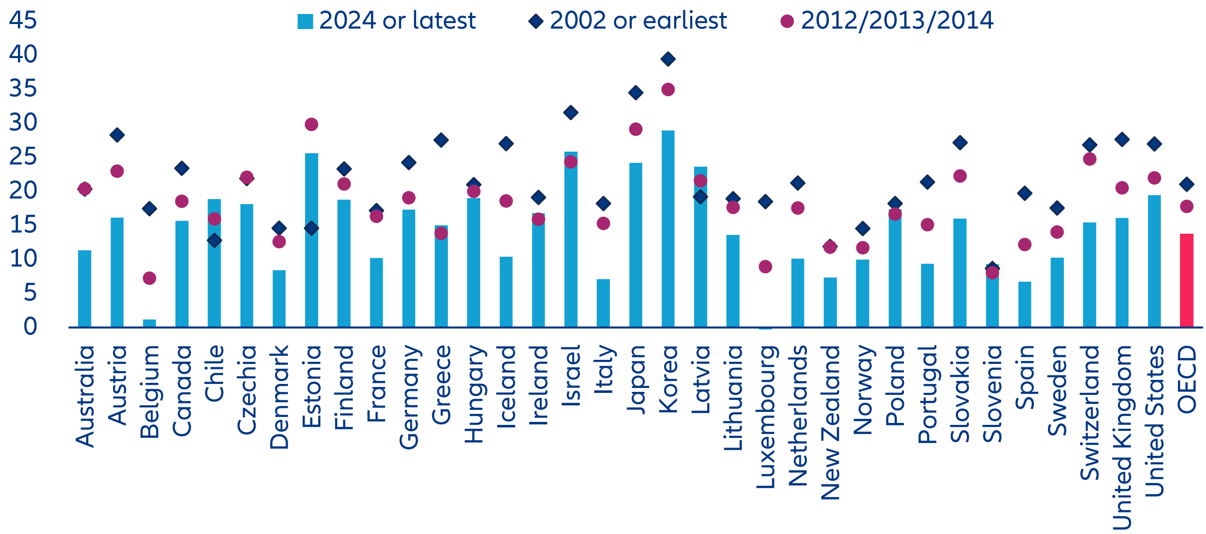
The gender pay gap has declined but hasn't disappeared

Over the past quarter century, gender pay equality has improved significantly across advanced economies. The go-to benchmark for policymakers and the public debate is the unadjusted gender pay gap, which measures the percentage difference between average gross hourly earnings of women and men working full-time relative to men's earnings. Across OECD countries, this gap declined from 21% in the early 2000s to 13.7% in 2024 (Figure 1). In the EU, it fell from 20.6% to 13.0%.[4] Despite this progress, the gap remains substantial.

The aggregate decline masks significant cross-country divergence. In France, Italy, Spain and Denmark, the unadjusted gender pay gap has declined to 10% or below, with Belgium nearing parity. Progress has been slower elsewhere. In Germany, the gap has decreased from 24.2% to 16% and in the US from 27.0% to 19.4%. In Hungary, reductions have been modest and in Chile the gap has widened.

[4] Source: OECD (2025a), "Gender Equality in a Changing World: Taking Stock and Moving Forward, Gender Equality at Work Series", OECD Publishing, Paris. https://doi.org/10.1787/e808086f-en.

Figure 1: Global evolution of unadjusted gender pay gaps, in %



Sources: OECD, Allianz Research.

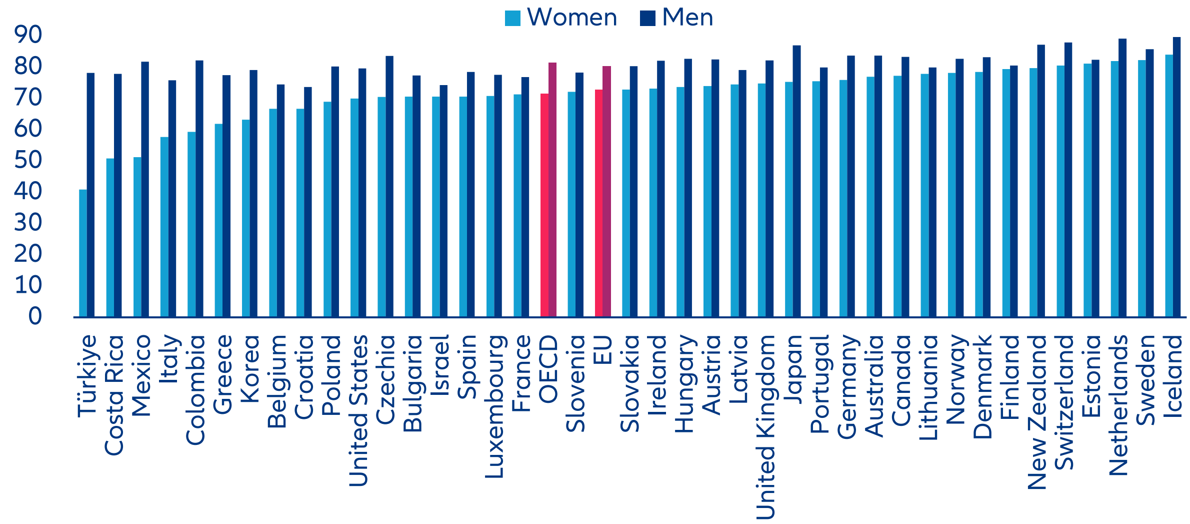
While widely used thanks to its simplicity and comparability, the unadjusted gender pay gap captures only part of the picture. It compares all working women and men without adjusting for differences in occupation, sector, education, seniority, number of working hours or career interruptions. Its key limitation is that it reflects how labor markets are structured rather than how pay is set within identical jobs. As a result, it captures the effects of part-time work, occupational segregation and motherhood-related career breaks as much as pay differences within comparable positions. To isolate within-job wage differences, economists estimate the adjusted gender pay gap, controlling for role, qualifications and experience. This typically reduces the headline gap by 50% or more, but does not eliminate it. In Germany, for example, the unadjusted gap of 16% narrows to around 6% once comparable roles are considered, but it is still present. Hourly pay says little about how much women work, how continuous their careers are or how earnings accumulate over time. From a lifecycle perspective, overall earnings – combining wages, working hours and employment rates – provide a more comprehensive view.

The gender work gap has narrowed but differences in work patterns persist

Women have made substantial gains in labor market participation. Across the OECD, the remaining gender labor force participation gap stood at 10pps in 2023 (71% for women versus 81% for men). In the EU, the gap has narrowed further to 7pps (73% versus 80%).[5] What once represented a structural divide has become a much smaller gap. Convergence is visible in many large economies (Figure 2). Germany now records a female participation rate of 76.5% compared to 83.9% of men. France stands at 71% for women, just six points below men. In the US, 70% of women participate in the labor force compared to 80% of men. A generation ago, these figures were significantly lower. But progress is uneven. Italy's female participation rate remains at 58%, nearly 20pps below men. In Türkiye, the gap exceeds 35 points. The overall picture is therefore one of strong structural progress, combined with persistent national divergence.

[5] Source: OECD (2025a), "Gender Equality in a Changing World: Taking Stock and Moving Forward, Gender Equality at Work Series", OECD Publishing, Paris. https://doi.org/10.1787/e808086f-en.

Figure 2: Global labor force participation rates by gender in 2023, ages 15-64, in %

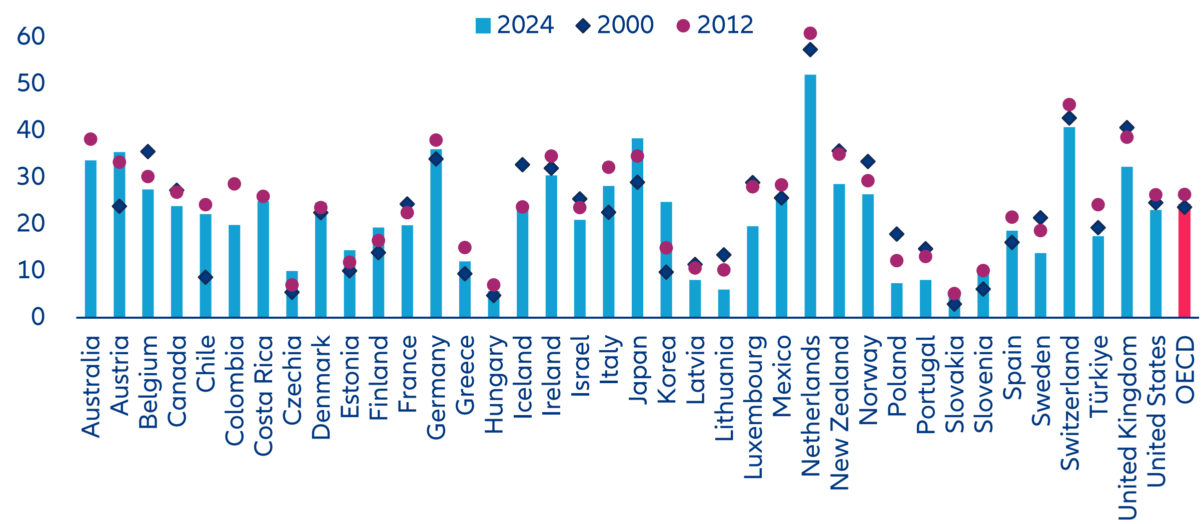


Sources: OECD, Allianz Research.

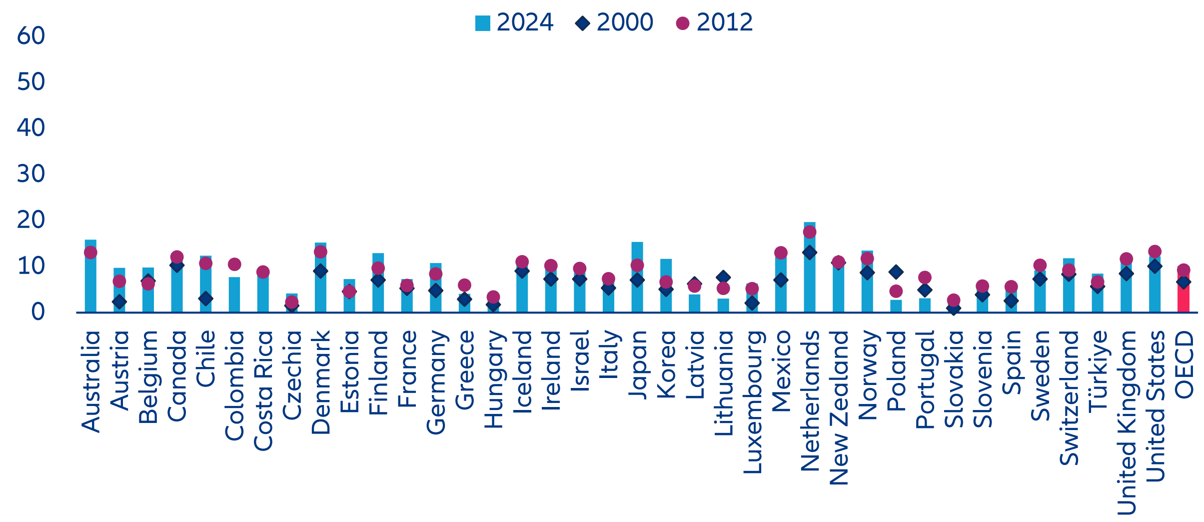
As employment gaps have narrowed, a different divide has moved to the forefront: not whether women work, but how much they work. While the participation gap has narrowed substantially across most OECD countries, this convergence has not translated into equal working patterns. Women remain disproportionately represented in part-time employment. Across the OECD, 24% of employed women work part-time, compared with 10% of men. In the EU, the figures are similar (23% versus 8%). The gender divide has thus shifted: women are more present in the labor market, but often on different terms. In some countries, part-time work defines female employment patterns. In the Netherlands, more than half of working women (52%) are employed part-time, compared to 20% of men. In Germany, 36% of women work part-time versus 11% of men. In the UK, nearly one-third of women (32%) work part-time, compared to 12% of men. Male part-time rates have increased over the past 25 years as well, but from a much lower base. In Germany, the male rate rose from 5% in 2000 to 11% in 2024. Across the OECD, male part-time employment moved from roughly 7% to 10%. This reflects gradual change in work culture, but the structural gap remains wide.

Figure 3a and 3b: Global evolution of part-time employment shares of women (Panel A) and men (Panel B) in 2024, in %

Panel A: Women.



Panel B: Men.



Sources: OECD, Allianz Research.

The reasons behind part-time employment also reveal structural asymmetries. Evidence from Germany illustrates patterns that are common across many OECD countries. Among those working part-time, 29% of women cite childcare or caregiving as the main reason for reduced hours, compared with only about 7% of men.[6] By contrast, men more often report education or training as the reason for part-time work (22% vs. 8%). The parental split is particularly striking. Among employed parents, 69% of mothers work part-time, compared with just 8% of fathers.[7] These figures suggest that part-time work is not simply a lifestyle preference but closely linked to caregiving structures, household specialization, and gender norms. Similar patterns are documented across OECD countries, where mothers' labor supply adjusts far more strongly to family responsibilities than fathers'.[8]

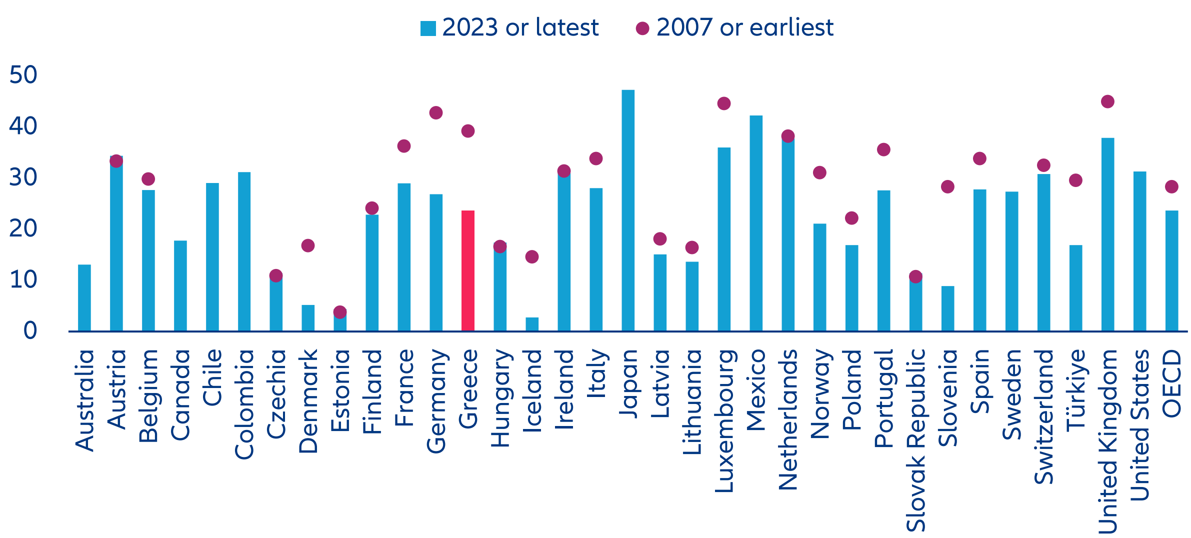
The result is a new configuration of labor market inequality. Women have largely closed the access gap to paid work, but not yet the gap in paid hours. Fewer hours mean lower earnings, slower advancement and weaker pension accumulation. The past quarter century has delivered convergence in participation but not yet in work intensity.

[6] Source: German Federal Statistical Office (2026), „28 % der Teilzeitbeschäftigten arbeiten auf eigenen Wunsch reduziert", https://www.destatis.de/DE/Presse/Pressemitteilungen/2026/01/PD26_N007_13.html.
[7] Source: German Federal Statistical Office (2025), „Fast jede zweite erwerbstätige Frau arbeitet in Teilzeit", https://www.destatis.de/DE/Presse/Pressemitteilungen/2025/05/PD25_175_13.html.
[8] Source: OECD (2025b), "Gender Gaps in Paid and Unpaid Work Persist", Policy Brief, Paris, https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/09/gender-gaps-in-paid-and-unpaid-work-persist_de465299/25a6c5dc-en.pdf.

Gender pension gaps reflect lifetime labor market inequality

If wage gaps capture inequality within working years, pension gaps reveal its cumulative effect. Across OECD and EU countries, women receive lower public pension income than men because pension entitlements reflect lifetime earnings, working hours and contribution histories, where the gender differences accumulate over decades. Across the OECD, the average gender pension gap declined from 28.3% in the late 2000s to 23.7% in 2023 (Figure 4). Progress has been real but retirement income disparities remain large. In several major economies, women still receive 30-40% lower pension income than men. The gap stands at 34% in Austria, 38% in the UK and 38% in the Netherlands. In some countries pension inequality has decreased substantially. In Germany, the gap fell from 43% to 27%, in Greece from 39% to 24%, and in Denmark from 17% to just 5%. These declines reflect rising female employment as well as reforms strengthening minimum pension floors and redistributive elements.

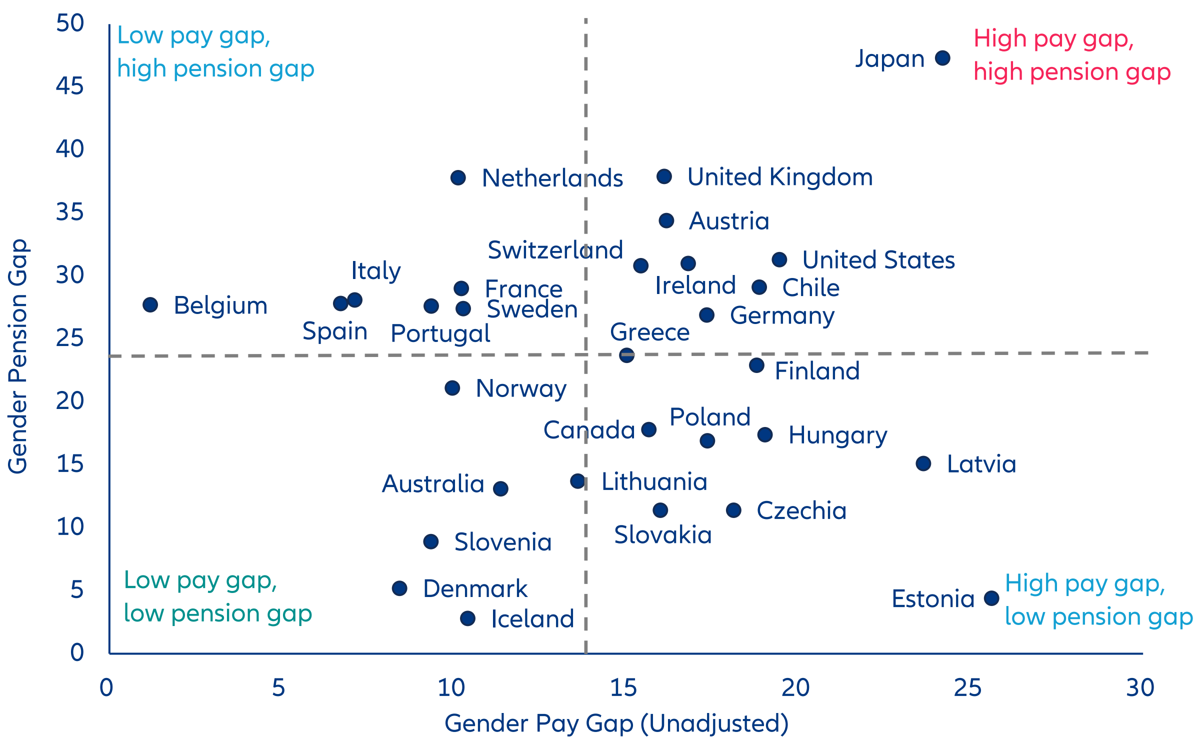
Figure 4: Global evolution of gender pension gaps, in %



Sources: OECD, Allianz Research.

Institutional design mitigates the transmission of labor market gender inequality into retirement. Figure 5 compares the unadjusted pay and pension gaps across countries relative to OECD averages (13.7% and 23.7%), revealing four distinct structural regimes. The first cluster combines high pay gaps with relatively low pension gaps, including countries such as Czechia, Slovakia and Estonia. In these systems, redistributive public pension pillars, minimum benefit floors or contribution credits compress lifetime income differences. Institutional design therefore mitigates the transmission of labor market gender inequality into retirement. A second group displays the opposite pattern: low pay gaps but high pension gaps, including countries such as the Netherlands, Spain, Sweden and France. In these countries, retirement income is closely tied to lifetime contributions. As a result, part-time employment and career interruptions – even when wage gaps appear limited – translate directly into sizeable pension disparities. Moderate pay gaps can therefore coexist with pronounced retirement inequality. A third cluster combines high pay gaps and high pension gaps, reflecting persistent gender inequality across the lifecycle, including countries such as the UK, Austria and the US. Here, labor market disparities during working years are transmitted almost one-to-one into retirement income. Pension systems with strong links to lifetime earnings amplify structural differences in employment patterns, reinforcing inequality rather than compressing it. Finally, a fourth group achieves low pay gaps and low pension gaps, including Denmark, Iceland, Slovenia and Australia. In these economies, relatively compressed wage structures combine with pension systems and employment patterns that limit the accumulation of lifetime disparities. High female participation and supportive family policies often reinforce this balance.

Figure 5: Global Clustering of Unadjusted Gender Pay and Pension Gaps, in %



Sources: OECD, Allianz Research.

Unequal care responsibilities remain a central driver of gender gaps

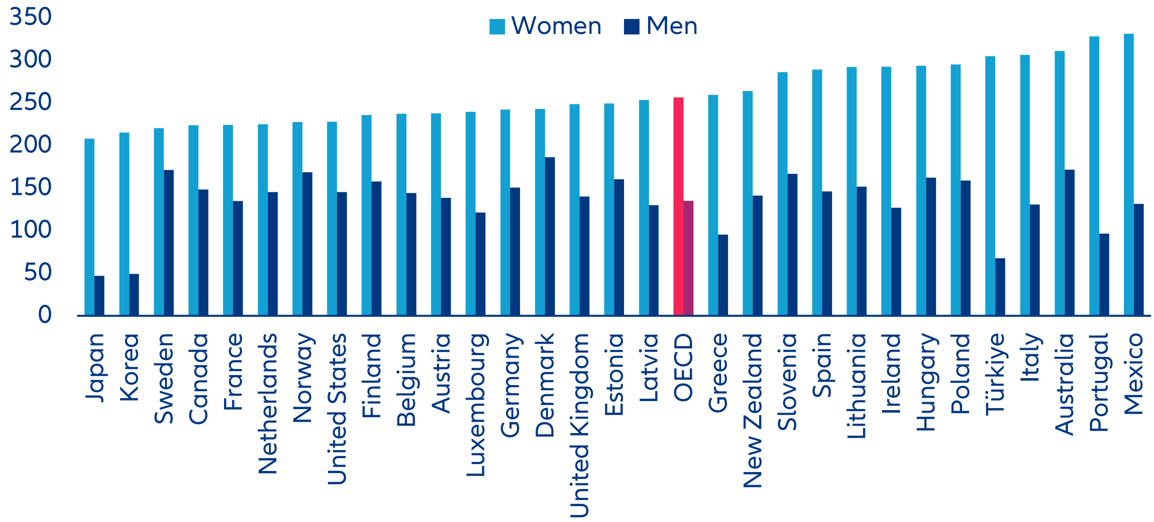
Across the OECD, women perform on average 256 minutes of unpaid work per day, compared with 135 minutes for men – a gap of more than two hours daily (Figure 6). Over a year, that amounts to the equivalent of several additional full-time working weeks. While the imbalance varies across countries, it remains substantial everywhere.

These time differences translate directly into economic outcomes. Time spent on unpaid care reduces paid working hours, career continuity and experience accumulation. While female labor force participation has risen markedly, the distribution of unpaid work has changed far more slowly. Mothers are more likely to reduce hours or interrupt their careers; fathers typically are not. Women are 55% more likely to take career breaks, and those breaks last longer on average – 19.6 months compared to 13.9 months.[9] (World Economic Forum 2025). Access to affordable and reliable childcare remains a critical constraint shaping these decisions.[10] Women are also more likely to provide informal eldercare later in life, further affecting labor supply and lifetime contributions.

Institutional design reinforces these gender inequalities. In many OECD countries, tax-benefit systems are structured around a primary earner model. Joint taxation or means-tested benefits can increase the marginal tax burden on the second income in a household, reducing financial incentives for full-time employment of the secondary earner, which are typically women (OECD 2025b). When childcare costs are high and net returns to additional working hours are limited, households often adjust through reduced female labor supply. What appears as an individual choice is frequently shaped by economic incentives. Unpaid care work therefore remains the hinge of gender inequality. As long as care responsibilities are unevenly distributed, convergence in participation will not translate into convergence in earnings or pensions.

[9] Source: World Economic Forum (2025), "Global Gender Gap Report 2025", WEF Insight Report.
[10] Source: OECD (2025b), "Gender Gaps in Paid and Unpaid Work Persist", Policy Brief, Paris, https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/09/gender-gaps-in-paid-and-unpaid-work-persist_de465299/25a6c5dc-en.pdf.

Figure 6: Global gender gaps in unpaid care work, in minutes per day



Sources: OECD, Allianz Research.

Box 1: The motherhood penalty

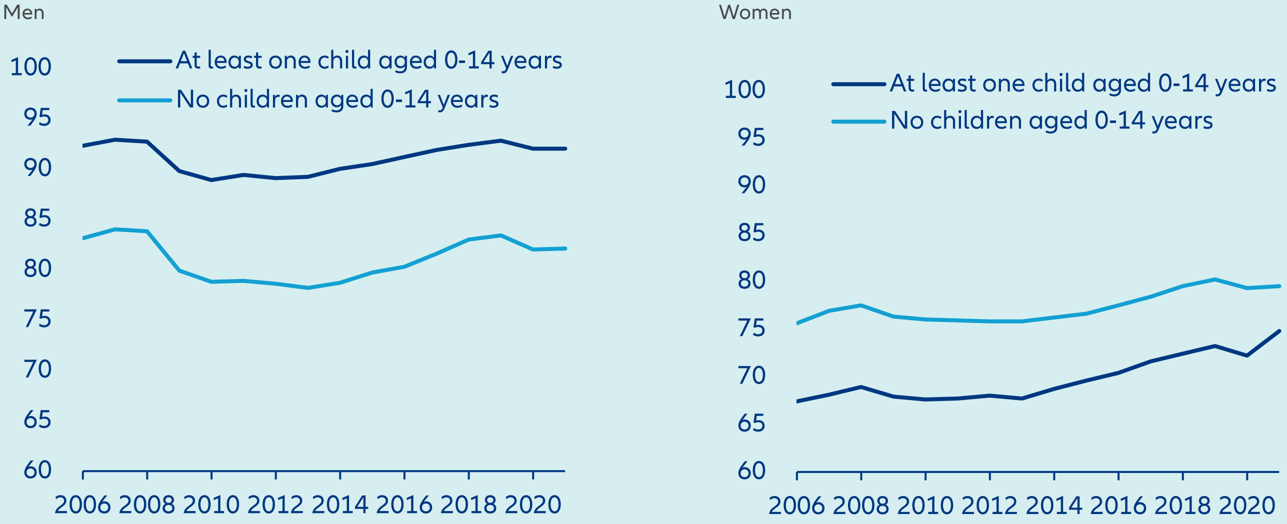
Parenthood is one of the most powerful structural drivers of gender disparities in labor markets. A large and growing body of research shows that the arrival of the first child leads to a persistent divergence in employment trajectories between women and men. Using event-study approaches around the birth of the first child, recent work by Kleven et al. (2025) constructs a global "Child Penalty Atlas" covering more than 130 countries.[11] While fatherhood has little effect on employment, motherhood is associated with large and lasting reductions in employment. Deeply embedded gender norms around care and paid work reinforce this divergence by shaping household decisions and employer expectations.

Descriptive OECD evidence already illustrates this pattern (Figure 7). Across countries, employment rates of men and women without children are broadly similar. After parenthood, however, trajectories diverge sharply. Fathers' employment rates are on average higher than those of men without children, while mothers' employment rates fall well below those of women without children. The charts make the asymmetry clear: parenthood strengthens men's labor market attachment but weakens women's. Survey evidence shows that in many countries a substantial share of respondents believe that children suffer when mothers work or that men should have stronger claims to jobs, highlighting how social norms continue to legitimize unequal care responsibilities.

Formally, the child penalty can be defined as the post-birth divergence in employment probabilities between mothers and fathers in the ten years after the first child. The size of the penalty varies substantially across countries (Figure 8). From around 10% in Norway and Sweden, and 25% in the US and France, it exceeds 40% in Germany, South Korea and Czechia. In several countries, only a partial recovery of female employment occurs even ten years after childbirth. In advanced economies, child penalties explain a large share – and in some cases almost all – of the remaining gender employment gap. Cross-country variation in penalties mirrors differences in social norms and institutional settings, underscoring that policy reform alone may be insufficient without shifts in expectations about work and care.

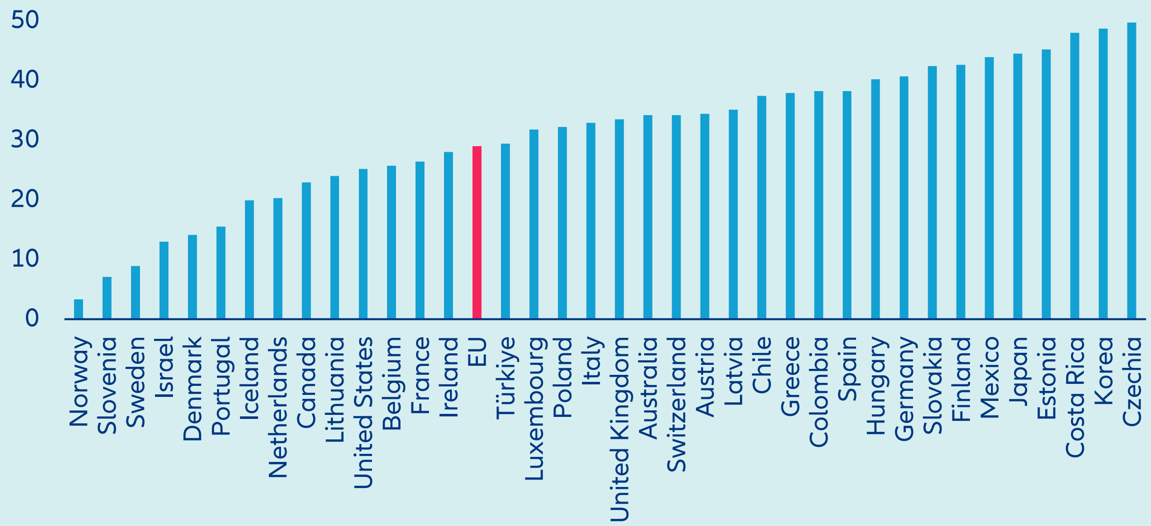
[11] Source: Kleven, H., C. Landais and G. Leite-Mariante (2025), "The Child Penalty Atlas. The Review of Economic Studies", 92(5), 3174-3207.

Figure 7: Employment rates of fathers vs. non-fathers (left) and mothers vs. non-mothers (right), 2006-2021, OECD averages in %



Sources: OECD, Allianz Research.

Figure 8: Child penalties around the world, in %



Sources: Kleven et al. (2025), Allianz Research.

The importance of child penalties increases with economic development. In low-income countries, gender gaps often predate parenthood and reflect broader structural factors. As economies shift toward salaried employment in industry and services, family formation becomes the central mechanism through which gender inequality persists. In high-income countries, reducing gender gaps in employment increasingly means reducing child penalties. Recent evidence also suggests that the effect is not limited to realized career interruptions. A study by Camille Landais and co-authors analyzes women with MRKH syndrome, who know from early life that they will never have biological children.[12] For this group, the typical widening of the gender pay gap in the 30s and 40s largely disappears: their earnings trajectories resemble those of men. This indicates that part of the gender gap reflects decisions made in anticipation of future motherhood, not only the post-birth interruptions.

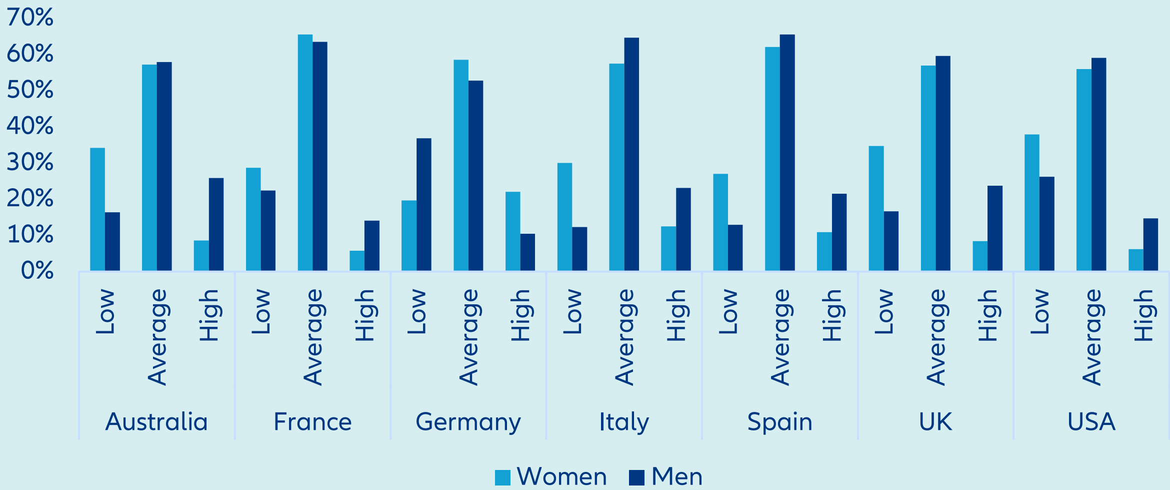
[12] Source: The Economist (2026), "What Drives the Wage Gap between Men and Women?", https://www.economist.com/finance-and-economics/2026/02/11/what-drives-the-wage-gap-between-men-and-women

The magnitude of the motherhood penalty may also evolve in the future as fertility patterns change. Across almost all OECD countries, total fertility rates have fallen below replacement level, in many cases since the mid-1970s. Fewer births could mechanically reduce aggregate motherhood-related career interruptions. The fertility decline has been particularly pronounced among highly educated women, whose opportunity costs of career interruption are highest. In economies where modern labor markets coexist with persistent traditional gender roles, such as South Korea, fertility rates are among the lowest globally. This suggests that when women anticipate unequal childcare burdens, they adjust not only employment after birth but also fertility decisions themselves.[13] Changes in fertility therefore do not automatically imply a weakening of the motherhood penalty; they may instead reflect its anticipation. Together, this evidence shows that the motherhood penalty operates not only through employment losses after childbirth, but also through anticipatory adjustments in career and family decisions. Because these choices are made early and persist over time, their effects accumulate across the lifecycle, shaping lifetime earnings, savings capacity and ultimately pension outcomes. Without parallel changes in social norms, progress in narrowing child penalties is likely to remain gradual even where formal policy frameworks are supportive.

Box 2: Financial literacy and investment behavior

Closing gender gaps in financial literacy can be a powerful lever in closing overall gender income gaps. According to the OECD/INFE 2023 survey, the average financial literacy score across participating OECD countries is 63 out of 100, and only 39% of adults reach the minimum target score.[14] Within this already modest performance, women systematically score lower than men across knowledge, behavior and confidence dimensions. In all countries except Germany (Figure 9), women are overrepresented in the low-literacy category and underrepresented among highly literate respondents.[15]

Figure 9: Financial literacy gender gaps, in %



Source: Allianz Research (2023).

[13] Source: Goldin, C. (2025), "The Downside of Fertility", NBER Working Paper No. 34268. https://www.nber.org/system/files/working_papers/w34268/w34268.pdf

[14] Source: OECD (2023), OECD/INFE 2023 International Survey of Adult Financial Literacy. OECD Publishing, Paris.

[15] Source: Allianz Research (2023), "Playing the Squared Ball: the Financial Literacy Gender Gap", https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/financial-literacy.html

These differences translate into measurable financial outcomes. Simulations based on stylized portfolios show that moving from low to average financial literacy increases annual real returns by 0.8-1.5pp. Over long investment horizons, this gap compounds substantially. Based on average household financial assets, the associated annual surplus investment income ranges from roughly EUR1,750 to EUR5,000 per household. Lower financial literacy reduces participation in capital markets, increases reliance on low-yield savings products and weakens retirement planning. In pension systems that increasingly shift responsibility to individuals, literacy gaps amplify existing income and career inequalities. The cumulative effect is clear: lower investment participation, lower long-term returns and, ultimately, wider gender wealth and pension gaps. Financial literacy is therefore not a marginal skill. It is a structural driver of wealth accumulation and retirement security.

Modelling gender lifetime income gaps across countries and over time

Individual indicators, such as labor force participation rates, hourly pay gaps or pension differences, describe important aspects of gender disparities. But economically, what ultimately matters is total income over the lifecycle. Lower labor income reduces savings capacity and investment income, and weaker contribution histories translate into lower pensions. To capture this cumulative dynamic, we developed an integrated lifecycle income model that measures total gender lifetime income gaps across countries and generations. The framework covers 14 OECD economies – Germany, France, Italy, Spain, Austria, Switzerland, Belgium, the Netherlands, Denmark, Sweden, Czech Republic, Poland, the UK and the US – and compares the three birth cohorts 1975, 2000 and 2025. This cohort-based design allows us to assess structural progress over time and to project how gender lifetime income gaps may evolve under current trends. By embedding country-specific labor-market patterns, tax systems, savings behavior and pension benefit levels, the model reflects the institutional environments in which income gaps evolve over time. Future dynamics are derived from observed structural trends since 1995. The development of labor force participation rates[16], part-time shares, the number of hours worked as well as the hourly wages in each 10-year age group between 20 and the respective retirement age is extrapolated forward, but in some cases with an identical dampening factor. Given the structural catch-up of women in recent decades, we assume that future growth of both female and male labor force participation rates and part-time shares continues at half the historical compound annual growth rate and remains within empirically observed bounds. In addition, we limit the "reverse" gender pay gap to 5%, i.e. as soon as the hourly wages of women exceed those of men in the same age group by more than 5% due to different growth dynamics in our model, we extrapolate the wages of women using the lower average growth rate of the wages of men. The projections therefore reflect a continuation of current structural trends rather than ambitious reform scenarios – enabling an assessment of whether existing momentum is sufficient to meaningfully narrow gender lifetime income gaps.

[16] Sources: Federal Reserve Bank of St. Louis (2025), Data, https://fred.stlouisfed.org/; Eurostat (2025), Database, https://ec.europa.eu/eurostat/web/main/data/database; International Labor Organization (2025), ILOSTAT, https://ilostat.ilo.org/data/, Office for National Statistics (2025), Gender pay gap in the UK 2025, released 23 October 2025, ONS website, statistical bulletin, Gender pay gap in the UK: 2025 (https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworking hours/bulletins/genderpaygapintheuk/2025); United Nations, Department of Economic and Social Affairs, Population Division (2024), World Population Prospects 2024, Online Edition; U.S. Bureau of Labor Statistics (2025), https://www.bls.gov/.

The overall result is clear: while women's lifetime income gaps have seen broad improvement across all countries from the cohorts 1975 to 2000, the trajectories for the youngest cohort born in 2000 are diverging across countries. Table 1 presents the evolution of lifetime income gender gaps across the 14 OECD countries analyzed and the three generations born in 1975, 2000 and 2025. The table ranks the countries based on the projected lifetime income gap of the youngest cohort, those born in 2025, and allows for a comparison of how this gap has evolved across three generations.

However, for the youngest cohort of women and men born in 2025, country trajectories diverge if current structural trends continue. While some countries continue to close the gap, others show signs of stagnation. At the top of the ranking, Sweden is projected to reach near parity for the 2025 cohort, with a slightly negative gap of -2 %. The US and Spain follow, with gaps of around 7%, reflecting sustained progress in employment and earnings. France, the Czech Republic and Belgium also move towards low double-digit levels. By contrast, Italy, Germany and Switzerland continue to exhibit sizeable gaps above 24%.

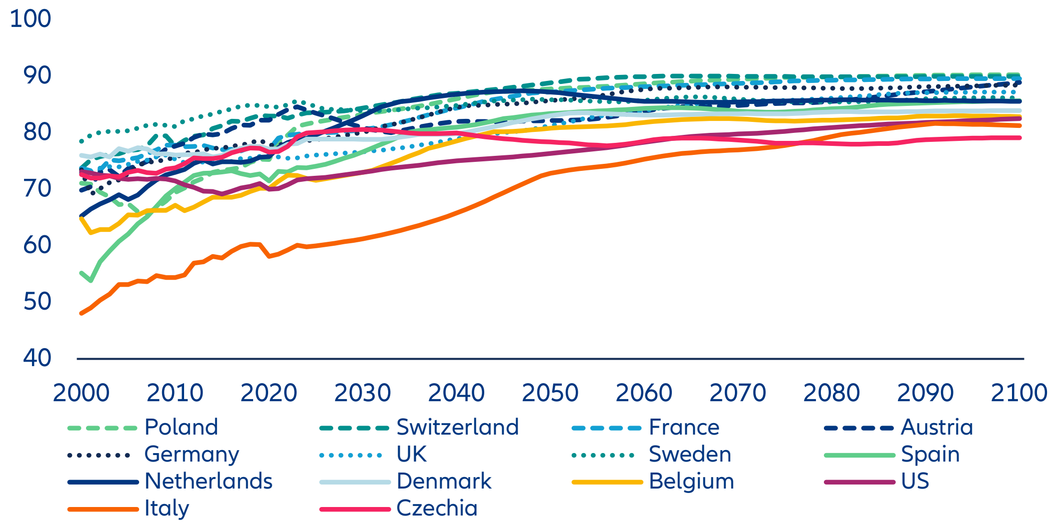
Table 1: Country ranking of lifetime income gender gaps*

Country	Ranking Based on Gap of 2025 Cohort	Lifetime Income Gender Gap 1975 Cohort	Lifetime Income Gender Gap 2000 Cohort	Lifetime Income Gender Gap 2025 Cohort
Sweden	1	14%	2%	-2%
United States	2	23%	16%	7%
Spain	3	13%	5%	7%
France	4	20%	11%	10%
Czech Republic	5	17%	5%	11%
Belgium	6	15%	8%	12%
Poland	7	25%	16%	14%
Netherlands	8	31%	17%	15%
Denmark	9	23%	18%	16%
UK	10	33%	19%	17%
Austria	11	31%	22%	20%
Italy	12	21%	21%	24%
Germany	13	34%	26%	26%
Switzerland	14	33%	28%	32%

*For the calculation of the lifetime income gap of the age cohort 2025 only labor and interest income is taken into account.
Sources: Eurostat, ONS, BLS, UN, OECD, Allianz Research.

The marked reduction in lifetime income gender gaps between the 1975 and 2000 cohorts across most countries reflects substantial gains in female labor-force participation and earnings over recent decades (Figure 10). Rising participation, particularly among women aged 50 and over, has been central to this progress. Not least due to pension reforms that have increased statutory retirement ages. This expansion in working lives largely explains the improved outcomes observed for the 2000 generation compared with those born in 1975.

Figure 10: Female labor force participation rates, 2000-2100, in %



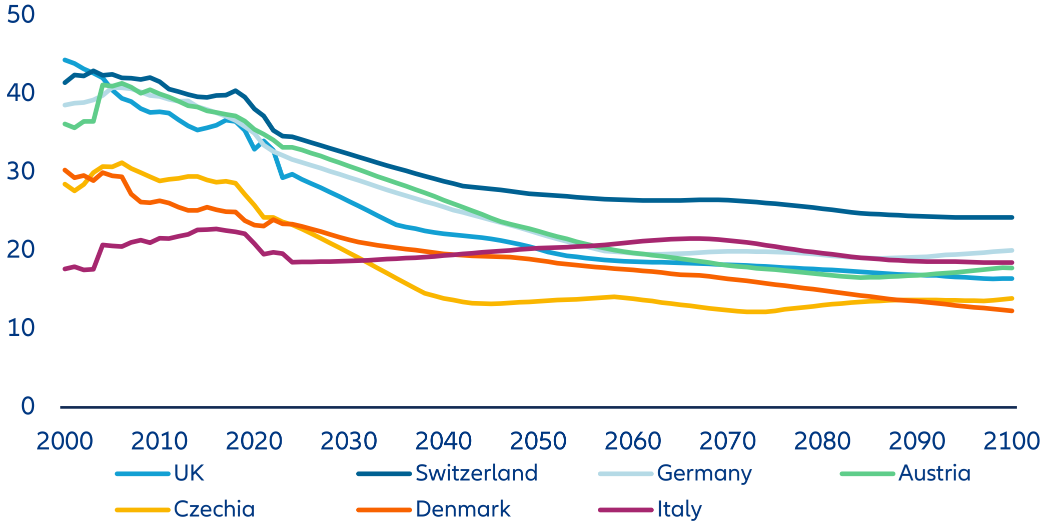
Sources: Eurostat, ILOSTAT, Federal Reserve Bank of St. Louis, OECD, ONS, UN Population Division (2024), Allianz Research.

A key driver of the diverging country trajectories for the 2025 cohort are trends in female part-time work. Since around 2010, these trends have begun to diverge. In countries with narrowing gaps, female part-time rates peaked between 2010 and 2019 and have since stabilized or declined. By contrast, in countries with higher gaps, the part-time shares of women are not only higher today but are projected to remain substantially higher if current trends persist. Among men, part-time employment has increased modestly, though from low levels. Pension systems, combined with women's longer life expectancy, mitigate some lifetime income disparities. Yet where lifetime gaps stagnate or widen, structural differences in work intensity and hours worked dominate. This explains why the momentum is uneven and, in some cases, stalling.

While the generational comparison reveals how lifetime income gaps evolve across cohorts, it does not capture the speed of convergence at any given moment. To assess this dynamic, Figure 11 shifts from a cohort-based view to an annual perspective on total labor income, combining participation, hours worked and hourly wages. This lens makes visible the underlying trajectory of gender earnings differences on the labor market: whether progress is accelerating, slowing or stalling. It complements the lifecycle results by indicating whether current structural trends are strong enough to sustain convergence in the decades ahead – or whether lifetime gaps are likely to persist despite generational improvements. The projections show that structural factors, particularly trends in part-time employment and work intensity, increasingly shape future outcomes. As a result, the annual labor income gap is expected to decline further in most countries, but at different speeds – explaining the divergent results projected for the 2025 cohort.

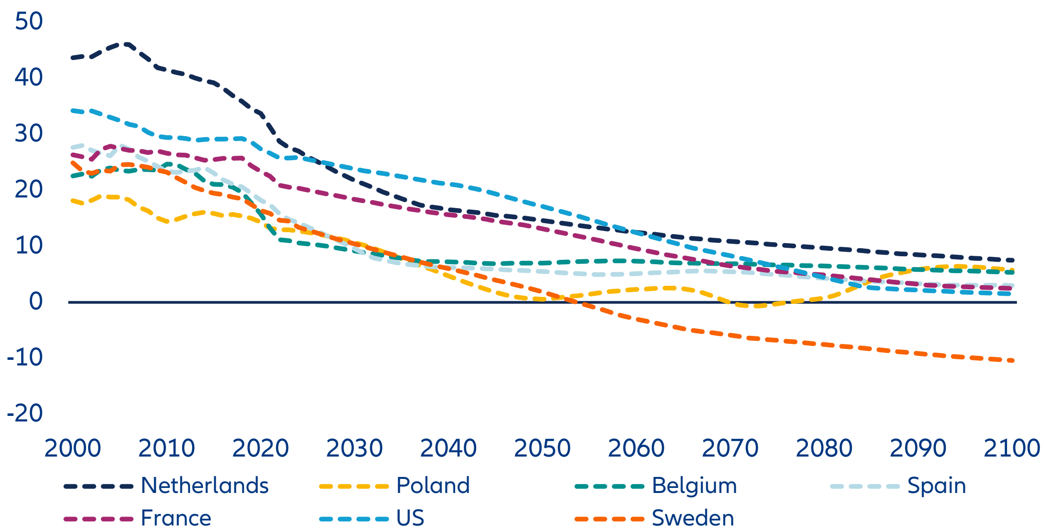
The leading group – Sweden, the US, Spain, France, Czech Republic, Belgium, Poland and the Netherlands – is projected to see annual labor income gaps fall below 10% by the end of the century, effectively approaching income parity (Figure 11). A lagging group achieves gradual but incomplete convergence. The lagging group – Germany, Italy, the UK, Denmark, Austria, Switzerland and Czechia – remains constrained by persistent differences in work intensity and part-time employment, leading to higher and stagnating lifetime income gender gaps (Figure 12).

Figure 11: Annual labor income gender gap 2000-2100, countries where the gap remains above 10%, in %



Sources: Eurostat, ILOSTAT, Federal Reserve Bank of St. Louis, OECD, ONS, UN Population Division (2024), Allianz Research. .

Figure 12: Annual labor income gender gap 2000-2100, countries where the gap remains above 10%, in %



Sources: Eurostat, ILOSTAT, Federal Reserve Bank of St. Louis, OECD, ONS, UN Population Division (2024), Allianz Research.

Country Insights

#1 Sweden: It's the low part-time share

Sweden tops the ranking among the 14 analyzed countries. The projected lifetime income gender gap narrows sharply from 14.2% for women born in 1975 to just 2.2% for the 2000 cohort. For those born in 2025, the gap turns negative (-2.4%), making Sweden the only country in the sample where the youngest generation is projected to achieve effective lifetime income parity.

A Swedish woman born in 2000 is projected to earn around 5% less over her lifetime than her average male peer – equivalent to approximately EUR 130,000 when accounting for all income streams from first employment earnings through pension income. As in most countries, the gap is primarily rooted in labor income. However, Sweden's pension system with its capital-based pillar as well as women's longer life expectancy partly offsets this disparity. With retirement at 69, women are expected to spend roughly two additional years in retirement, resulting in total pension income that is about 5% higher than that of men. This cushions roughly half of the underlying labor income gap, leaving a relatively small residual difference over the lifecycle.

Hourly wage differences are already limited. Women working full-time earn close to male levels, and under current trends the gender pay gap is expected to close by the mid-2050s. Over time, women's hourly wages reach parity and are projected to slightly exceed those of men.

The decisive structural factor, however, is working time. Sweden combines high female labor-force participation with comparatively low and steadily declining female part-time employment. The share of women working part-time fell from 41.5% in the mid-1990s to 24% in 2025, while the male rate increased from 8% to 12%. If this trajectory continues, female part-time employment declines further to around 12%, while male part-time rises toward 18%, effectively reversing the traditional gender gap in reduced hours.

Female labor-force participation is already high at 85% and remains close to male levels over the long run. Taken together – participation, hours worked and wage convergence – these dynamics drive the annual labor income gap down from 13% today to slightly negative territory in the long term, with women's average annual earnings eventually exceeding men's by around 10%.

Sweden illustrates how sustained convergence in working-time patterns, alongside wage equality, can translate into lifetime income parity. Maintaining policies that support both work and family as well as balanced care responsibilities will be essential to preserve this trajectory.

#2 US: Strong and sustained convergence

The US ranks second among the 14 studied countries and stands out for a strong and continued convergence process across generations. The projected lifetime income gender gap declines from 23.1% for those born in 1975 to 15.9% for the 2000 cohort – and then more than halves to 7.4% for those born in 2025. The downward trajectory is clear and sustained.

For the 2000 birth cohort, the total lifetime gap corresponds to roughly USD1.13mn. The decomposition shows that 73.6% of the gap stems from work earnings, 3.7% from investment income and 22.7% from pensions – one of the highest labor-income shares among the countries studied. In the comparatively lean US welfare state, where public redistribution is more limited and pension systems are more closely tied to individual earnings and savings decisions, labor-market disparities translate more directly into lifetime income differences. As a result, women have stronger economic incentives and often greater necessity to maintain full-time employment over the life course. Labor market disparities therefore translate more directly into lifetime income differences, but convergence in employment and wages also feeds through more quickly into overall gender income equality.

Participation dynamics support this trend. Today, 70.3% of women participate in the labor market compared with 79.7% of men. Under current structural projections, this gap is expected to narrow further, driven in particular by rising employment among women aged 60 and above.

Another crucial driver is part-time employment. Unlike in many European countries, female part-time shares in the US are declining from already moderate levels and are projected to converge toward male rates. Among 25-49-year-olds, 16.5% of women work part-time today compared with 6.4% of men; by 2100, these shares are projected to fall to 4.8% and 2.7%, respectively. A similar compression is visible among 50–59-year-olds (15.9% vs.

5.4% today, declining to 4.6% and 2.2%). The narrowing part-time gap significantly strengthens income convergence.

Hourly wages present a more mixed picture. In part-time employment, women already earn as much as men on an hourly basis. However, the wage gap in full-time employment remains material. Under current structural trends, women are projected to reach around 85% of men's hourly pay in full-time roles by the end of the century – leaving a persistent residual gap. As discussed in the gender pay section, however, this measure does not fully control for occupational structure, experience or sectoral composition, which remain important explanatory factors.

An annual perspective confirms the strong convergence path. The total annual labor income gap stands at 19.7% in 2026, is projected to fall below 10% by 2067 and to shrink further to around 4% by 2100. The key lever for faster closing of the remaining lifetime income gap in the US therefore lies in narrowing hourly pay differences in full-time employment, alongside continued gains in participation and work continuity.

#3 Spain: Gender lifetime income parity in sight

Spain ranks third among the 14 studied countries and stands out for sustained progress toward gender income parity. The projected lifetime income gap declines sharply from 13.0% for those born in 1975 to just 4.9% for the 2000 cohort. For those born in 2025, the gap rises modestly to 7.4%, but remains in the single digits. Spain is therefore among the countries closest to lifetime income equality.

For the 2000 birth cohort, the total lifetime income gender gap corresponds to roughly EUR126,000. The decomposition reveals an unusual pattern: while work earnings generate a sizeable gap (122.4% of the total), this is more than offset by pension income (-28.3%), with investment income contributing only marginally (6.0%). Spain is one of the few countries where pension entitlements act as a net equalizer, effectively counterbalancing labor market disparities and keeping the overall lifetime gap low.

Labor market dynamics support this convergence. Participation rates are relatively high and narrowing (71.8% for women versus 79.9% for men) and are projected to continue converging, although not fully equalizing. Part-time employment remains more prevalent among women, but the gap is modest compared with many other countries and is projected to close further. Among 25–49-year-olds, 19.8% of women work part-time today compared with 5.7% of men; by 2100, the gap narrows significantly (15.6% versus 11.5%). In several age groups, female part-time shares are projected to decline while male rates increase, compressing differences in work intensity. Wage dynamics further reinforce convergence. In full-time employment, hourly wages are already close to parity today and women are projected to overtake men over time. In part-time employment, women are expected to earn more per hour than men as early as 2046. Working hours among full- and part-time employees are currently similar across genders. An annual perspective confirms this trajectory. Spain's annual labor income gap fell from 27.7% in 2000 to 12.7% today and is projected to decline further to around 3% by 2100. Spain therefore represents one of the clearest cases where structural labor market changes and pension design together translate into near equality in gender income across the lifecycle.

#4 France: Structural convergence supported by the pension system

France ranks fourth among the 14 analyzed countries. The projected lifetime income gender gap narrows markedly from 20.5% for women born in 1975 to 10.5% for the 2000 cohort. For those born in 2025, the gap remains broadly stable at 10.0%, placing France among the countries closest to long-term convergence.

Mirroring the dynamics of most countries, the lifetime income gender gap in France is primarily driven by differences in labor income. A woman born in 2000 is projected to earn around 16% or approximately EUR393,000 less than her male peers over her life, including income from labor, pension and interest income. While this originates from work earnings over the lifetime, France's pension system partly offsets these disparities. Thanks to higher life expectancy, women are expected to spend more years in retirement. If the statutory retirement age remained at 64, a woman retiring in 2064 would spend 28.3 years in retirement, compared with 24.6 years for a man. As a result, total pension income over the retirement phase is projected to be marginally higher for women (-0.4%), cushioning the overall lifetime income gap.

Hourly wage differences explain only part of the labor income gap. Women working full-time earn between roughly 80% and 100% of male hourly wages across age groups, with somewhat wider variation in part-time employment. Under current trends, the gender gap in hourly pay is projected to close gradually from the 2070s onward. Over time, women's full-time wages approach parity in most age groups.

The main structural driver of lifetime income gender gaps is working hours, as French women remain significantly more likely to work part-time. In 2025, around 25% of women were employed part-time, down from a peak of 31% in 1999, while the male part-time rate has risen to around 7%. If current structural trends continue, female part-time employment could decline further to 17% in the long run, while the male rate stabilizes near 11%. At the same time, female labor-force participation is projected to increase from 80% today to around 90% by 2090, broadly converging with male participation.

Under these assumptions, the total annual labor income gap – combining participation, working hours and wages – is projected to fall from around 20% today to just 2% in the long run, pointing to near parity in annual earnings.

Further progress in closing gender income gaps in France will therefore depend on structural change. Reliable childcare provision and a more balanced distribution of paid and unpaid work remain essential to sustain higher female participation and reduce remaining gender differences in part-time work.

#5 Czech Republic: Near-parity reached but risk of partial reversal

The Czech Republic ranks 5th among the 14 studied countries and represents a case of rapid convergence followed by renewed divergence. The projected lifetime income gender gap falls sharply from 17.0% for those born in 1975 to just 4.5% for the 2000 cohort, approaching parity. However, for the 2025 cohort, the gap widens again to 11.3%, interrupting the convergence trend.

For the 2000 birth cohort, the lifetime gap corresponds to roughly EUR248,000. The decomposition reveals a striking structure: work earnings account for 147.5% of the total gap, while pensions contribute -59.5%, more than offsetting labor market disparities. Investment income plays a moderate role (12.%). As in Spain, the pension system acts as a strong equalizer, compressing lifetime inequality. The core driver remains the labor market.

Participation rates are relatively high and stable (71.7% for women versus 83.1% for men) and are projected to remain broadly unchanged. Today, the Czech Republic stands out for comparatively low female part-time employment, supporting income equality. 13.3% of women aged 25-49 work part-time compared with 2.4% of men – one of the smallest gender gaps among the countries studied. However, under a continuation of current structural trends this would change notably, with female part-time employment on a clear upward trend.

Meanwhile hourly wages are on a clear convergence path. Women are projected to catch up with men in part-time hourly pay by 2040 and in full-time employment by 2080. As in other countries, the emerging challenge is therefore not wage discrimination per hour, but structural shifts in working time.

An annual perspective confirms this dynamic. The total annual labor income gap stands at 22.2% in 2026 and is projected to shrink to around 13% by 2045. Thereafter, progress stalls as gains in wages and participation are offset by rising female part-time employment.

The policy implication for the Czech Republic is therefore clear: maintaining the traditionally high share of women in full-time employment will be critical. Preventing a structural drift toward extensive part-time work will determine whether near parity can be sustained rather than reversed.

#6 Belgium: Convergence in sight – but part-time work keeps the gap alive

In Belgium, which ranks sixth among the 14 analyzed countries, the lifetime income gender gap is expected to narrow from 15.4% for a woman born in 1975 to 7.6% for one who was born in 2000 and has just started her working life. Given population aging and a shrinking labor force population, for those born today, labor market conditions and the institutional framework might improve further, so that the labor income gap might at least remain rather constant or narrow even slightly further in the long run.

As in most countries, the lifetime income gender gap is driven by labor income. A woman born in the year 2000 is set to earn around 13% less than her male peers over the course of her working life. However, due to the higher average further life expectancy of women, the total retirement income is likely to be around 4% higher than that of a male born in the same year, which reduces the overall gap markedly. But nevertheless, in our model, the sum of labor, pension and interest income of a woman born in 2000 is going to be around EUR568,000 lower than that of her male peer.

The labor income gap does not primarily stem from differences in hourly wages. In fact, women working full-time earned in every groups more than their mail peers with the difference ranging between 1% in the age group 60 and older up to 14% in the age group 20 to 29. However, the situation looked markedly different when comparing average hourly wages for those working part-time: Here the average wages of women were up to 18% lower than those of their male peers. Furthermore, the average hourly part-time wage of women was merely between 75% and 80% of the full-time wage in the respective age group, while the difference between hourly wages paid in full-time and part-time was between 93% and 100% in the case of men. Due to the assumed continuation of the observed trends, the gap between the average full-time and part-time wage is not set to close neither in the case of women nor in the case of men. However, in the long run the gap in the average hourly part-time wage of women and men is going to close, with the average hourly part-time wages ranging between 96% and 105 % in most age groups from mid-century onwards.

Therefore, it is the markedly higher share of women working part-time that is a key driver of the average labor income and hence lifetime income gender gap. In 2025, the share of women working part-time had declined to around 37% after peaking at 43% in 2012. At the same time the share of men working part-time has slightly increased to around 11%, mainly driven by the developments in the age groups 15 to 29 and 25 to 49. If we assume that the observed trends are going to continue, the share of women working part-time is set to decline to 27%, while that of men is going to increase to around 20% in the long run. During the same period the labor force participation rate of women is expected to increase from 72% today to 83%, i.e. almost the same level as that of men, which is going to increase from 80% to 84% in our model. Given these assumptions, the gap in the total annual labor income per capita – taking into account the labor participation rate, part-time and full-time shares, the number of average hours worked and not least the respective average hourly wage in each age group – is set to decline from 11% today to 5% in the long run.

However, further increasing the labor force participation rate of women will require reliable childcare, stronger second-earner incentives and a more balanced distribution of paid and unpaid work.

#7 Poland: Strong catch-up, slowing dynamic

Poland ranks seventh among the 14 studied countries and has achieved substantial generational progress, although convergence is slowing. The projected lifetime income gender gap declines from 24.5% for those born in 1975 to 15.7% for the 2000 cohort and 14.2% for the 2025 cohort. The improvement is significant, but the comparison of the latter two cohorts shows that the momentum has weakened.

For the 2000 cohort, the lifetime gap of 15.7% corresponds to roughly EUR455,000. The majority of 59.7% of the gap stems from labor income, 5.6% comes from investment income and a comparatively high share of 34.8% from pensions. Pension disparities therefore weigh more heavily in Poland than in its peer countries, such as Czech Republic, reflecting how lifetime employment patterns translate differently into retirement income.

Labor-force participation is the central driver Poland. Today, 69.4% of women participate compared with 80.1% of men. Under current structural trends, this gap is projected to persist. The gap is particularly evident for prime-age workers: among 60–64-year-olds, female participation stands at just 25.8% compared with 64.2% for men. In the international comparison, Poland stands out for its very low gender differences in part-time employment. For instance, among 25-49-year-olds, 7.2% of women work part-time versus 2.2% of men, expected to decline further to 5.2% and 1.1% by 2100. Stable full-time employment shares of women help narrow gender income gaps.

Hourly wages are nearing male levels but are not projected to fully converge. Together with persistently lower female participation, this explains why progress is slowing despite the relatively high full-time share of women. Unlike most other countries, where women are projected to overtake men in hourly earnings, wage convergence in Poland remains incomplete, and pension design provides less of an equalizing effect.

From an annual perspective, the labor income gap stands at 12.2% in 2026 and is projected to fall below 1% by 2048 before stabilizing. Raising female employment particularly among older individuals represents the most powerful lever to accelerate gender income convergence.

#8 Netherlands: Rapid early convergence, then stalls at structural plateau

The Netherlands ranks eighth among the 14 studied countries and illustrates a case of strong initial progress followed by structural stagnation. The projected lifetime income gender gap declines sharply from 30.7% for those born in 1975 to 17.4% for the 2000 cohort. For those born in 2025, the gap narrows further to 15.0%, but the pace of convergence slows markedly. After substantial early gains, further progress becomes incremental.

For the 2000 birth cohort, the lifetime gap of 17.4% corresponds to roughly EUR982,000. The decomposition shows that 64.9% of the gap stems from labor earnings, 7.2% from investment income and 27.9% from pensions. The Netherlands stands out for a comparatively high retirement replacement rate of nearly 75%, combined with a gradual increase in the statutory retirement age to 70. While this cushions lifetime disparities, labor income remains the dominant driver of inequality.

Labor-force participation is already among the highest in the sample. Today, 82.1% of Dutch women participate in the labor market compared with 88.9% of men. Under current projections, this gap is expected to narrow slightly further. In terms of hourly wages, convergence is well advanced: women already earn as much as men in part-time employment, and in full-time roles they are projected to reach parity by around 2030.

The central structural constraint, however, is the exceptionally high prevalence of female part-time employment – the highest in the EU. Among 25-49-year-olds, 56.9% of women work part-time today compared with 14.3% of men. Although this gap is projected to narrow significantly by 2100, it remains substantial. A similar pattern holds across older age groups, where female part-time rates remain persistently elevated despite gradual declines.

As a result, differences in work intensity rather than participation or hourly pay continue to drive income disparities. The Netherlands represents a clear case where gender equality in wages coexists with large differences in hours worked.

An annual perspective confirms this dynamic. The total annual labor income gap stands today at 25.1% and is projected to decline steadily to around 7.5% by the end of the century. The downward trend is visible but slower than in countries with lower structural reliance on part-time employment.

The policy implication is therefore straightforward: faster convergence would require either a further reduction in female part-time employment or a more equal distribution of part-time work between men and women. Without such structural adjustments, lifetime income equality will improve only gradually despite high participation and wage parity.

#9 Denmark: High participation but persistent hours gap slows progress

Denmark ranks ninth among the 14 studied countries and represents a Nordic model of high participation and strong institutions – yet with slower income convergence than might be expected and is visible in neighboring Sweden. The projected lifetime income gender gap declines from 23.3% for those born in 1975 to 18.2% for the 2000 cohort and further to 15.9% for the 2025 generation. Progress is steady but gradual, and a gap remains.

For the 2000 birth cohort, the lifetime gap corresponds to roughly EUR2.07mn, reflecting the high income level. The decomposition shows that 65.0% of the gap stems from labor earnings, 3.5% from investment income and 31.5% from pensions. Denmark combines a strong public pension pillar with broad occupational pension coverage, helping to cushion disparities over the life cycle. Still, labor income remains the dominant source of inequality.

Denmark stands out for its extensive welfare architecture. Universal, heavily subsidized childcare and generous parental leave schemes – including earmarked leave for fathers – support high female labor-market attachment. As a result, participation rates are among the highest in the sample: 79.7% of women are active in the labor market compared with 85.1% of men, with only limited change projected under current trends.

Unlike in the Netherlands or Germany, part-time employment is not structurally dominant, but gender differences persist. Among 25-49-year-olds, 29.0% of women work part-time compared with 11.6% of men. In older age groups, the gap remains wider. These differences are projected to narrow gradually over time. However, in full-time employment, Danish men work on average around three hours more per week than women – a gap that is expected to persist under current structural projections.

The more decisive constraint lies in wage dynamics. Denmark is characterized by relatively compressed wage structures but persistent occupational and sectoral segregation, with women overrepresented in public-sector and care professions. While part-time hourly wage convergence is projected around 2070, parity in full-time hourly pay is not expected until after 2090 – significantly later than in many peer countries. In other words, Denmark achieves high participation, but full wage convergence unfolds slowly. The projection of how annual income differences on the labor market evolve complete the picture. The total annual labor income gap stands at 22.8 % in 2026 and is projected to decline to 12.3 % by the end of the century. The downward trajectory is clear, but income differences remain persistent due to higher female part-time shares despite strong institutional support for employment.

The projection of how annual income differences on the labor market evolve complete the picture. The total annual labor income gap stands at 22.8 % in 2026 and is projected to decline to 12.3 % by the end of the century. The downward trajectory is clear, but income differences remain persistent due to higher female part-time shares despite strong institutional support for employment. Denmark therefore illustrates a distinct case: extensive childcare provision and high female labor-force participation successfully anchor women in the labor market, but differences in wages and working hours continue to sustain a sizeable lifetime income gap. Further convergence will depend less on participation gains and more on reducing occupational segregation and closing full-time gender differences.

#10 UK – The gaps won't close

In the UK, which ranks 10th among the 14 analyzed countries, the lifetime income gender gap is expected to narrow from 32.7% for a woman born in 1975 to 19.2% for one who was born in 2000 and further to 16.9% for the 2025 generation.

As in most countries, the lifetime income gender gap in the UK is primarily driven by differences in labor income. A woman born in 2000 is projected to earn around 22% less than her male peers over the course of her working life. The resulting gap in retirement income, at 13%, is somewhat smaller, reflecting women's higher life expectancy. On average, a 68-year-old woman is expected to spend 22.3 years in retirement, compared with 20.8 years for her male counterpart. Nevertheless, even after including pension and interest income, the total lifetime income of a woman born in 2000 is projected to be approximately EUR 930,000 lower than that of a man born in the same year.

Differences in average hourly wages explain only part of the labor income gap. According to the latest data from Eurostat and the national statistical office, women working full-time earn between 84% and 96% of the hourly wages paid to men in the same age group. For part-time employment, the corresponding range is 75% to 98%. There are also notable differences between part-time and full-time wages. Within the same age group, women working part-time earn 80% to 86% of the hourly wages paid to full-time female employees, while for men the range is 81% to 92%. Assuming current trends in hourly wage growth continue, the gap between part-time and full-time wages is expected to narrow, particularly among younger age groups and for both genders. However, our model suggests that although the gender pay gap will continue to decline, it will not disappear across all age groups. If long-term trends persist without interruption, average hourly wages of women working full-time are projected to range between 65% and 105% of those of their male peers in the same age group. For part-time workers, the range is expected to be 58% to 105%, with the lowest relative wages in both cases observed among those aged 30 to 39.

A significantly higher share of women working part-time is another key driver of the labor income gap and, consequently, the lifetime income gender gap. Since 1996, the proportion of women in part-time employment has declined to 37% in 2025. Over the same period, the share of men working part-time has risen slightly, from 7% to 11%. Even if these trends continue, the gap will remain substantial: in the long run, 31% of women are projected to work part-time, compared with 16% of men - meaning the female share would still be nearly twice as high. At the same time, women's labor force participation rate is expected to increase from 76% today to 87%, reaching the same level as men. Male participation is also projected to rise, from 84% to 87%, in our model. Taking all these factors into account - labor force participation, full-time and part-time shares, average hours worked, and age-specific hourly wages - the per capita annual labor income gap is projected to shrink by almost half, from 30% today to 16% in the long run.

Further progress, however, will depend on continued improvements in access to reliable childcare and a more balanced distribution of paid and unpaid work. Such measures are essential to enable women in particular to better reconcile work and family life over the long term.

#11 Austria: Progress with continuing but slowing convergence

Austria ranks 11th among the 14 studied countries, showing strong progress over three generations but slowing convergence. The projected lifetime income gender gap declines from 31.4% for those born in 1975 to 22.4% for the 2000 cohort and 19.4% for those born in 2025. The long-term direction is encouraging, but the comparison between cohorts shows that the speed of convergence has decreased but not stopped.

As in most countries, the lifetime gap is primarily driven by labor income. For the 2000 birth cohort, a total gap of 22.4% implies that a representative 26-year-old woman can expect to earn roughly EUR1.24mn less over her lifetime than a man of the same age, in nominal terms, including earnings, savings and pensions. Decomposing the gap shows that 62% stem from labor income, 3% from investment income and a comparatively high 35% from pensions – a larger pension contribution than in many peer countries. Lower earnings during working life therefore not only reduce current income, but also amplify disparities in retirement.

The structural driver is again working hours, with gender differences in part-time employment in Austria being particularly pronounced. Among 25-49-year-olds, 52% of women work part-time today compared with 11% of men; among 50-59-year-olds, the figures are 50% and 8%. Under current structural trends, these gaps are projected to widen further, with female part-time employment rising toward 70%, while male part-time increases more moderately toward 20%. At the same time, labor-force participation continues to converge (currently 74.2% for women versus 82.1% for men), and hourly wages are projected to align further, with women expected to overtake men in both full- and part-time employment during the 2050s. As in Germany, the central issue is therefore not hourly pay, but hours worked.

Moving to an annual outlook highlights the forward trajectory. In 2026, women's total annual labor income in Austria – combining participation, hours and wages – is 32.5% lower than men's. Even under continued convergence, the model projects that this annual gap would fall to around 17% by 2080 and then stabilize at that level, reflecting persistent structural differences in working time.

The implication is clear: without a substantial expansion of women's full-time and near full-time employment, progress will slow further. Reliable childcare, stronger second-earner incentives and a more balanced distribution of paid and unpaid work will determine whether Austria can accelerate convergence again.

#12 Italy: Stagnation and structural persistence of gaps

Italy ranks 12th among the 14 studied countries and stands out for the absence of meaningful progress across generations. The projected lifetime income gender gap remains broadly unchanged between the 1975 and 2000 cohorts (21.1 % and 20.9 %), before slightly widening to 24.4 % for those born in 2025. Unlike most peers, Italy exhibits structural persistence of gender income gaps across generations.

For the 2000 birth cohort, a total lifetime gap of 20.9 % corresponds to roughly EUR480,000 less income over the lifecycle for a representative woman compared with a man of the same age. The decomposition underscores the dominance of the labor market: 82.5% of the gap stems from work earnings, while only 1.8% derives from investment income and 15.7% from pensions. Italian households have relatively low savings rates (around 4.2%), limiting the role of capital income in widening inequality. At the same time, comparatively high pension replacement rates (70.6%) and women's longer life expectancy act as partial equalizers in retirement. The core of the disparity therefore lies in employment.

Labor-force participation gaps are exceptionally large in Italy but are projected to close. Today, 57.6% of women participate in the labor market compared with 75.6% of men. Under current projections, women are expected to broadly catch up with men by the 2050s, which reduces one major source of income differences. However, participation is only part of the story. Part-time employment remains heavily gendered and is projected to diverge further. Among 50–59-year-olds, just 4.8% of men work part-time today compared with 29.3% of women. While hours worked among full-time employees and hourly wages are broadly similar, the combination of lower participation (for now) and rising female part-time intensity structurally constrains women's lifetime labor income.

Shifting from the cohort perspective to an annual outlook confirms the stagnation. Italy's total annual labor income gap stands at 18.6% today and, under current structural trends, is projected to remain virtually unchanged through 2100. The projected catch-up in participation is offset by widening differences in part-time employment. Without a structural shift toward stronger and more continuous female full-time employment, Italy's gender income gap is unlikely to narrow.

#13 Germany: Clear Progress, Stalling Progress, Fading Stalling momentum

Germany ranks 13th among the 14 studied countries, showing clear progress over three generations but fading momentum. The projected lifetime income gender gap declines from 34% for those born in 1975 to 26.2% for the 2000 cohort and 25.9% for those born in 2025. The long-term direction is encouraging, but the comparison between the 2000 and 2025 cohorts suggests that convergence has largely stalled.

The gender gap in total lifetime income is driven primarily by labor income. For the 2000 birth cohort, the total gap of 26.2% means that a representative 26-year-old woman today can expect to earn roughly EUR1.15mn less over her lifetime than a man of the same age, in nominal terms, factoring in earnings, savings and pension income. Breaking this total lifetime gap down into its components reveals that 78% stems from work earnings gap, 3% from investment income and 19% from pension income. Lower earnings during working life are the key driver, as they translate into weaker wealth accumulation and pension entitlements. The consequences are already visible in retirement: one in five women of retirement age in Germany is at risk of poverty.[17]

The reasons for the lower lifetime labor income are now structural rather than wage-based. While hourly wages between women and men are projected to continue to converge, differences in hours worked remain substantial, as more women work part-time. Among 25- to 49-year-olds, 47% of women work part-time compared with just over 10% of men. Even with a forecasted increase in men's part-time share in Germany, the hours gap remains substantial. Labor-force participation rates are much closer (76.5% for women versus 83.9% for men) and hourly wages are projected to converge. In part-time employment, women in Germany earn already today slightly more than men on an hourly basis. Under current structural trends, the gender pay gap per hour in full-time

[17] Source: German Federal Statistical Office (2025), "Jede fünfte Person im Ruhestand hat maximal 1 400 Euro netto pro Monat zur Verfügung", https://www.destatis.de/DE/Presse/Pressemitteilungen/2025/10/PD25_N054_12_13.html

employment is projected to fall below 10% by 2037 and to fully close by 2062. The central issue is therefore not hourly pay, but hours worked.

Shifting from a cohort-based lifetime perspective to an annual view provides an outlook on future dynamics. In 2026, women's total annual labor income in Germany – factoring in participation, hours and wages – is 30.9% lower than men's. If current structural trends continue, the model projects that this annual labor income gap would not close but still stand at around 19% in 2100.

Therefore, the policy implications for Germany are clear: without a meaningful expansion of women's full-time and near full-time employment, further progress will remain limited. Reliable childcare, stronger incentives for second earners and a more equal sharing of paid and unpaid work will determine whether convergence regains momentum.

#14 Switzerland: Early gains, reversing trend

Switzerland ranks last among the 14 studied countries and stands out as a rare case where convergence first improved but is now reversing. The projected lifetime income gender gap declines from 33.1% for those born in 1975 to 27.8% for the 2000 cohort – but then widens again to 32.1% for those born in 2025. After initial progress, momentum fades and the gap begins to reopen.

For the 2000 birth cohort, the composition of the lifetime gap reveals a distinct structure. Around 59.6% stems from labor income, a comparatively low share given Switzerland's high earnings levels. Investment income accounts for 5.1%, a relatively elevated contribution in international comparison, reflecting the importance of capital income. Pension income explains 35.3% of the total gap, also above average. In Switzerland, therefore, lifetime inequality is not only a labor-market story; it is strongly amplified by capital accumulation and retirement income dynamics.

Labor market fundamentals are not the main weakness. Participation rates are already high by international standards (80.8% for women versus 87.4% for men) and are projected to converge further, particularly in older age groups where gaps remain today. Hourly wages are also narrowing. Despite a current wage gap of around 15%, projections suggest that women's hourly pay in part-time employment could overtake men's in the 2060s, with parity in full-time employment reached by 2100. The issue is therefore again working time.

Switzerland combines high employment with exceptionally high part-time rates. Among women aged 25-49, 60.5% work part-time compared with 19.5% of men; among 50-59-year-olds, the figures are 68.7% versus 17.5%. Even by 2100, the structural gap remains pronounced: female part-time shares stay above 50% in prime working age, while male rates remain near 20%. Across almost all age groups, women work two to three hours fewer per week, a persistent work-intensity gap that helps explain the lifetime income divide.

Shifting from the cohort-based lifetime view to an annual outlook illustrates the dynamics. The annual labor income gap declined from 41.4% in 2000 to 33.8% in 2026 and is projected to fall further to around 27% by 2050. Thereafter, however, progress effectively stalls. The initial narrowing reflects wage convergence and rising participation; the later plateau reflects enduring differences in working hours.

Switzerland's trajectory highlights a structural paradox: high employment and strong wage convergence are not sufficient to ensure lifetime income equality. Without a significant shift in working-time patterns – particularly among women in prime and older age groups – gender income gaps may not only persist but widen again across generations.

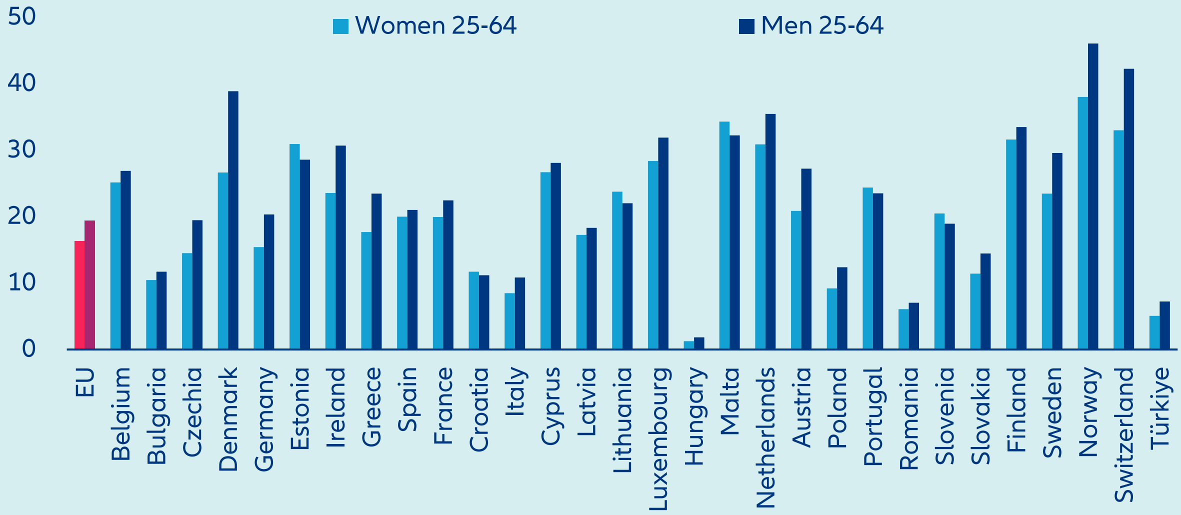
Box 3: How artificial intelligence could affect gender gaps

The lifecycle projections in this study are based on current labor market structures and therefore do not fully capture disruptive technological shifts such as artificial intelligence (AI). However, AI is likely to reshape employment, productivity and earnings trajectories in ways that could either narrow or widen gender gaps.

AI adoption today is already widespread, but uneven. In 2025, 32.7% of people aged 16-74 in the EU reported using generative AI tools, though primarily for personal rather than professional purposes.[18] Usage at work remains lower and is gendered. Among working-age adults (25-64) in the EU, 19.4% of men use generative AI tools for professional purposes compared with 16.4% of women (Figure 13). This implies a gender gap of 3.0pps in professional use (≈ 16% relative gap).

In large economies such as Germany, the gap is wider: 20.4% of men versus 15.4% of women use AI at work, a difference of 5.0pps (around 24% in relative terms). Italy shows a similar relative divide despite lower overall use (10.9% vs. 8.5%). Denmark combines high adoption with a pronounced gap (38.9% vs. 26.7%). By contrast, Spain is close to parity (21.1% vs. 20.0%), and in Estonia women slightly outpace men (31.0% vs. 28.6%). These latter cases, however, remain exceptions rather than the rule.

Figure 13: AI adoption at work gender gaps, GenAI usage in %



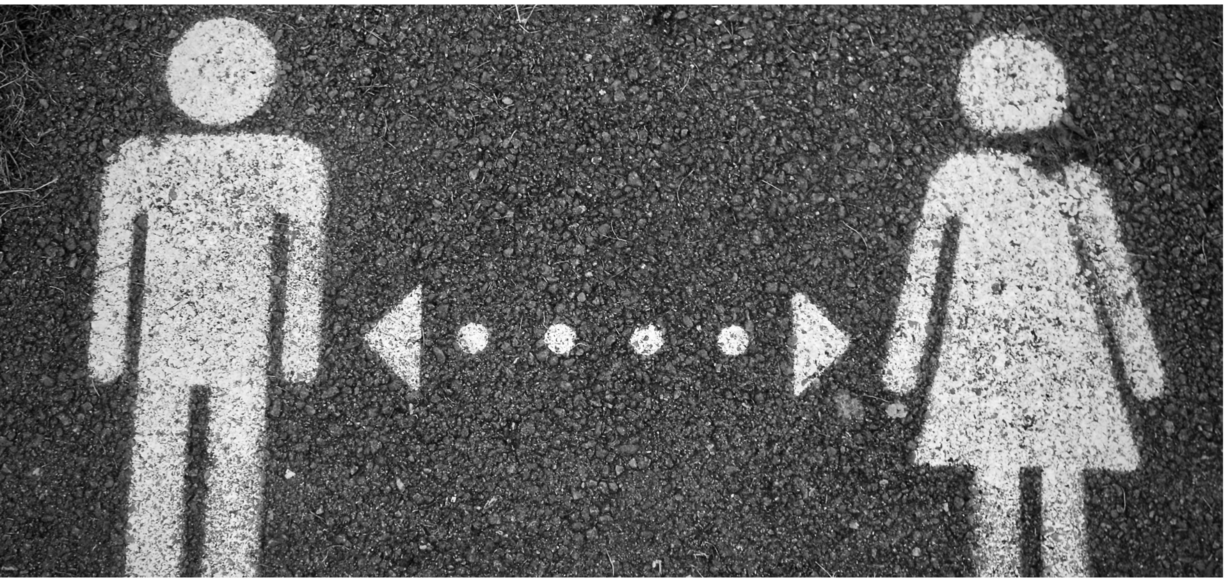
Source: Eurostat (2025), Allianz Research.

The relevance of these gaps goes beyond technology use itself. As AI becomes a general-purpose productivity tool, differences in professional adoption today may translate into differences in future skill accumulation, performance evaluations and wage growth. Early adopters are more likely to learn how to integrate AI into workflows, redesign tasks and move into AI-complementary roles. A persistent 3-5pps gap in usage at work may therefore compound over time.

At the same time, the impact of AI will depend on how it is deployed across occupations. Research shows that AI can function either as an automation technology that substitutes for routine tasks or as a labor-augmenting technology that enhances human expertise.[19] Given existing occupational segregation, women may be more exposed to automation risks in some clerical and administrative roles, while remaining underrepresented in high-growth, AI-intensive technical occupations. In that case, differential exposure and differential take-up would reinforce each other.

Whether AI ultimately narrows or widens gender gaps will therefore hinge on two factors: closing adoption gaps and steering deployment toward augmentation rather than substitution. Without both, technological change risks amplifying, rather than correcting, existing gender inequalities.

[18] Source: Eurostat (2025), Individuals - Use of Generative AI Tools, https://doi.org/10.2908/ISOC_AI_IAIU.
[19] Source: Acemoglu, Daron, David Autor, and Simon Johnson, „Building Pro-Worker Artificial Intelligence," NBER Working Paper 34854 (2026), https://doi.org/10.3386/w34854.



What can be done to close gender gaps faster?

Gender gaps narrow when work intensity, career continuity and capital accumulation align. The remaining disparities are no longer primarily about access to employment, but about hours worked, career progression, savings behavior and retirement design. Labor market incentives matter. Women who work more hours, remain continuously employed and move into higher-paying roles accumulate substantially higher lifetime income. Reducing structural drivers of involuntary part-time work and ensuring that flexibility does not come at the expense of advancement are central to closing earnings gaps.

To close the labor income gender gap, childcare availability and parental leave design are decisive. Where care responsibilities are shared more evenly, employment trajectories diverge less. Systems that discourage second earners or concentrate leave uptake among mothers entrench lifetime inequality.

At the same time, women must not fall behind in emerging growth areas on the labor market such as AI. Closing gaps in AI skills, adoption and representation will be critical to ensure that women fully participate in future productivity gains and technological transformation.

Regarding the investment income gender gap, Capital accumulation is the another lever. Small differences in financial literacy, investment participation and portfolio allocation compound over decades. Strengthening financial capability and encouraging long-term investment can materially reduce wealth and pension gaps.

Finally, pension systems determine whether labor market inequalities are amplified or compressed. Designs that include minimum floors and care credits mitigate lifetime disparities. Systems tightly linked to uninterrupted full-time careers tend to reproduce them.

The implications differ across cohorts. Women born in 1975 can still influence retirement outcomes through later-career earnings and investing in high-return assets. Women born in 2000 face critical decisions about career continuity and investment participation, as they approach the typical age. For women born in 2025, the institutional environment they are born into will shape early opportunities, but most will depend on their own choices in education, work, and finance.

Closing gender gaps is therefore not a single reform agenda. It requires alignment between labor markets, family policy, financial behavior and social norms. Without such alignment, convergence will remain slow – and lifetime income gaps will persist across generations.

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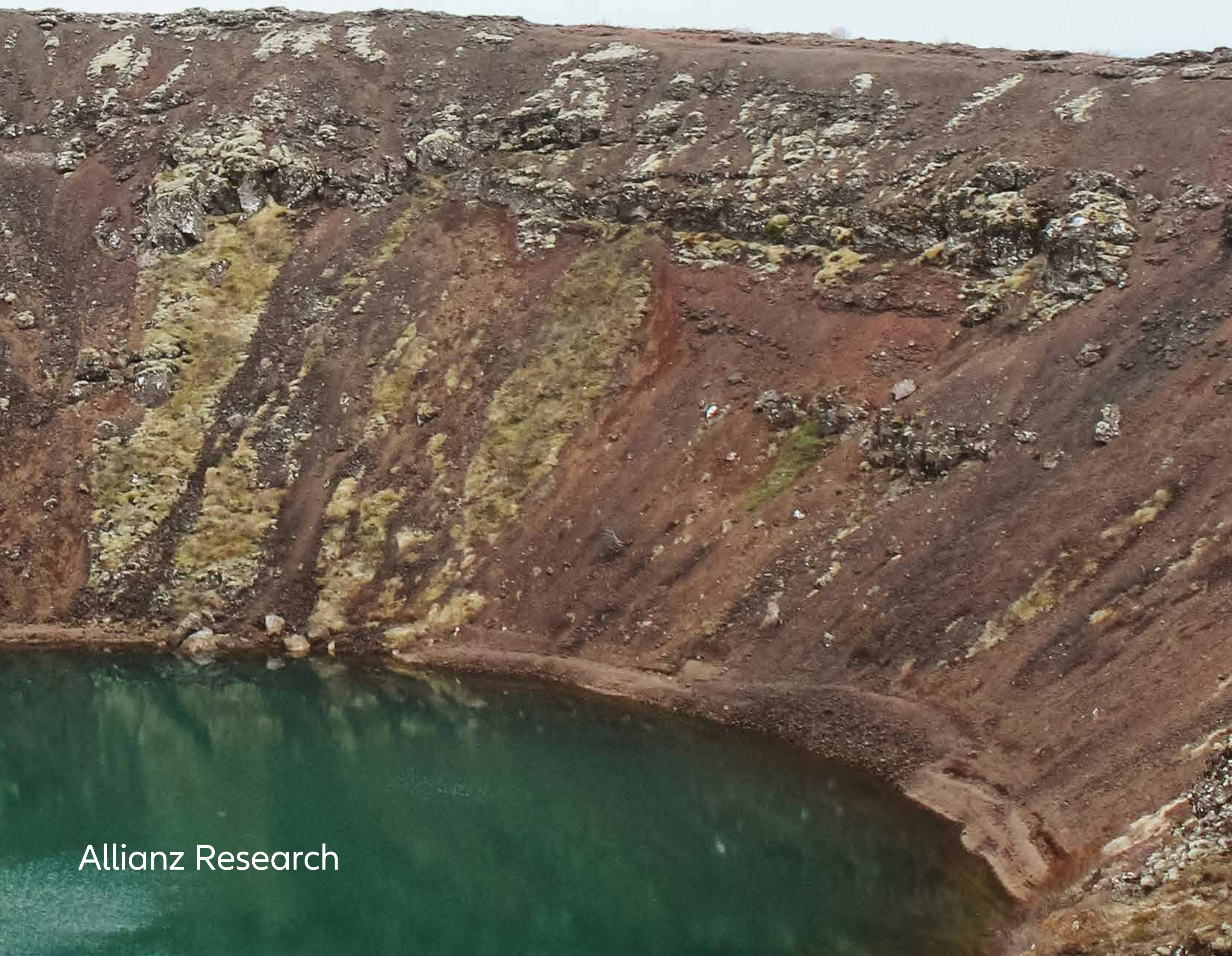
Mining for the future: Addressing liabilities and unlocking sustainable transition opportunities

25 February 2026



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Content

25 February 2026

Executive Summary

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- **A twin transition is driving a surge in mineral demand as decarbonization and digitalization are reshaping the industrial base simultaneously.** Energy systems are shifting toward electrification, renewables, storage and expanded grids, while artificial intelligence is driving heavy investments in data centers and computing infrastructure. Demand for critical minerals is set to rise accordingly. The International Energy Agency projects that by 2040 lithium demand could increase fivefold; graphite and nickel may double; cobalt and rare earth elements could rise by 50–60%; and copper demand by around 30%. Meeting this surge will require not only new mines, but expanded processing capacity, infrastructure and skilled labor. Meanwhile data-center electricity demand alone is expected to more than double by 2030, reinforcing the case for grid investment and additional materials.

- **Supply is struggling to respond.** Although a more circular economy may eventually ease reliance on primary extraction, supply chains for key transition minerals remain slow to scale. New projects take years to permit and finance, and face rising environmental and social standards. The pace at which mining and refining capacity can grow is therefore a central constraint; failure to scale responsibly risks prolonging fossil-fuel dependence.

- **Environmental pressures reinforce these constraints.** Mining contributes around 2–4% of global GDP, supports millions of livelihoods and accounts for approximately 4–7% of global greenhouse gas emissions. Although less significant than agriculture or urban expansion in driving forest loss, mining directly deforested nearly 20,000 km^2 between 2001 and 2023, generating roughly 0.75 Pg of CO_2 - slightly more than Germany's annual emissions in 2024. Stronger operational practices, rehabilitation and credible compensation mechanisms such as biodiversity offsets and reforestation could reduce these impacts. While debated, effective implementation could help limit net forest loss as mineral demand rises.

- **Bridging this gap will require substantial capital.** We estimate cumulative investment needs of around USD 1.1trn to 2040 across mining, processing and circularity. Sustainability has become a binding production constraint: permitting, insurance and community consent now shape capacity as directly as technical factors. Company disclosures suggest roughly USD 450bn may be required for decarbonization, tailings management, recycling and environmental controls. After accounting for overlap with the roughly USD 800bn needed to develop new supply, total capital requirements remain close to USD 1.1trn.

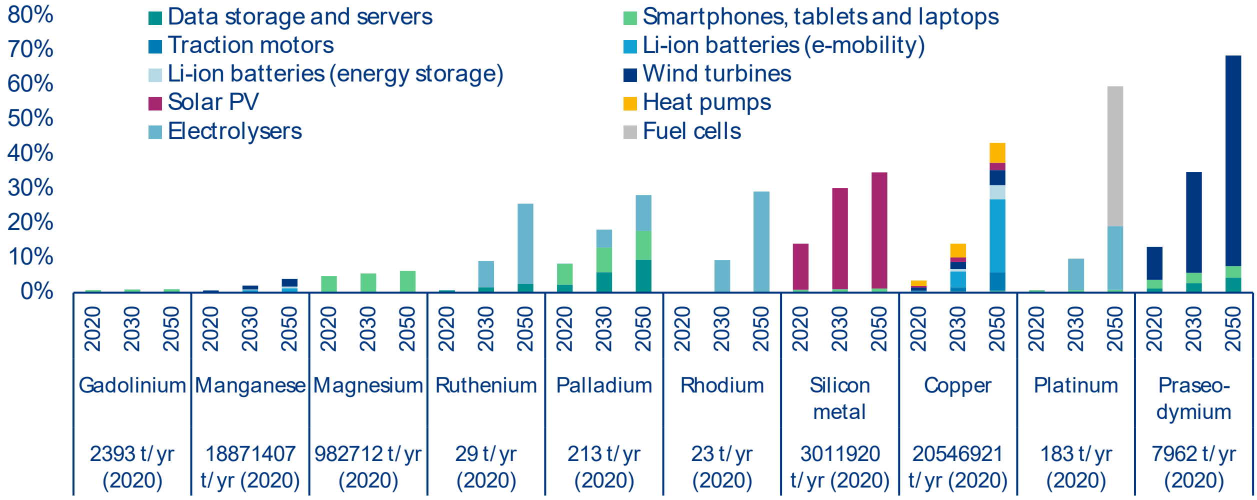
- **Sustainable mining is integral to the transition.** Underinvestment risks leaving decommissioning liabilities to society and raising project risk. This is where improved sustainable practices and operations can be a win-win. Clear, early expectations from regulators, communities and customers would allow costs to be priced in upfront, streamline engagement and improve the credibility of approvals.

Why the transition depends on mining done right

The twin transitions in energy and digital systems are driving a structural surge in material demand. The energy system is being rebuilt around electrified end-use, renewables, storage and expanded grids; in parallel, AI is accelerating investment in data centers, network equipment and high-performance computing. Both of these shifts are often framed as "clean" or "virtual", but ultimately they are physical. They require large volumes of metals and minerals – and the industrial capacity to refine and process them into usable inputs to manufacturing. As a result, global mineral extraction has surged to unprecedented levels in the 21st century. Krausmann et al. (2018)[1] provide a landmark account of this shift, documenting a 23-fold increase in global socioeconomic material stocks, from just 6 gigatons (Gt) in 1900 to over 1,300 Gt by 2015. According to the IEA, lithium demand is set to jump by around fivefold by 2040, graphite and nickel will roughly double, cobalt and rare earth elements will rise by around 50-60% and copper will increase by around 30%.[2] These multipliers matter because scaling supply is not just about opening new mines; it also requires increasing processing capacity, expanding transport infrastructure and growing a skilled workforce. AI introduces an additional demand vector by accelerating the build-out of data centers and associated power infrastructure. In the IEA's base case scenario, global data center electricity demand rises from about 415 TWh in 2024 (about 1.5% of global electricity consumption) to about 945 TWh in 2030.[3] Meeting this demand implies more grid investment and, indirectly, additional demand for construction materials and critical metals (e.g. copper for cabling and transformers). From a critical-minerals perspective, rising data-center load matters for two reasons. First, it reinforces the need for faster grid expansion and resilience investments, which are copper- and aluminum-intensive. Second, it accelerates demand for hardware supply chains (servers, networking, semiconductors) that rely on a range of metals and high-purity inputs. The result is an additional layer of demand uncertainty on top of already steep growth from EVs and renewables.

[1] From resource extraction to outflows of wastes and emissions: The socioeconomic metabolism of the global economy, 1900–2015 - ScienceDirect
[2] International Energy Agency (IEA). 2025. Global Critical Minerals Outlook 2025. Paris: IEA.
[3] International Energy Agency (IEA). 2025. Energy and AI. Paris: IEA

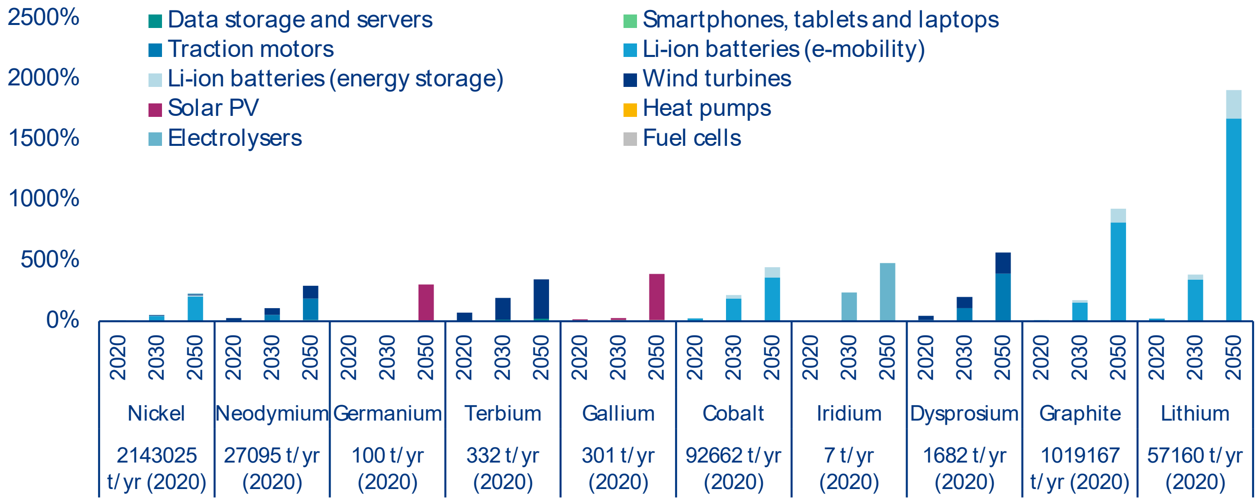
Figure 1: Global raw material demand by transition technology (in % of 2020 total production): EU strategic materials (high demand scenario), shares between 1% and 100% in 2050



Sources: JRC[4]; Allianz Research

For several Strategic Raw Materials (SRMs), projected demand growth is large enough to absorb most of today's global output - and in some cases exceed it. SRMs are a subset of Critical Raw Materials (CRMs) essential to strategic technologies underpinning the green and digital transition, as well as defense and aerospace applications[5]. Figures 1 and 2 translate technology deployment into annual raw material demand relative to current (2020) global production levels. When projected demand approaches or surpasses 100% of today's output, supply chains must expand rapidly—or demand must adjust through efficiency gains, substitution, recycling or higher prices. The challenge is particularly acute for materials produced in only a handful of countries, or primarily as by-products, such as gallium and germanium.[6]

Figure 2: Global raw material demand by transition technology (in % of 2020 total production): EU strategic materials (high demand scenario), shares above 100% in 2050.



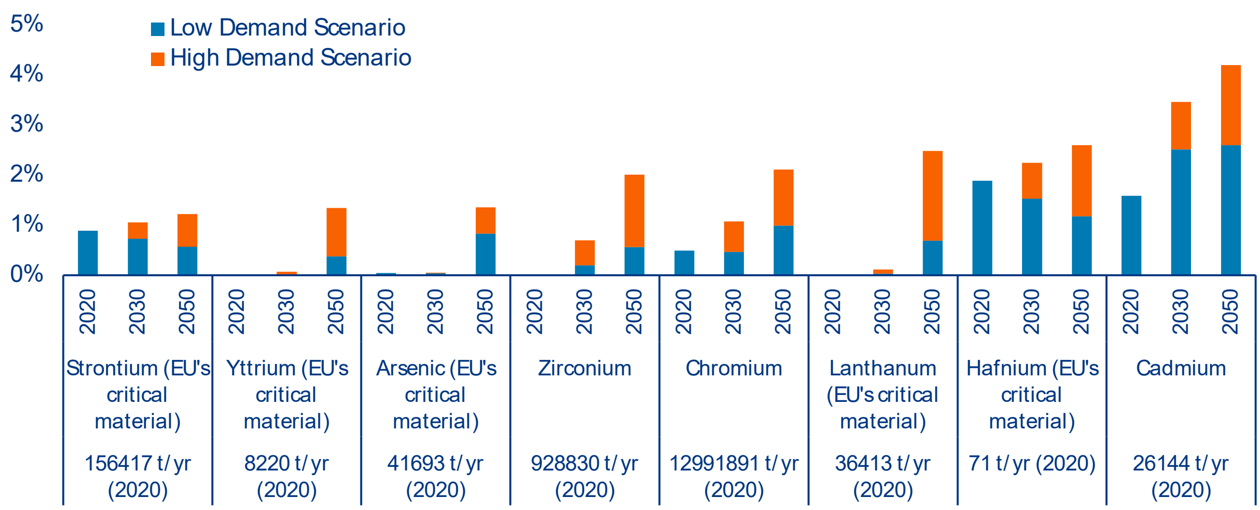
Sources: JRC[7]; Allianz Research (Note: Germanium demand estimates for solar PV are from Carrara et al., 2020) [8]

[4] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission.
[5] European Commission. 2025. "RESourceEU Action Plan: Accelerating our critical raw materials strategy to adapt to a new reality." Communication from the Commission. COM(2025) 945 final, December 3, 2025.
[6] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission.
[7] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission
[8] Carrara, S., et al. 2020. "Raw Materials Demand for Wind and Solar PV Technologies in the Transition Towards a Decarbonised Energy System." EUR 30091 EN. Joint Research Centre, European Commission. Accessed February 1, 2026.

Focusing policy on a narrow list of strategic materials risks overlooking two additional bottlenecks. First, some `ordinary' materials—such as steel, concrete, copper and aluminum—are required in very large volumes. Even in deep and diversified markets, rapid scale-up can strain production capacity, raise prices and create permitting or energy-supply constraints. Second, certain minor metals are needed only in small quantities but at very high purity, and are often produced as by-products. This makes supply slow to adjust and vulnerable to trade disruptions. Figures 3-5 illustrate projected demand for these materials relative to 2020 global production. Two patterns stand out. First, some non-strategic materials show large percentage increases but remain a small share of today's production because the underlying markets are already very large (e.g. steel and aluminum). Second, a subset of non-strategic but critical inputs – including indium, tellurium and phosphorus – can, by 2050, reach demands that exceed current production, largely driven by specific technologies such as photovoltaics and batteries. The implication for industrial planning is clear: materials need not be formally classified as 'strategic' to become binding constraints. Rapid demand growth, limited refining capacity or by-product dependence can generate price volatility and project delays.[10]

Phosphorus illustrates how transition demand can spill into essential commodity markets. It is used in batteries' lithium iron phosphate (LFP) cathodes, which are increasingly deployed because they avoid nickel and cobalt and offer cost and safety advantages. As EV and stationary storage markets expand and LFP gains market share, the battery sector can become a meaningful additional source of phosphate demand on top of its dominant use in fertilizers.[11] From a risk perspective, this creates a ‚competition for molecules' problem: a material that is essential for food security also becomes relevant for electrification. Supply is not only a mining issue; it also depends on chemical conversion and on the ability to manage environmental impacts from phosphate mining and processing.
For Europe, this strengthens the case for recycling phosphorus-containing waste streams, and for technology innovation that reduces phosphate intensity where possible.[12]

Figure 3: Global raw material demand by transition technology relative to total 2020 production: EU non-strategic materials, shares below 4.5% in 2050



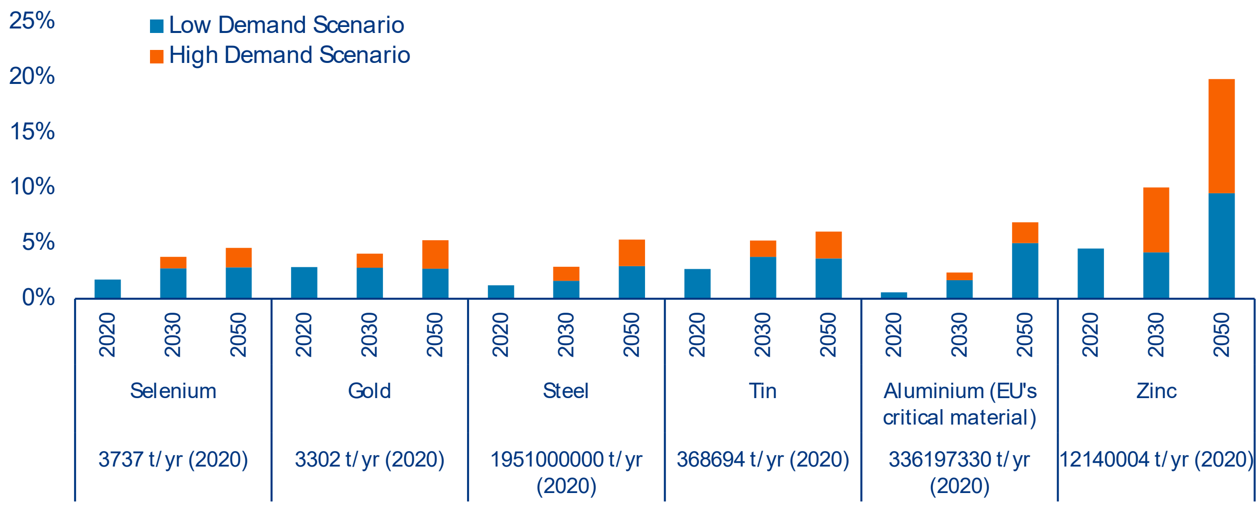
Sources: JRC[13]; Allianz Research

[10] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission.
[11] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission.
[12] European Commission. 2025. "RESourceEU Action Plan: Accelerating our critical raw materials strategy to adapt to a new reality." Communication from the Commission. COM(2025) 945 final, December 3, 2025.
[13] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission

Rising material demands create a practical tension at the heart of the transition. On the one side, the transition offers the opportunity to move to a circular economy, one where the materials that we mine enter near infinite loops of use and re-use. In this scenario, once sufficient materials are within human control, the overall quantity of mining precipitously drops – especially of fossil fuels. However, on the other side, supply chains for several transition-relevant minerals are slow to scale: New mines and refineries currently take years of planning, studies, permitting, financing and construction. Further, projects rightly face rising performance expectations around water use, biodiversity impacts, community benefits, labor conditions, tailings safety and end-of-life liabilities. The pace at which extraction and refining capacity can be expanded is a primary bottleneck for building a decarbonized economy. If we do not get mining right to unlock the full potential of our industrial might and know-how, we risk prolonging our fossil fuel economies well past the time horizon that science and insurers have indicated will be catastrophic.[14]

What will it take to scale the "right" projects: those that are deliverable on time, financeable at scale, insurable against severe loss and seen as legitimate in the eyes of communities and regulators? Mining's legacy can often translate into lower societal trust; the challenge is therefore not only to do more and do it faster, but to keep operations safe, beneficial to communities and with acceptable levels of remediable environmental impact. It argues that "faster" only works when it is "faster because better": history teaches us that acceleration without stronger performance and governance backfires through delays, accidents, litigation and long- duration liabilities. Looking ahead, competition for batteries, grids, magnets and semiconductor inputs will intensify; midstream capacity, permitting and acquiring social license will often be the binding constraints.

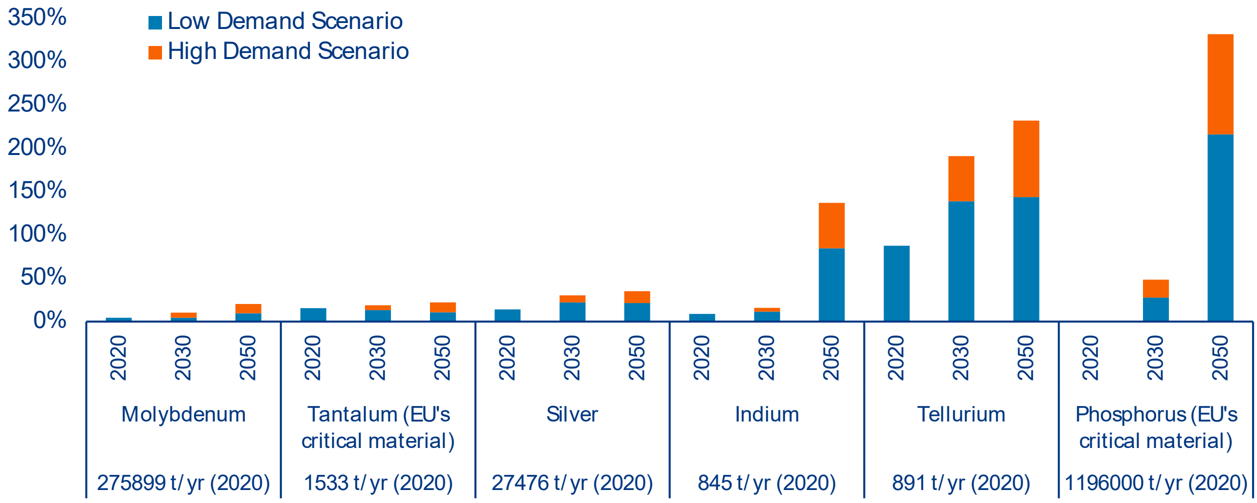
Figure 4: Global raw material demand by transition technology relative to total 2020 production: EU non-strategic materials, shares between 4.5% and 20% in 2050



Sources: JRC[15]; Allianz Research

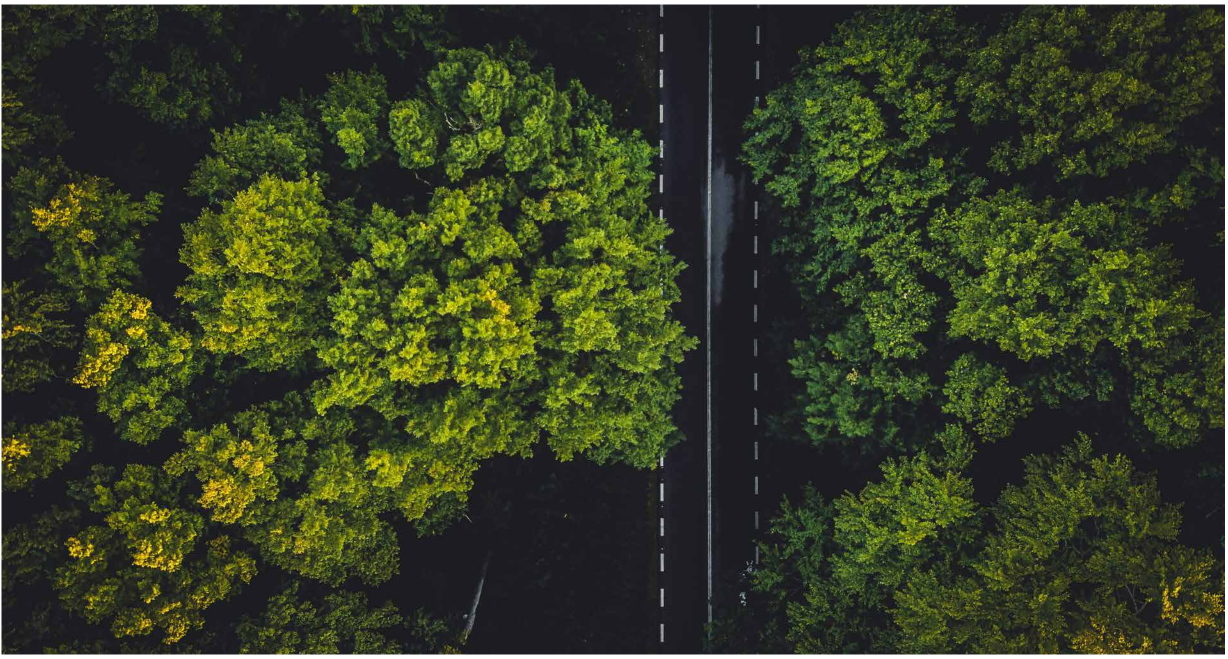
[14] IPCC (2023). CLIMATE CHANGE 2023 - Synthesis Report. Summary for Policymakers.
[15] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission.

Figure 5: Global raw material demand by transition technology relative to total 2020 production: EU non-strategic materials, shares above 20% in 2050



Sources: JRC[16]; Allianz Research

[16] Carrara, S., et al. 2023. "Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study." Joint Research Centre, European Commission.



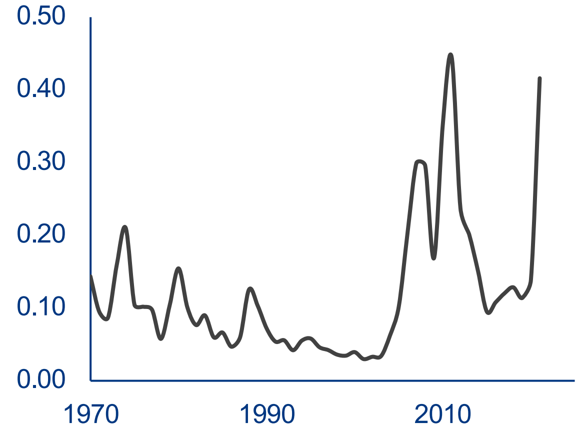
The sustainability bottleneck: The social license to deplete, environmental limits and project disruption risk

Material stocks - buildings, transport infrastructure, machinery and other long-lived assets - embody the cumulative footprint of industrialization and urbanization, particularly in high- and middle-income economies. Their construction and maintenance now consume nearly half of all raw materials extracted annually, underscoring the structural link between economic development and material throughput. Despite the rise of service-based economies, there has been no absolute decoupling of growth from resource use or environmental pressure. Expanding built environments lock societies into long-term flows of energy consumption, emissions and waste, raising questions about the sustainability of prevailing development models.

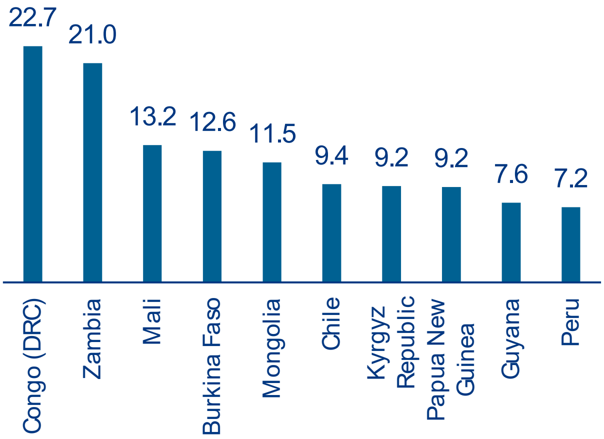
Figure 6a illustrates this trajectory, showing that mineral extraction as a share of global Gross National Income (GNI) more than tripled between 2000 and 2021, reaching its highest level in over five decades. However, while mineral depletion is rising globally, it is highly concentrated geographically. Figure 6b shows that in 2021, several resource-rich countries, such as the Democratic Republic of Congo (22.7%), Zambia (21.0%), and Mali (13.2%), saw depletion levels exceeding 10% of their GNI, underscoring how extractive activity dominates their economies. In contrast, Figure 6c reveals that most OECD countries exhibit depletion rates below 1%, highlighting a stark imbalance: the material basis of global development is disproportionately shouldered by a small group of lower-income, resource-exporting nations. This growing dependence on extractive rents places these countries at risk of a resource trap, where short-term economic gains come at the cost of long-term sustainability and structural vulnerability.

Figure 6: Mineral depletion

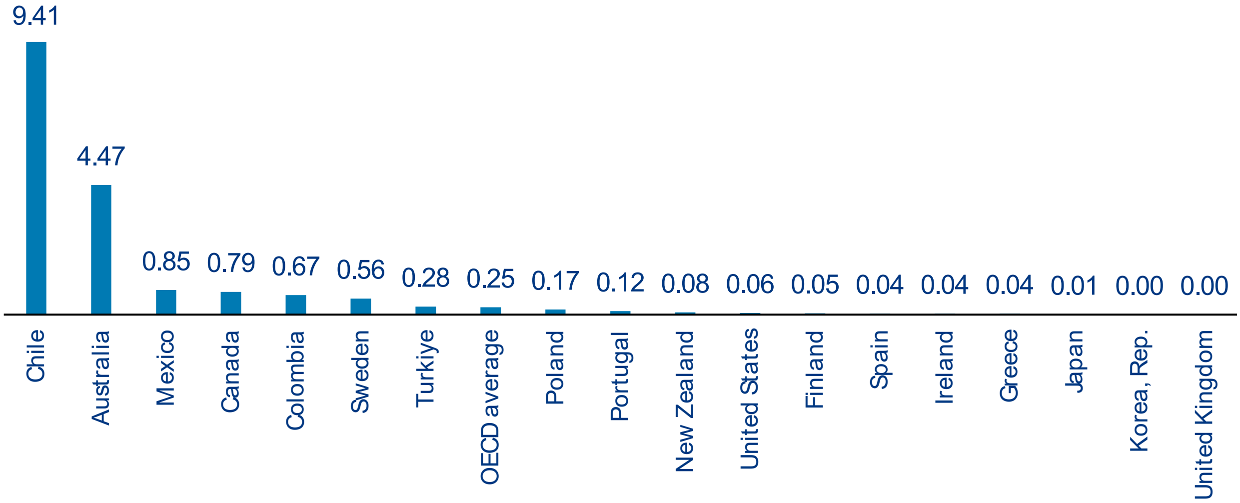
a) Global development (1970 – 2021) as a share of Gross National Income (GNI, %)



b) Top 10 countries with highest depletion rate in % in 2021



c) Depletion rate in OECD countries in % in 2021



Sources: World Bank Data, Allianz Research

The unequal distribution of mineral depletion as a share of national income reveals deep patterns of environmental injustice in the global resource economy. While raw material extraction remains essential to infrastructure, energy systems, and global trade, the environmental burdens of these activities fall disproportionately on resource-rich but economically vulnerable countries. According to Arendt et al. (2022)[17], the total global environmental costs from these activities are estimated to range between EUR0.4trn and EUR5trn annually, depending on the valuation method used.

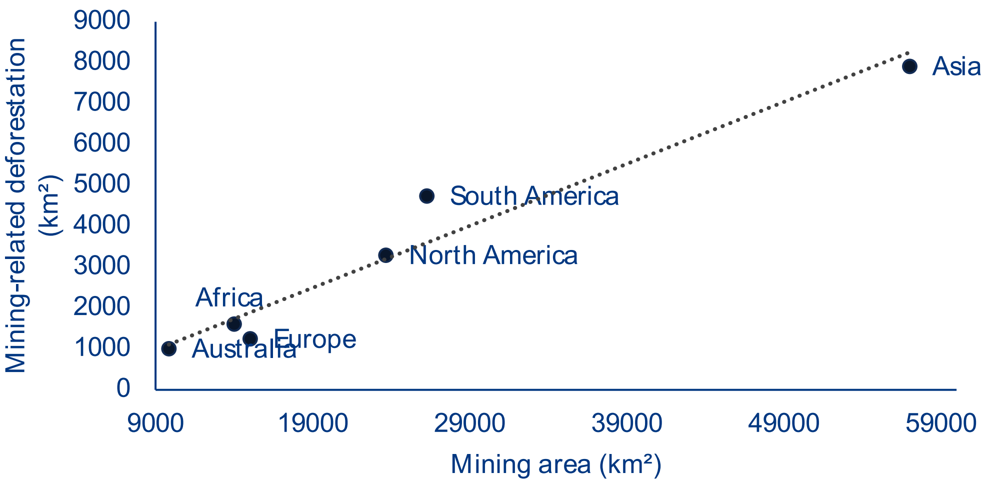
The analysis considers over 190 nations and shows that in many lower-income countries, domestic environmental costs, such as air pollution, acidification and land degradation, often outweigh the GDP benefits generated by mining and processing. For instance, Rwanda exhibits the highest ratio of environmental damage to economic gain, driven by informal tantalum mining with minimal local value retention. In monetary terms, several countries, among them Gabon, Ghana, Madagascar and Afghanistan, incur environmental costs that exceed their mining-related GDP gains, indicating a net-negative

[17] The global environmental costs of mining and processing abiotic raw materials and their geographic distribution - ScienceDirect

welfare impact from extraction. Conversely, high-income economies like Germany, Japan and South Korea benefit from favorable cost–benefit ratios by importing raw materials and externalizing the most damaging stages of extraction. Countries such as China and India, while significant producers, also show low benefit-to-cost ratios, particularly due to high domestic pollution from iron, steel and aluminum production. For example, China's mining sector accounts for a disproportionate share of global particulate matter and greenhouse gas emissions per euro of GDP gain. These disparities make clear that current global supply chains structurally externalize environmental harm, concentrating damage where economic returns are lowest. As demand for critical minerals intensifies, especially for the green and digital transitions, the risk is not only ecological but developmental, trapping producer countries in high-extraction, low-benefit paths, i.e., the so-called "commodity trap".

One of the major social impacts of mining stems from the environmental harm associated with the sector, such as deforestation. While mining contributes around 2–4% of global GDP, supports millions of livelihoods and accounts for approximately 4–7% of global greenhouse gas emissions, it also is a significant driver of forest loss, although well behind leading causes of agriculture, infrastructure and urban expansion. By overlaying high-resolution global forest-loss datasets with mining areas over the period 2001–2023, Zhang et al. (2025)[18] find that 19,765 km^2 of forested land were directly deforested by mining activities. This deforestation resulted in gross carbon emissions of 0.75 Pg CO_2[19], slightly higher than annual emissions of Germany in 2024, with unrecorded mining activities accounting for 66.5% of total mining-related forest loss. Spatially, Asia experienced the largest absolute deforestation in mining areas, followed by South America, North America, Africa, Europe and Australia (Figure 7). Overall, 175 countries experienced mining-related deforestation during this period, with tropical forests disproportionately affected, emerging as global hotspots of carbon emissions.

Figure 7: The mining areas and mining-related deforestation across different regions (cumulative 2001 – 2023)



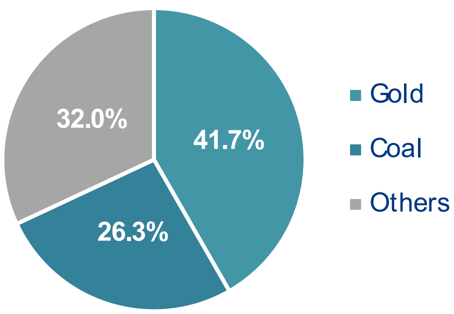
Sources: Zhang et al. (2025), Allianz Research

[18] Overlooked deforestation from global mining activities in the 21st century | Nature Communications
[19] 0.75 Pg CO_2 = 0.75 Gt CO_2 = 750 million tonnes of CO_2

An analysis of mining-related deforestation by primary commodity reveals a strong dominance of gold and coal extraction at the global scale (Figure 8). For instance, gold mining accounts for 41.7% of total mining-driven deforestation, followed by coal mining with 26.3%, corresponding to carbon emissions of 0.39 Pg CO_2 and 0.18 Pg CO_2, respectively. Gold-related deforestation is highly concentrated geographically, with Brazil, Russia, Indonesia, Peru, Ghana and Suriname emerging as the largest contributors. Here unrecorded mining activities account for over 73% of gold mining-related deforestation between 2001 and 2023, with particularly high shares observed in Brazil, Peru, Myanmar and Guyana. Coal mining also exhibits a substantial unrecorded component, responsible for nearly half of coal-related deforestation, with Indonesia alone contributing close to half of the global total. Temporally, deforestation linked to coal and other commodities increased until 2012 before declining thereafter, reflecting reduced coal demand amid the global energy transition. The increase in mining to drive the transition will, ultimately, lead to a reduced impact of coal mining as operations wind down.

While the shift toward renewable energy has contributed to declining coal-related deforestation, it has simultaneously increased demand for mineral inputs critical to renewable technologies and infrastructure. When categorizing mining activities by their relevance to renewable versus non-renewable energy production, deforestation has increasingly shifted toward regions extracting minerals associated with renewable energy supply chains. After 2012, nearly three-quarters of mining-related deforestation occurred in areas producing minerals used for renewable energy applications, underscoring the risk that accelerating energy transitions may intensify deforestation pressures if mineral supply chains remain weakly regulated.

Figure 8: Deforestation-related to different mining commodities (%, cumulative 2001 – 2023)



Sources: Zhang et al. (2025), Allianz Research

Importantly, the environmental footprint of mining is increasingly being addressed through compensation mechanisms. In several jurisdictions and corporate strategies, mining expansion is now accompanied by commitments to reforest or conserve ecosystems in other locations, following “no net loss” or “net gain” biodiversity frameworks. These approaches rely on biodiversity offsets, large-scale reforestation programs or conservation finance aimed at balancing deforestation induced by extractive activities. While such compensation schemes remain debated. If credibly implemented, these mechanisms could partially decouple continued mineral extraction from net forest loss, especially in the context of rising demand for transition minerals.

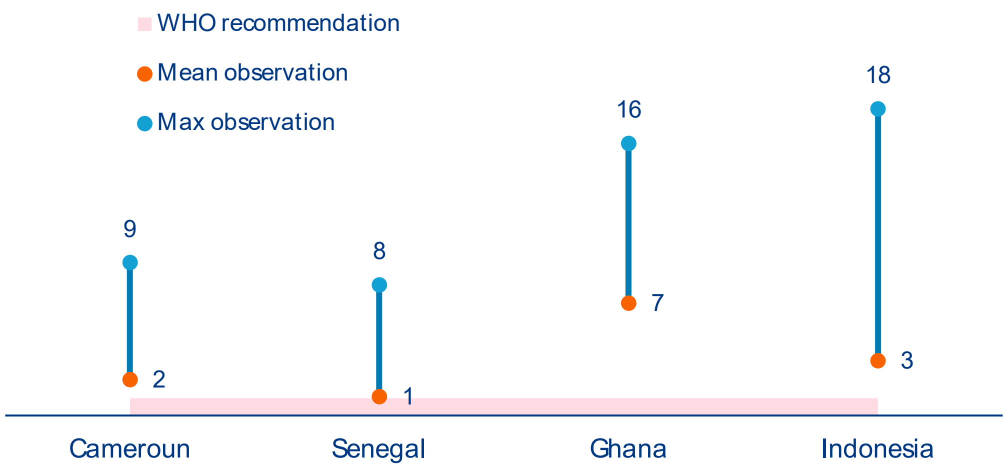
Beyond its environmental and climate externalities, mining is frequently associated with concerns over labor conditions and worker safety[20]. However, any assessment of labor outcomes in the sector must clearly differentiate between large-scale industrial mining (LSM) and artisanal and small-scale mining (ASM), as the nature of risks, governance and employment relations differs substantially between the two. In industrial mining, rising levels of mechanization, stronger regulatory oversight and the gradual adoption of occupational safety and health (OSH) standards have contributed to measurable improvements in workplace safety over recent decades. Fatal accident rates, while still non-negligible, have declined in many jurisdictions, reflecting better risk-management practices. The increasing reliance on automated and remotely operated machinery has also reduced direct worker exposure to physical and chemical hazards, as many high-risk tasks are now performed from control rooms rather than on site. By contrast, the labor challenges most commonly highlighted in public debates, such as unsafe working conditions, and exposure to chemical substances (e.g., mercury, cyanide etc.), are far more prevalent in ASM, where informality, limited (intellectual and economic) capital and weak institutional oversight remain dominant.

Despite its largely informal nature, ASM plays a central role in global mineral value chains. The sector represents one of the largest sources of employment in the mining industry, providing livelihoods to approximately 44.75mn workers across more than 80 countries worldwide. In many ASMs children, sometimes as young as ten years old, continue to participate in mining-related activities, highlighting persistent social and regulatory challenges (ILO 2024). ASM also contributes a substantial share of global mineral production. Estimates indicate that the sector accounts for 15–20% of total global mineral output, including around 80% of sapphires, 20% of gold and 20% of diamonds. Beyond precious stones and metals, ASM is also an important supplier of minerals that are strategically critical for modern manufacturing and electronics, notably tantalum and tin.

These structural characteristics come with significant occupational health risks, as illustrated in Figure 9, which reports mercury concentrations in the hair of mining workers across selected ASM-producing countries. Measured exposure levels substantially exceed the WHO recommended maximum of 1 mg/kg in all cases, with particularly elevated mean and maximum values observed in Ghana and Indonesia. In these contexts, recorded mercury concentrations reach levels an order of magnitude above international health thresholds, underscoring the widespread and chronic nature of toxic exposure in ASM settings.

[20] Chemical exposures in mining | International Labour Organization

Figure 9: Mercury hair levels for mining workers (mg/kg)

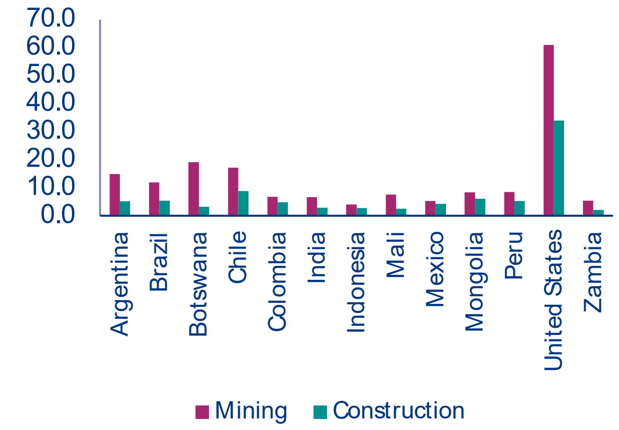


Sources: Harianja et al. (2020), Opoku et al. (2019), Birane et al. (2014), Allianz Research

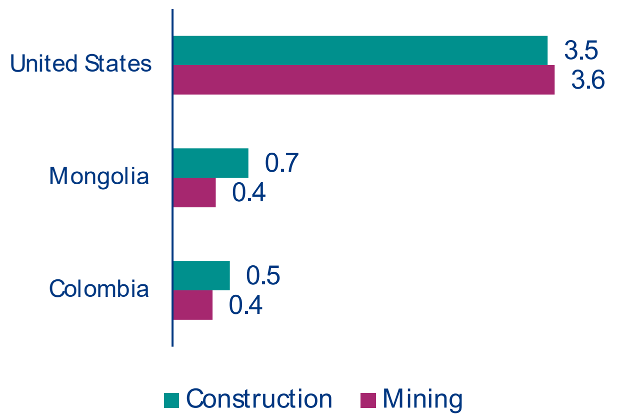
Figure 10 illustrates the contrast in mining labor outcomes between developed and developing economies. Across the full country sample, average hourly wages in mining exceed those observed in several other sectors, including construction (Figure 10a). This pattern is consistent over time and remains robust when mining wages are compared with those in agriculture, services and public administration, suggesting a persistent wage premium associated with employment in the mining sector. This apparent advantage, however, masks an important structural trade-off. Higher remuneration in mining partly reflects a risk premium, compensating workers for the elevated probability of physical harm and injury. Unfortunately, although the goal should always be for injuries to be zero, this physical harm can also be life altering severe or fatal. To determine the extent of this risk premium, Figure 10b presents wages adjusted by the number of fatalities, offering a risk-weighted comparison across sectors. Once fatal risk is incorporated, the relative attractiveness of mining wages declines markedly in several developing economies. In countries such as Colombia and Mongolia, mining wages fall below those observed in construction, indicating that monetary compensation does not fully offset the level of occupational risk borne by workers. By contrast, in advanced economies such as the US, mining retains a modest wage advantage even after adjusting for fatality risk, suggesting more effective risk management, stronger enforcement of safety standards or higher compensation for residual risk.

Figure 10: Compensation in mining compared to construction

a) Average hourly wage in 2023 (USD, 2021 PPP)



b) Wage per unit of fatal risk in 2023

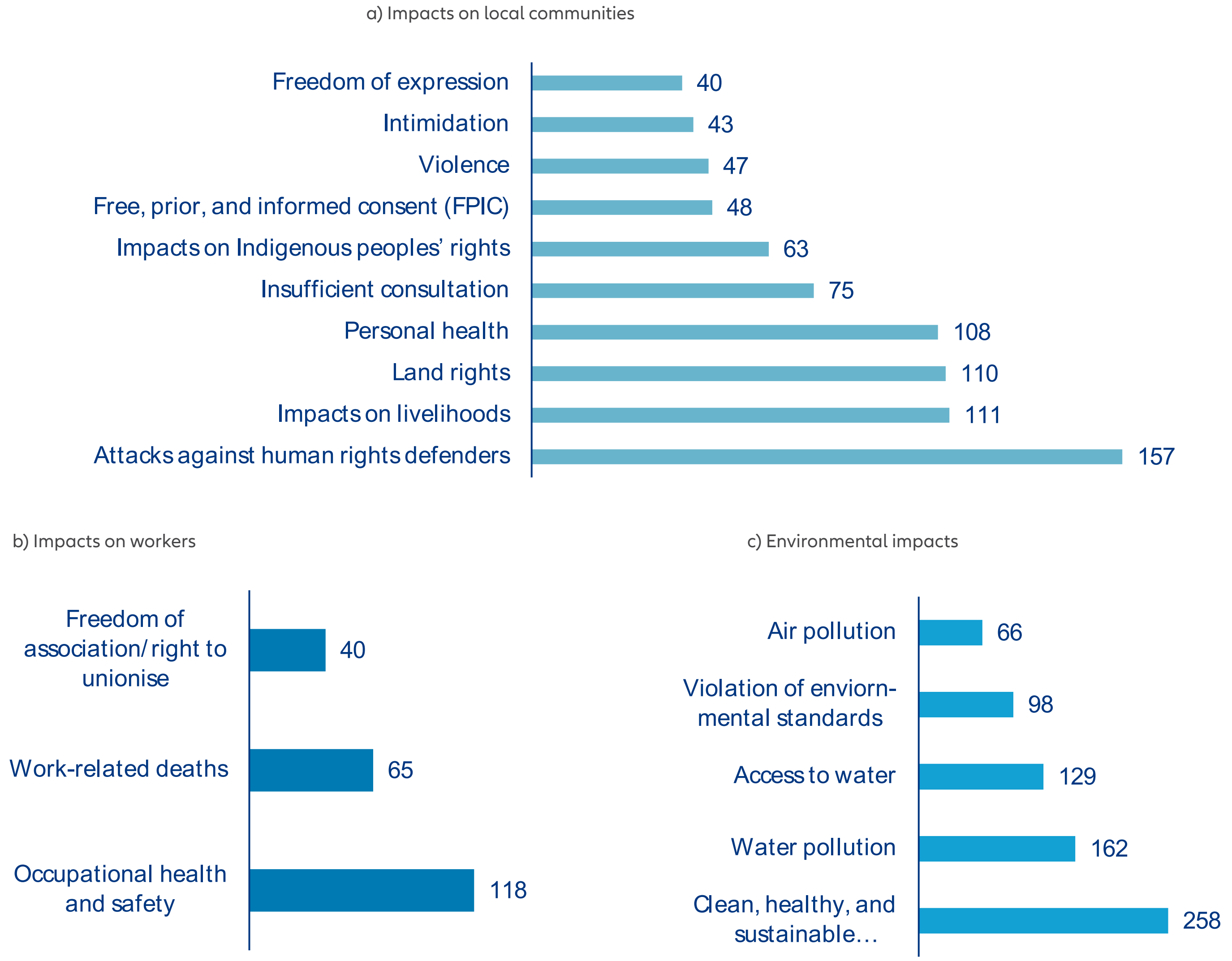


Sources: ILOSTAT, Allianz Research

The mining activity has been accompanied by several reported incidents, as documented by the Transition Minerals Tracker between 2010 and 2024[21]. Figure 11 provides a comprehensive overview of the allegations associated with mining expansion between 2010 and 2024, highlighting the breadth and persistence of social, labor and environmental challenges linked to the sector. Impacts on local communities dominate the reported allegations (802 allegations, Figure 11a), with particularly high incidence related to attacks against human rights defenders, loss of livelihoods, land rights violations and land grabbing. These patterns point to recurring tensions between mining projects and surrounding communities, often rooted in inadequate consultation processes, weak protection of customary land rights and uneven distribution of economic benefits. Allegations related to workers' conditions remain substantial (223 allegations, Figure 11b), with occupational health and safety issues emerging as the most frequently reported concern. Work-related deaths and restrictions on freedom of association further underscore persistent deficits in labor protection, especially in contexts characterized by weak enforcement and high levels of informality. Environmental allegations are the second most numerous, with 713 allegations (Figure 11c), where water pollution, restricted access to water and the creation of toxic or uninhabitable environments accounting for a large share of reported cases (77% of the allegations). The high incidence of water-related allegations has direct and severe implications for human health, while simultaneously exacerbating existing water scarcity pressures. In regions already experiencing water stress, mining-related water pollution and restricted access to clean water can intensify competition among users, undermine local livelihoods and increase vulnerability to water-borne diseases. These impacts are particularly acute in rural and peri-urban areas where communities rely heavily on surface and groundwater resources for drinking, agriculture and sanitation, and where alternative water supplies are limited.

[21] Transition Minerals Tracker: 2025 Global Analysis - Business and Human Rights Centre

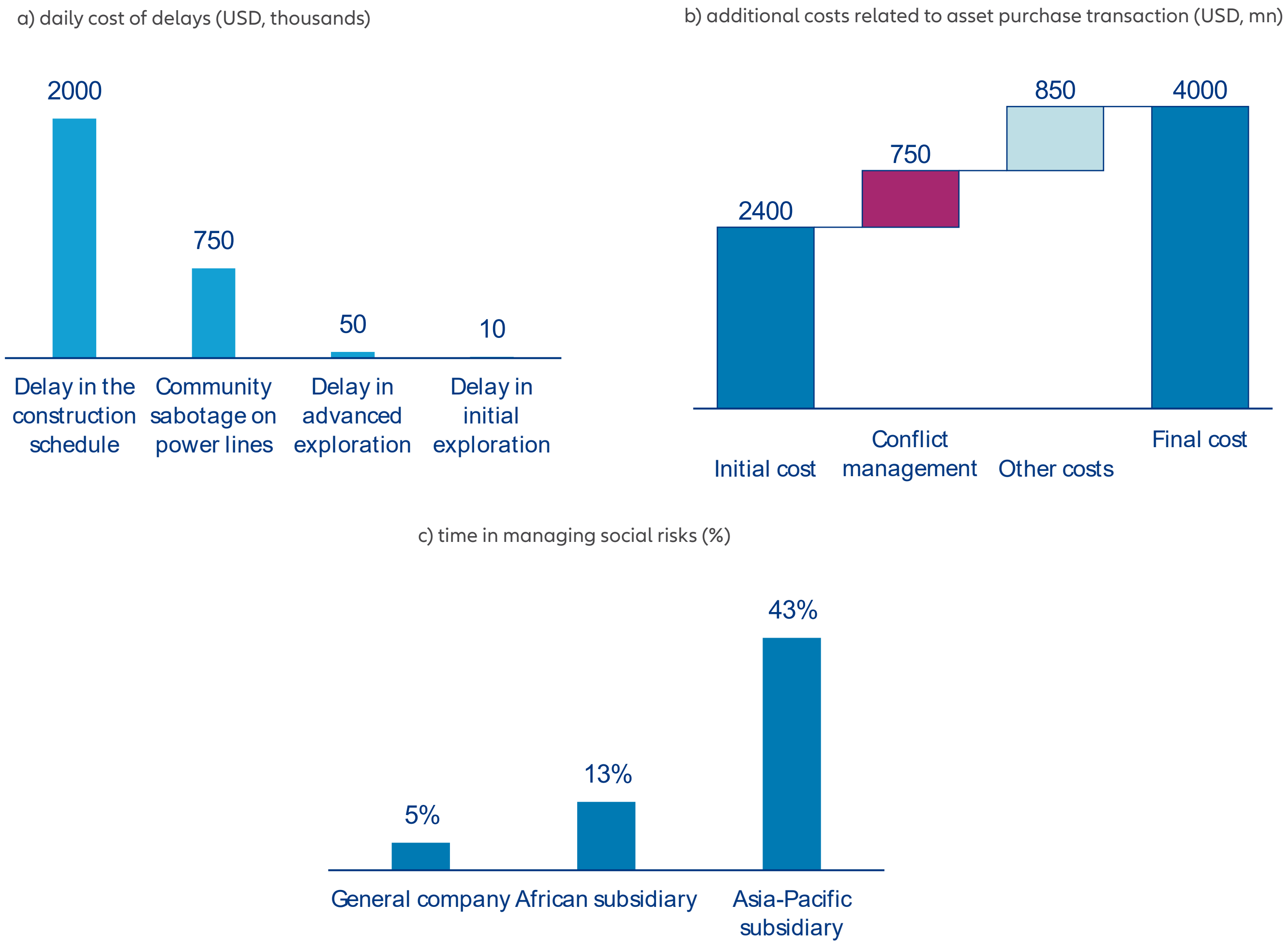
Figure 11: Number of allegations associated with mining expansion (2010 – 2024)



Sources: Business and Human Rights Centre, Transition Minerals Tracker, Allianz Research

Social and environmental controversies in the mining sector entail substantial economic costs that extend well beyond reputational damage. Evidence from the sustainability consultant Environmental Resources Management (ERM) indicates that 46% of major mining capital projects failed to meet planned delivery timelines between 2008 and 2016. Importantly, these delays were largely attributable to non-technical factors, with community opposition accounting for 42% of delayed projects and environmental concerns for 35%[22]. This highlights the growing role of social acceptance and environmental governance as binding constraints on project implementation. More recent ERM findings further confirm this pattern, showing that 62% of mining projects delayed at the permitting stage were affected by stakeholder opposition or concerns related to environmental impacts, rather than administrative or engineering limitations. The financial implications of such delays can be severe. Another joint study[23] estimates that social conflict can cost large-scale mining operations up to USD20mn per week of delayed production (see Figure 12 below), primarily through lost revenues. Beyond direct production losses, mining companies also face indirect costs that are often underestimated, including the diversion of senior management time as well as higher insurance premiums and increased legal expenses.

Figure 12: Economic costs of mining-related conflicts as reported by expert assessments



Sources: Davis and Franks (2014), Allianz Research

[22] Critical Mineral Security and the Social License to Operate | Oxfam
[23] Costs_of_Conflict_Davis-Franks.pdf

These aggregate cost estimates are further illustrated by the expert-based evidence presented in Figure 12, which decomposes the economic impacts of mining-related conflicts across different channels. Delays in construction schedules emerge as the most financially consequential outcome, with expert assessments indicating losses of up to USD2mn per day of delay (Figure 12a). These costs are driven not only by additional operational and financing expenses, but more importantly by postponed revenue streams once production is delayed. Disruptions linked to community opposition, such as sabotage of infrastructure or interruptions to power supply, add further costs, while delays during advanced exploration and initial production phases, though smaller in absolute terms, compound overall project risk.

Conflict-related costs extend beyond project timelines. In asset transactions, unresolved social issues can significantly inflate the expected total cost of ownership beyond the agreed acquisition prices. Expert evidence suggests that, in some cases, up to USD750mn in additional costs have been incurred to address community-related liabilities that were not fully identified at the time of purchase (Figure 12b). Moreover, the managerial burden associated with social risk management is substantial. While firms often assume that around 5% of senior management time is sufficient to address social risks, this share rises to 10–15% in African subsidiaries and 35–50% in parts of the Asia-Pacific region, reflecting the intensity and persistence of local conflicts (Figure 12c). These findings show how social and environmental conflicts constitute material economic risks for mining projects and should therefore be treated as core components of project risk assessment rather than peripheral externalities.

The trillion-dollar build out: Capital needs across mining, processing and circularity

Mining companies are being asked to do two contradictory things at once: raise output fast enough to meet an AI-and-electrification demand shock, while shrinking the footprint per ton as ore grades fall and scrutiny tightens. What is asked from the mining sector is quite explicit: roughly USD800bn of mining investment between now and 2040 in a net-zero pathway, with copper alone needing about USD490bn. Yet, the financing pipeline is behind schedule as less than half of the additional investment needed for net-zero critical-material demand is "on track", with long development lead times. Furthermore, to green mining operations, the sector also needs to take concrete action: electrify mines and lock in water systems for scarcity, treat biodiversity as critical and raise tailings and closure liabilities. Ore-grade decline makes this non-negotiable. For instance, copper ore grades in Chile have fallen about 30% over 15 years, mechanically lifting energy, emissions and water intensity (see Figure 13).

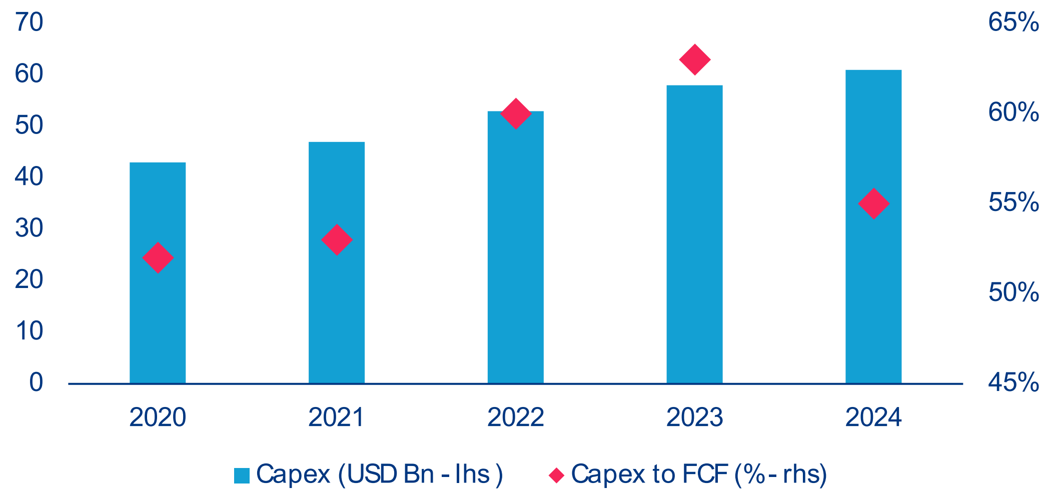
Figure 13: Concentrate copper ore grade in Chile (in %, 2005-2019)

Sources: Davis and Franks (2014), Allianz Research

Demand is no longer a single-cycle story but a stacked set of pull-throughs and it changes what "enough supply" means for corporate strategy. As mentioned above, metals demand is significantly boosted by transition needs. AI adds a second-order but fast-moving vector: global data-center electricity demand could rise from about 415 TWh in 2024 to about 945 TWh in 2030. That argues for capital allocation that looks unfashionable from a pure upstream lens: mine-to-chemical and mine-to-metal chains where economics allow, more long-dated offtake structures with end-users who need certainty and selective investments in midstream capacity in the US and EU to reduce single-point-of-failure exposure, especially where policy is explicitly pulling for it.

The investment program has to be treated as an industrial mobilization problem rather than incremental growth capex. Investment growth slowed in 2024 as prices fell, even though critical-minerals mining investment still grew. Looking at top mining firms, we also notice that 2024 marked a fall in capex-to-free cash flows, underlining that miners scaled back investments relative to their capacities despite headline numbers pointing towards more investments (see Figure 14). It also stresses that low prices and weaker cash flow constrain the industry's ability to fund future growth. Copper illustrates the shape of the problem: It is ubiquitous in grids and electrification, and ore grades are falling, which raises capital per ton and stretches timelines. As a potential copper supply shortfall could materialize alongside the reality that top projects can take around 17 years from discovery to production, decisive action is required sooner rather than later. The implication is that miners cannot wait for perfect price signals: they need to lock in capital through counter-cyclical investment and risk-sharing structures. For example it could be prepayments, streaming where appropriate, and co-investment with automotive manufacturers and suppliers, utilities and even tech companies. At the same time, they should stop treating "ESG capital" as a marketing category and instead treat it as a cost-of-capital constraint.

Figure 14: Capex in the largest 15 global mining companies



Sources: LSEG Workspace, Allianz Research

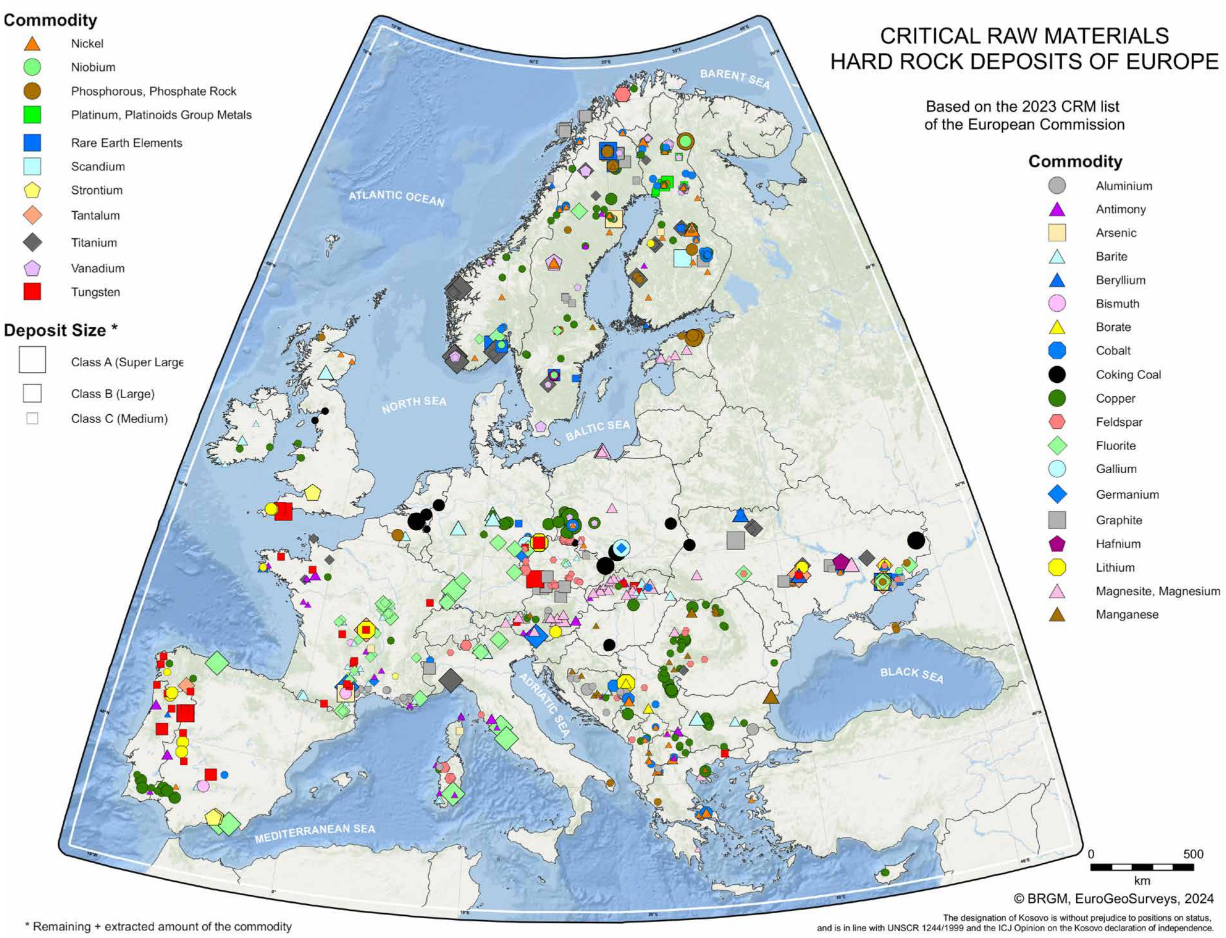
Sustainability performance is now a production input because permitting, insurance and community consent are binding constraints, and miners should manage them like they manage grade and recovery rates. The sustainability review points to the operational reality that miners' experience can worsen as grades decline, growing energy use, emissions and water use even with good intentions. That is precisely why "do more" has to mean "do differently", with measurable unit-intensity reductions and credible transition plans. Nature and biodiversity are not peripheral either: mining's impacts span land-use change, ecosystem disruption, pollution and waste through the full value chain, and sector frameworks converging into a clearer expectation in terms of risk management. Companies should assume that "partial" reporting will be treated as risk. Going forward, firms might need to go towards location-specific impact accounting (e.g. total spatial footprint, disturbed versus restored land, water withdrawals, pollutants etc.). This will be increasingly asked in a form that can be audited and compared. The same logic applies to tailings and closure: provisions for closure and rehabilitation should be treated as capital allocation, not footnotes, and structured so they survive commodity cycles. Social license must be operationalized: Robust consultation and benefit-sharing with indigenous communities where relevant is no longer reputational hygiene but a schedule-control variable, as evidenced by permitting reversals and court-driven requirements for deeper community input in multiple jurisdictions. In essence, miners should stop arguing that sustainability slows growth and start showing that sustainability is what makes growth deliverable: fewer stoppages, fewer tail risks, lower sovereign-act exposure and more stable long-term returns.

Circularity and low-carbon production are the only credible relief valves, but they would take the total bill to over a trillion dollars to 2040. The circular-economy discussion on mining illustrates both promise and constraint: resource-efficiency strategies can materially reduce primary copper demand by mid-century in an upside scenario, yet the time lag is structural because in-use metals have multi-decade lifetimes. For copper, average use-life is cited around 23 years, meaning secondary supply responds slowly to today's demand shock. That pushes miners towards a two-track strategy. First, build "low-carbon tons" as a differentiated product: electrify fleets where feasible, secure renewable PPAs for power-hungry processing, invest in heat and process electrification and disclose emissions and energy in ways that allow customers to book credible Scope 3 reductions. Second, invest selectively in recycling, urban mining and scrap-processing capacity as a growth business and a hedge against ore-grade decline: some meaningful existing recycling capacity imply that industrial-scale models exist but require upstream collection and downstream refining partnerships to grow. For critical and strategic materials with concentrated refining footprints (e.g. graphite, rare earths, cobalt), value-chain map makes the point that de-risking is often about chemicals, separation and purification rather than digging alone, and those are precisely where sustainability missteps (wastewater, hazardous by-products, local air pollution) can kill permits and social acceptance. Based on financial and transition disclosures from large mining companies which committed to strong sustainability targets, we can assume that 6% of capex should be spent for operational decarbonization, 8% for tailings/waste/closure, 3% for recycling, 2% for water & pollution controls and 1% for nature/community. Scaling this to the industry would mean about USD450bn in investments to 2040. However, some of this overlaps with the headline USD800bn. Indeed that number takes into account new mining projects that would be built in a sustainable way. With a conservative 1/3 overlap, it would mean USD1.1trn total investments to 2040: USD800bn for transition needs and another USD300bn to make mining companies sustainable.

From liabilities to legitimacy: Mining done right

Low-risk hard-rock and geothermal brine resources are essential for sustainable transition mining. Building a resilient supply of transition minerals requires balancing geological endowment with political and sustainability risks. Country-level indicators are a useful first filter, but extractive investments are additionally exposed to deal-level sovereign risk (including unexpected regulatory intervention) and to social-license risk when communities contest environmental impacts or benefit sharing. A pragmatic portfolio therefore combines domestic European resources and partnerships with stable jurisdictions, while reducing exposure to high-risk, high-concentration supply nodes. In Europe, the Geological Service for Europe (GSEU) has compiled a harmonized map of the largest medium-to-super-large hard-rock deposits for the EU's 2023 critical raw materials list (Figure 15). The map illustrates that potential critical raw material supply is geographically diverse across Europe, with prominent clusters in the Nordic shield, the Iberian Peninsula and parts of Central and South-Eastern Europe. Importantly, the dataset is a geological assessment and does not by itself imply economic, permitted or socially acceptable extractability; infrastructure, processing capacity, permitting timelines and environmental constraints remain decisive.

Figure 15: Critical Raw Materials hard-rock deposits of Europe



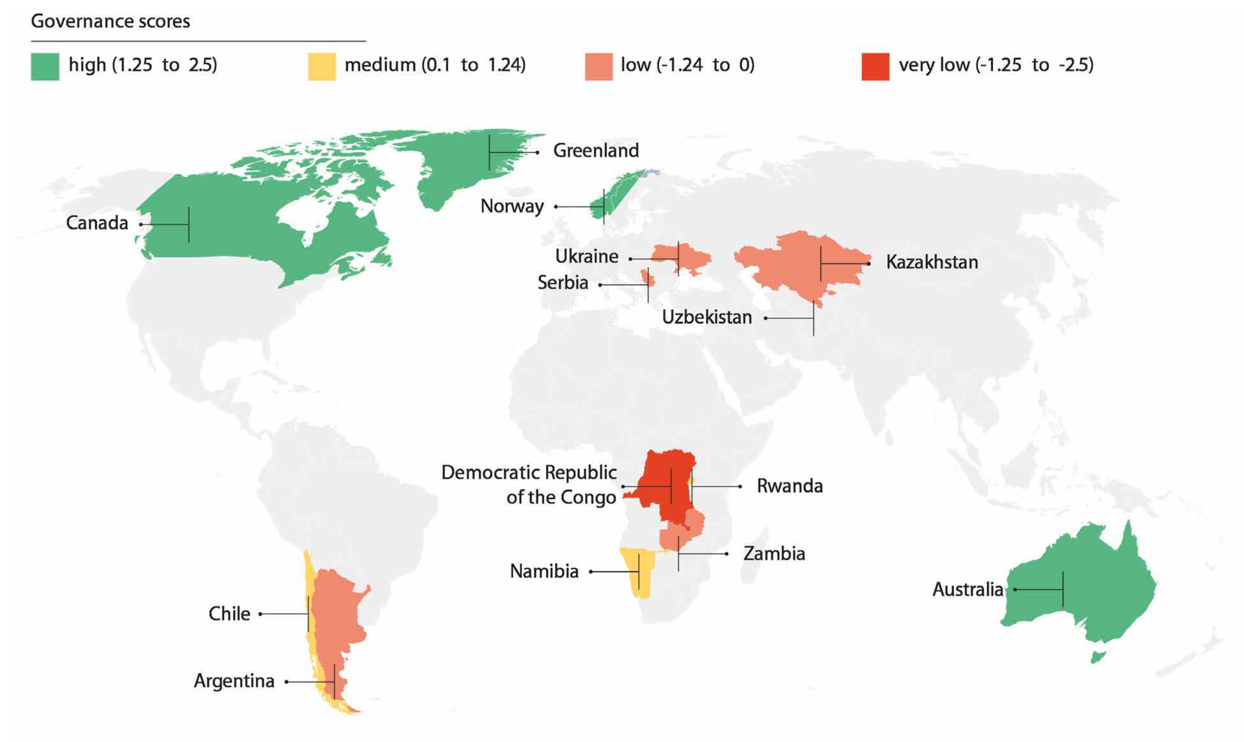
Sources: GSEU[24]; Allianz Research

Geothermal brines add a complementary co-production option. Many geothermal systems circulate large volumes of highly mineralized fluids that can be enriched in elements such as lithium, rubidium, antimony or tungsten and mineral recovery can be integrated with geothermal power and heat where brines are already produced and typically reinjected. Compared with conventional mining, this can reduce incremental land disturbance and avoid ore comminution and leaching, but sustainability is not automatic: site-specific life-cycle assessments show wide ranges in climate-change impacts for lithium carbonate from geothermal brines (Salton Sea, US, and Upper Rhine Graben, Germany), and impacts can exceed commonly used database values when drilling intensity is high or fossil energy is used. This reinforces the case for prioritizing projects in low-risk jurisdictions with enforceable environmental standards, and for transparent, site-specific assessment from the earliest project stages.

The EU's expanding network of strategic raw materials partnerships (e.g., MoUs/partnerships with Namibia, Zambia, the Democratic Republic of the Congo (DRC), Rwanda, Chile, Serbia, Norway, Uzbekistan and Australia) is intended to diversify supply and embed "sustainable mining" principles across upstream, processing and value chain development. However, where diversification occurs matters just as much as how much diversification happens: partner jurisdictions differ markedly in governance quality, which is a first order driver of both supply reliability and real world sustainability outcomes.

[24] Albert, C. & Bertrand, G., 2025. Map of Critical Raw Materials hard rock deposits of Europe 2024. Geological Service for Europe

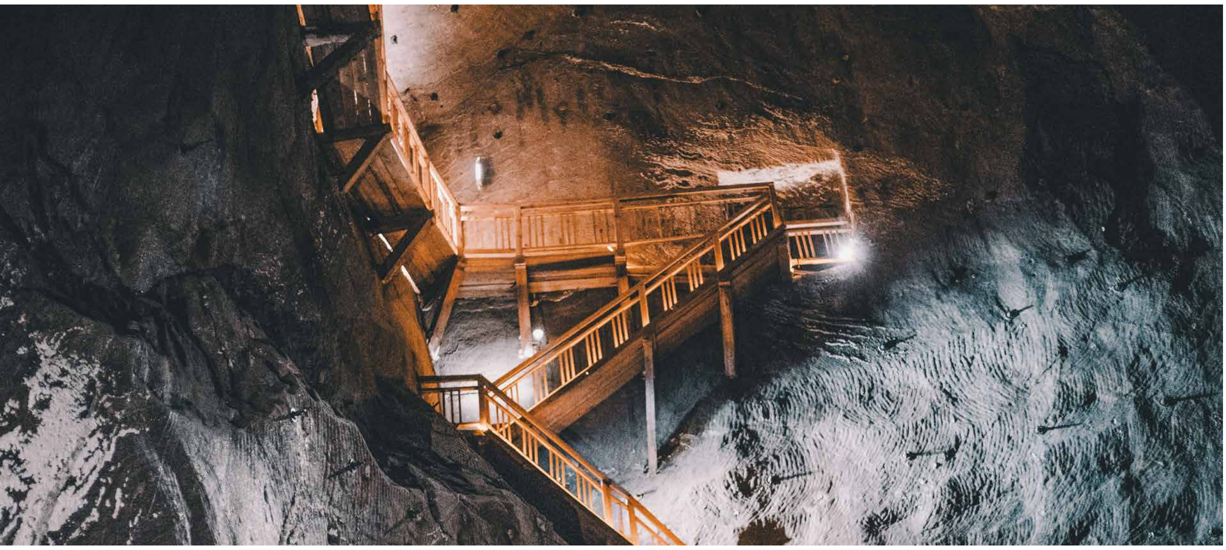
Figure 16: Governance quality of EU's strategic partnership countries



Sources: ECA[25]; Allianz Research

A useful "reality check" comes from the European Court of Auditors (ECA), which mapped the EU's 14 strategic partnership countries (January 2021–June 2025) against the World Bank's Worldwide Governance Indicators (WGI, 2023).[26] The results underline the heterogeneity of the partner set: seven of the 14 partnership countries fall into the "low governance" category, and the distribution spans from high governance partners (e.g., Canada, Norway, Australia) to very low governance jurisdictions (notably the DRC) (Figure 16).

[25] ECA (2026). Special report 04/2026: Critical raw materials for the energy transition – Not a rock-solid policy.
[26] ECA (2026). Special report 04/2026: Critical raw materials for the energy transition – Not a rock-solid policy.



Sustainable mining is not optional - it is essential

Conducting mining with poor sustainability performance is not a one-time trade-off; it is a compounding liability. Long-lived obligations include tailings stewardship, community harm, land rehabilitation, post-closure monitoring and, in many cases, long-duration water treatment (Table 1). Financial assurance research highlights that post-closure costs, especially water treatment, can materially increase total closure costs, and that conservative assumptions and independently secured financing instruments to cover closure and maintenance costs are needed.[27] Tailings failures are low-frequency but high-severity events; modern governance expectations emphasize full lifecycle accountability, independent review, emergency preparedness and disclosure.[28] Biodiversity and water impacts increasingly influence permitting, cost of capital and insurability, reinforcing the need for nature-related assessment and disclosure;[29] Underestimation and underfunding reserves for rehabilitation can create a 'decommissioning gap' whereby industrial sites are never remediated to an acceptable form. This adds risks to investors being left with budget exceedances, or in the event that the operator ceases to exist, can leave society with the costs of either cleaning up the sites, or dealing with the damages they create.[30] This is where more sustainable practices can be a win-win. As communities, regulators, governments and customers of mining companies set a higher, clearer, and more reliable expectation on mining operations – before projects enter the scoping phase – costs can better be planned, stakeholder engagement conducted efficiently and projects greenlit more quickly with higher levels of trust in expedited approvals being followed by best in class operations. This helps meet the growing demand by bringing online more mining, faster, with a high level of certainty that the practices will meet stakeholder expectations.

- Policy priority: require robust financial assurance and update closure plans regularly; avoid reliance on self-bonding or optimistic discounting.
- Market priority: treat tailings and nature risks as governance and disclosure issues, not only engineering issues.

27 Chambers, David M. 2024. Net Present Value Calculations for Mining Post-Closure Financial Assurance. Mine Water and the Environment.
28 International Council on Mining and Metals (ICMM), United Nations Environment Programme (UNEP), and Principles for Responsible Investment (PRI). 2020. Global Industry Standard on Tailings Management.
29 Taskforce on Nature-related Financial Disclosures (TNFD). 2023. Recommendations of the Taskforce on Nature-related Financial Disclosures
30 European Commission. 2023. "Commission Delegated Regulation (EU) 2023/2772 … Annex I: ESRS E4

Table 1: Estimated decommissioning costs for the mining sector in respective regions for the existing infrastructure

USDbn	Mining
Africa	32
Asia	555
Australia	37
Europe	189
North America	138
South America	252
Total	1,203

Sources: BNP Paribas[31]; Allianz Research

The good news: we know that better is possible. Mining operations that can benefit communities, keep workers safe, limit environmental harms and create value for shareholders while leaving a net-positive legacy already exist. Global standards and a plethora of national or regional laws and regulations, cumulatively, demonstrate best practices that are sufficient to making mining projects attractive and beneficial to many. Pulling together key approaches to the sector and standardizing best practices by making them requirements for expedited project approval would help ensure that good practices are followed and that they meet societal needs. Such notable standards include the Initiative for Responsible Mining Assurance (IRMA), the Global Standard on Mining Tailings and Safety (GISTM), the International Council on Mining and Metals (ICMM) member guidelines and principles and the developing Consolidated Mining Standard Initiative (CMSI).

All of these standards together create a safety net that covers some of the most critical topics for the sector: (1) integrated closure planning and conservative financial assurance, (2) tailings safety frameworks and independent assurance in line with the Global Investors Standard on Tailings Management[32] (Figure 17), (3) community participation and benefit-sharing, (4) biodiversity and pollution prevention, (5) due diligence and traceability, (6) electrification and automation to reduce operational emissions and safety risks and once materials enter the economic system, by design we should focus on (7) circularity and recyclability by design to reduce long-term primary demand.[33] [34] [35]

There is a risk that as the standards and expectations proliferate and overlap, they create an untenable web of expectations, making it difficult and frustrating for industry and its stakeholders to converse on a common language and set of expectations. That is why key players in the financial sector, with the success of GISTM providing a clear example, are now coming together again with a broad set of stakeholders within the Mining 2030 initiative. Mining 2030 has set a 10 year vision that is supportive of the industry while detailing what is needed to make mining attractive again to all its critical stakeholders and investors, and address the growing gap in raw materials available to meet the needs of economic growth and technological development. The aim is to raise the attractiveness of the mining sector for investors, community members, customers and consumers to support its expansion with stronger conviction of consistent adherence to best practices. As detailed by academics from the University of British Columbia, these sorts of best practices often start with the companies themselves, and then garner legitimacy via societal partnerships and further stakeholder involvement. They later need to become minimum expectations or regulated requirements to really ensure a new level, but higher, standard of operations. The best practices and expectations of investors being developed by Mining 2030 will ultimately start to inform the minimum expectations of mining companies in the future.

[31] BNP Paribas Asset Management (2023). Decommissioning Stranded Energy Assets: A USD 8 Trillion Challenge.
[32] GTMI. Making Mine Tailings Facilities Safer For People & The Environment.
[33] Environment Programme (UNEP), and Principles for Responsible Investment (PRI). 2020. Global Industry Standard on Tailings Management. November 28, 2020
[34] United Nations Environment Programme (UNEP). 2025. Financing the Responsible Supply of Energy Transition Minerals for Sustainable Development. October 9, 2025
[35] OECD. 2023. Handbook on Environmental Due Diligence in Mineral Supply Chains. Paris: OECD Publishing

Figure 17: GISTM global benchmark - comprising 6 topic areas and 15 principles

Sources: Source: GISTM[36]; Allianz Research

[36] GISTM. Global Industry Standard on Tailings Management.

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